PHY 104: General Physics IV (Vibration, Waves and Optics)

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Simple Harmonic Motion (SHM)

The Fundamental Vibrational Motion

- Restoring force proportional to displacement
- $F = -kx$ (Hooke's Law)
- Examples:
  - Mass-spring systems
  - Pendulums (small angles)
  - Molecular vibrations

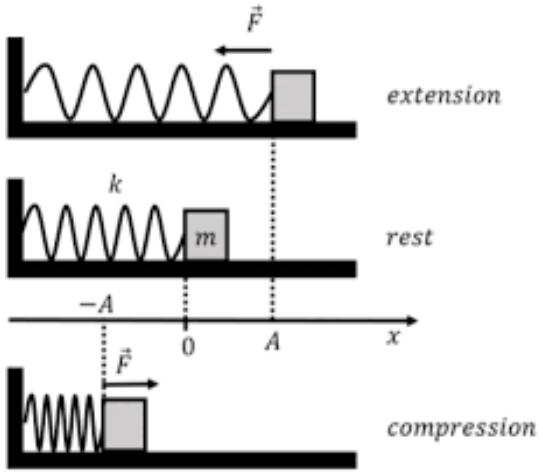


Figure: Mass-spring system

Equation of Motion

$$\begin{aligned} ma &= F \quad (\text{restoring force}) \\ &= -kx \end{aligned}$$

$$ma + kx = 0 \Rightarrow m\ddot{x} + kx = 0 \quad \Rightarrow \ddot{x} + \frac{k}{m}x = 0$$

SHM Fundamentals

Defining Equation

$$\ddot{x} + \omega_0^2 x = 0$$

$$\text{where} \quad \omega_0^2 = \frac{k}{m} \quad \text{or} \quad \omega_0 = \sqrt{\frac{k}{m}}$$

SHM Fundamentals

Q: I thought SHM was oscillatory motion (sine and cosine)?
A: Yes, when we solve the SHM equation

Recall:

$$\ddot{x} \propto -x$$

Q: What function can you think of which, when we differentiate it twice, we get the same function?
A: (1) Exponential and (2) Sine/Cosine
Actually, we don't want to get the same function but the same function with a negative sign. Sine/Cosine work

SHM Fundamentals

$$\frac{d\cos(qt)}{dt} = -q\sin(qt)$$

Differentiate again:

$$\frac{d^2\cos(qt)}{dt^2} = \frac{d[-q\sin(qt)]}{dt} = q[-q\cos(qt)] = -q^2\cos(qt)$$

So, we can set q^2 to ω_0^2

Exercise: What of $\sin(qt)$? or $\cos(qt+\phi)$ or $\sin(qt+\psi)$? Would any of these work?

Equation of Motion & Solutions

$$\ddot{x} + \omega_0^2 x = 0 \quad \text{where} \quad \omega_0 = \sqrt{\frac{k}{m}}$$

$$x(t) = A\cos(\omega t + \phi) \quad \text{or} \quad x(t) = A\sin(\omega t + \psi)$$

$$\omega = \sqrt{\tfrac{k}{m}}, \quad \omega = 2\pi f, \quad f = \tfrac{1}{T}, \quad \Rightarrow T = 2\pi\sqrt{\tfrac{m}{k}}$$

Exercise: How is ϕ related to ψ?

SHM Fundamentals

Defining Equation

$$\ddot{x} + \omega_0^2 x = 0 \quad \text{where} \quad \omega_0 = \sqrt{\frac{k}{m}}$$

Solution:

$$x(t) = A\cos(\omega_0 t + \phi)$$

$$v(t) = \frac{dx}{dt} = -\omega_0 A \sin(\omega_0 t + \phi)$$

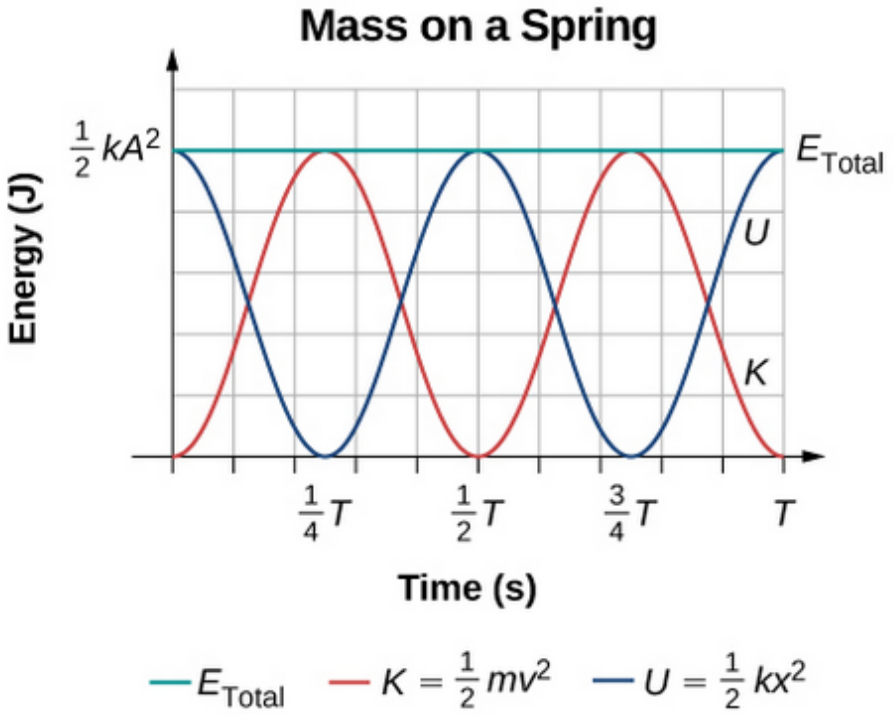
- A: Amplitude
- ϕ: Phase constant

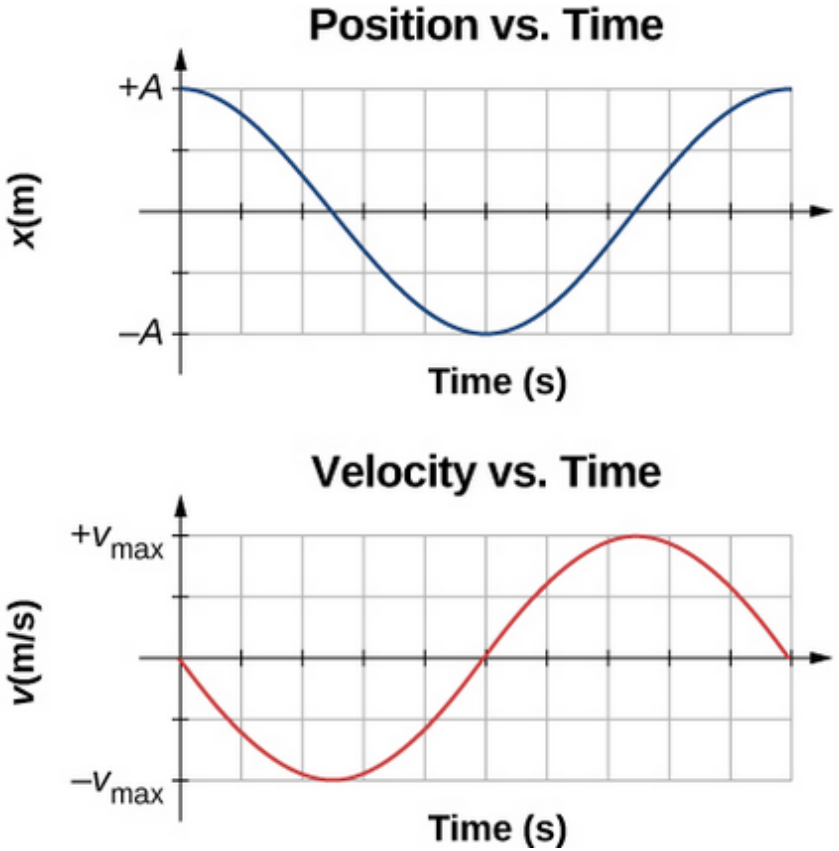
- Kinetic Energy:
  $\frac{1}{2}mv^2 = \frac{1}{2}m\omega_0^2 A^2 \sin^2(\omega_0 t + \phi)$
- Potential Energy:
  $\frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2(\omega_0 t + \phi)$

Total Energy:

$$E = \frac{1}{2}kA^2 = \text{constant. Prove it!}$$

SHM Energies, Position, and Velocity





Problems

Uche's resting heartbeat (pulse) is 60 beats per minute (bpm). While he is doing some house chores, it rises to 80 bpm.
A. Determine, for both cases:
i) the period of the (complex) motion
ii) the frequency of the (complex) motion

B. Is the beating of Uche's heart SHM?
Is the beating of Uche's heart Periodic?

PHY104 Week2: Simple Harmonic Motion and Beyond

From Pendulums to Resonance

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September 3, 2025



Overview

1. Simple Harmonic Motion (SHM)
2. Damped Oscillations
3. Resonance
4. Coupled Oscillations
5. Summary

Simple Harmonic Motion

Q: A motion that repeats itself at regular intervals is a Simple Harmonic Motion (SHM):

1. True
2. False



What is Simple Harmonic Motion?

The Defining Equation

$$F = -kx$$

- F is the force
- x is the displacement from equilibrium
- k is a constant (e.g., spring constant)
- The **minus sign** shows the force is always opposite the displacement $\Rightarrow$ (Restoring Force)

Question: Why is it called HARMONIC? (Check Internet)

The Mathematics of SHM

Using Newton's Second Law ($F = ma$), we combine it with $F = -kx$ and $a = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$:

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx$$

Rearranging gives the fundamental **equation of motion for SHM**:

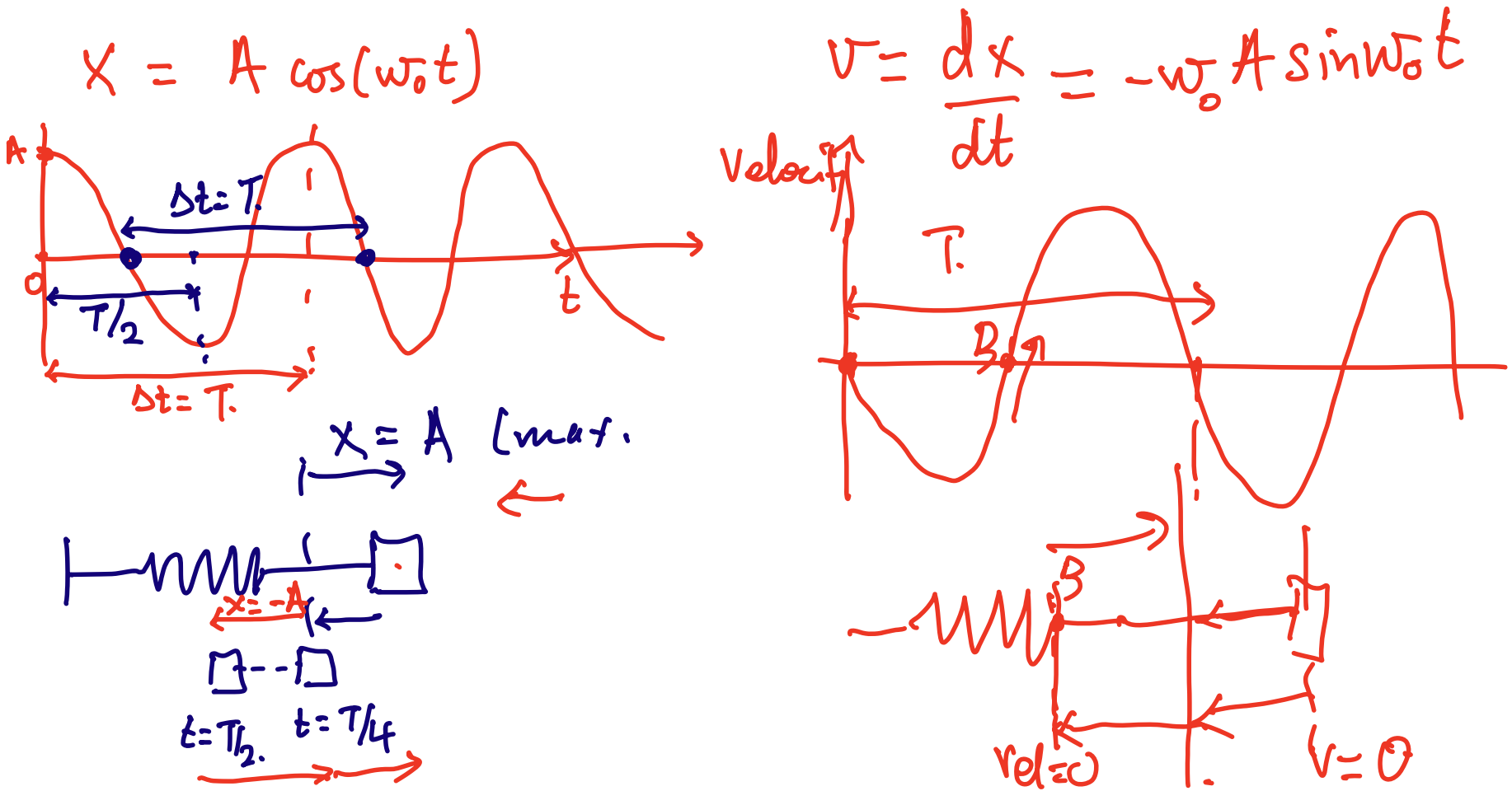
$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \omega_0^2 x = 0 \quad \text{where} \quad \omega_0 = \sqrt{\frac{k}{m}}$$

The solution to this differential equation describes the position as a function of time:

$$x(t) = A\cos(\omega_0 t + \phi)$$

- A: Amplitude (maximum displacement)
- ω_0: Angular frequency ($\mathrm{rad\,s^{-1}}$)
- ϕ: Phase constant (depends on initial conditions)

The Mathematics of SHM



The Mathematics of SHM: Max KE and Max PE

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$$

The Simple Pendulum

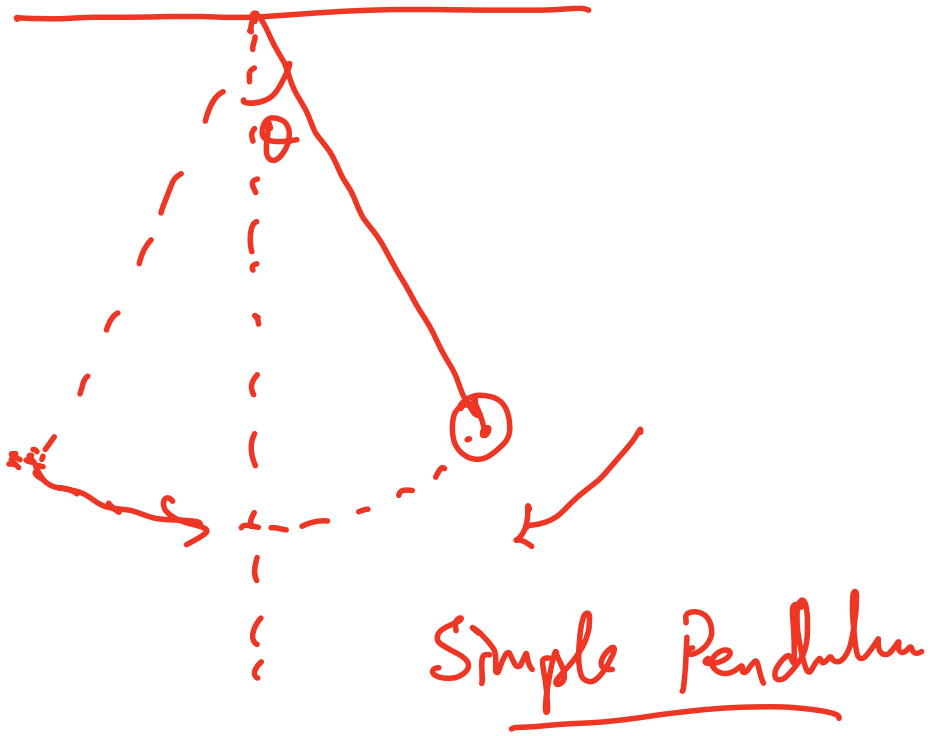
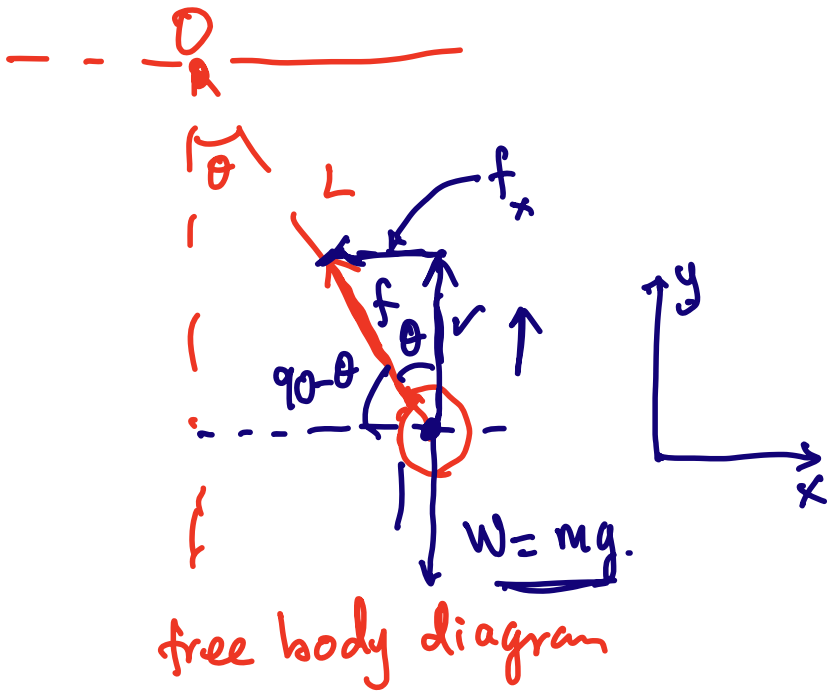


Figure: Forces on a simple pendulum.

For small angles ($\theta \lesssim 10^\circ \approx 0.2$ rad): The restoring force is proportional to displacement.

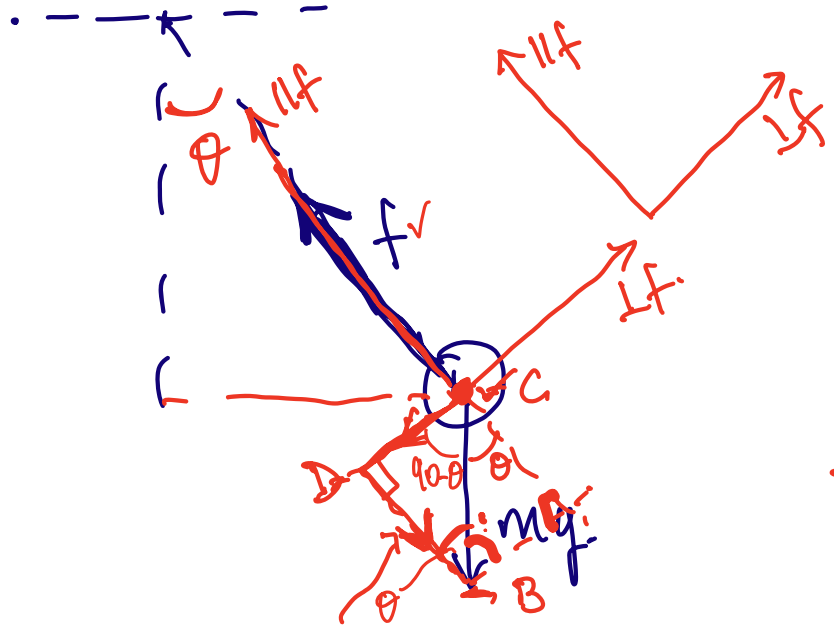
Simple Pendulum Derivation



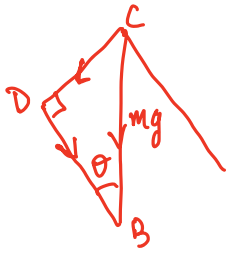
free body diagram

forces	accn
x: $0 - f_x$	
$0 - f\sin\theta$	ma_x
y: $f\cos\theta - mg.$	$ma_y.$

$$\vec{f} = m\vec{a}$$



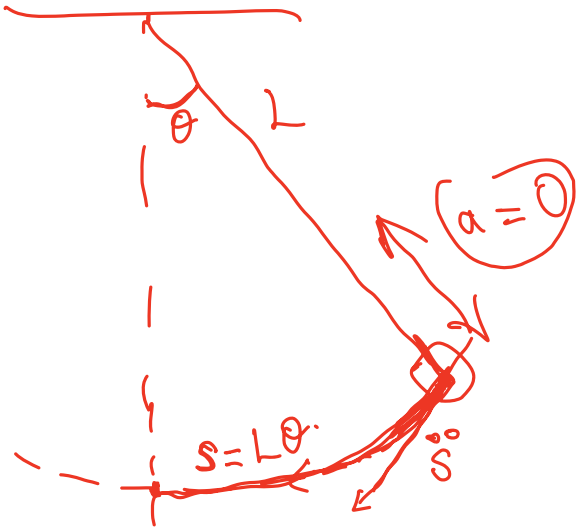
forces		accn
∥f:	$f - mg\cos\theta$	0
⊥f:	$0 - mg\sin\theta$	$\ddot{s} = L\ddot{\theta}$



$$\vec{CB} = \vec{CD} + \vec{DB} = mg\cos\theta \cdot$$

$\vec{CB}$ ↑ ⊥ to f

$\vec{CD}$ ↑ ∥ to f





$s = L\theta$ ← in radians

$s = L\theta$

$$\dot{s} = \frac{ds}{dt} = L\frac{d\theta}{dt} = L\dot{\theta}$$

$$\ddot{s} = \frac{d(L\dot{\theta})}{dt} = L\ddot{\theta}$$



$s = r\theta$

$s = L\theta$

$f - mg\cos\theta = m\ddot{0}$

$-mg\sin\theta = m\ddot{s}$

$= mL\ddot{\theta}$

$\Rightarrow mL\ddot{\theta} = -mg\sin\theta$

$\ddot{\theta} = -\frac{g}{L}\sin\theta \Rightarrow \ddot{\theta} = -\omega_0^2 \sin\theta \Rightarrow \ddot{\theta} \approx -\omega_0^2\theta$ if θ is small

$\ddot{\theta} + \frac{g}{L}\sin\theta = 0$

$\omega_0^2 = g/L$

$\ddot{x} = -\omega_0^2 x$

$\ddot{y} = -\omega_0^2 y$

$\ddot{\beta} = -\omega_0^2 \beta$

$\omega_0^2 = g/L$

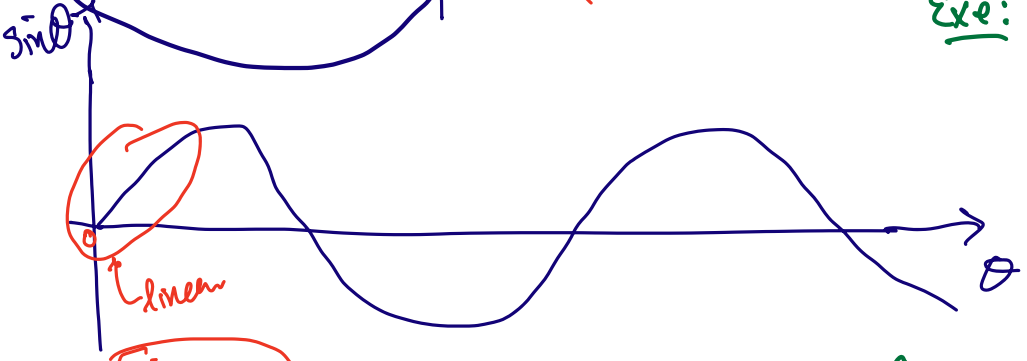
$\omega_0 = \sqrt{g/L}$

$\frac{2\pi}{T}$

$\Rightarrow T = 2\pi\sqrt{\frac{L}{g}}$ ✓

$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots$

$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots$

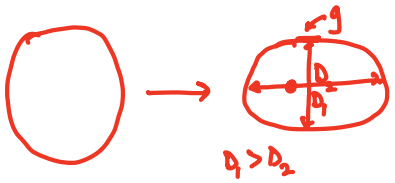


$\sin\theta \approx \theta$ for small θ

$\sin 0.1 = 0.099 \approx 0.1$

↳ radians

sin



Exe: If the clock on earth is taken to the moon, by how much should we adjust the length.

$\frac{L}{g} = \frac{L/6}{g/6}$

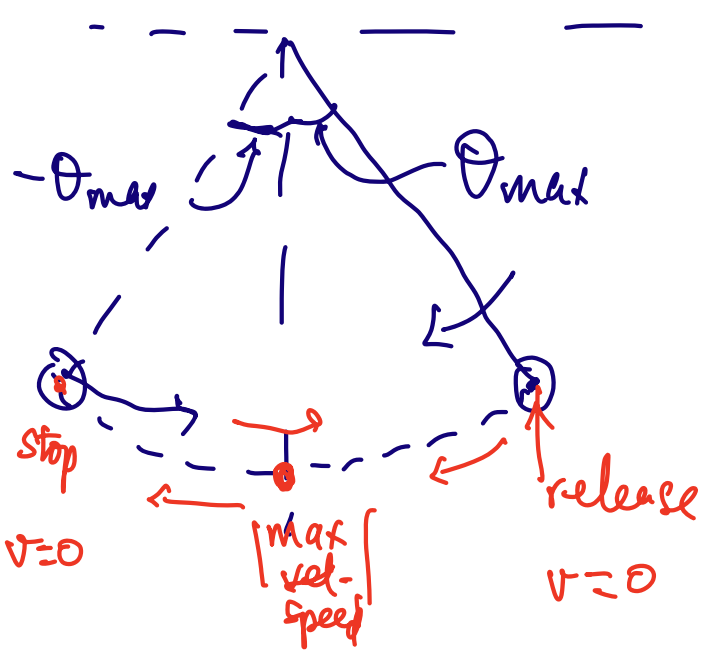
$g_{moon} \approx \frac{1}{6} g_{earth}$

$$\ddot{\theta} = -\omega_0^2 \theta \quad \text{or} \quad \frac{d^2\theta}{dt^2} = -\omega_0^2 \theta$$

$\omega_0^2 = g/L$

$$\Rightarrow \theta(t) = \theta_{max} \cos(\omega_0 t) \checkmark$$

$$\theta(0) = \theta_{max} \cos(0) = 1$$

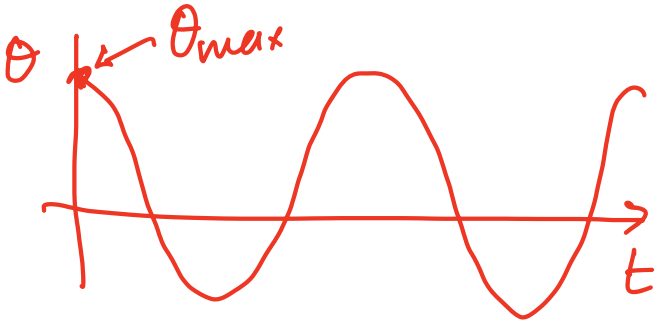


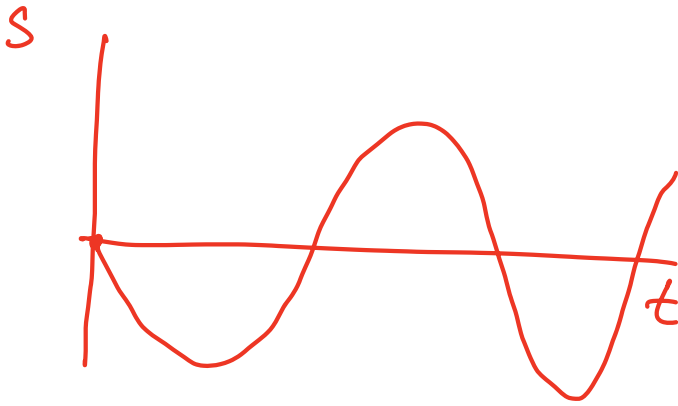
$$s = L\theta$$

$$v = \frac{ds}{dt} = \frac{d}{dt}(L\theta) = L\frac{d\theta}{dt}$$

$$= L(-\omega_0 \theta_{max} \sin\omega_0 t)$$

$$= -\omega_0 L \theta_{max} \sin\omega_0 t.$$





Simple Pendulum: Total Energy is constant

$$E_{tot} = KE + PE$$

$$= \frac{1}{2}mv^2 + mgh.$$

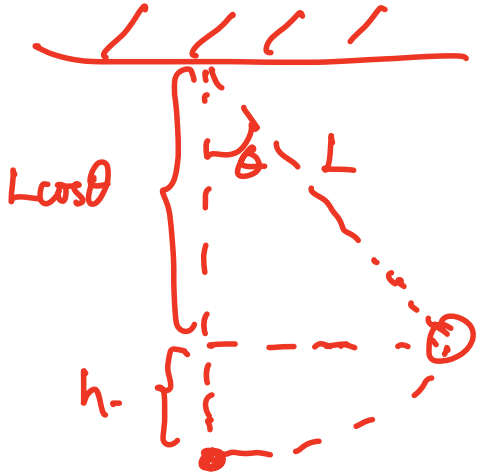
$$= \frac{1}{2}mv^2 + mgL(1-\cos\theta)$$

$$\approx (1-\frac{\theta^2}{2})$$

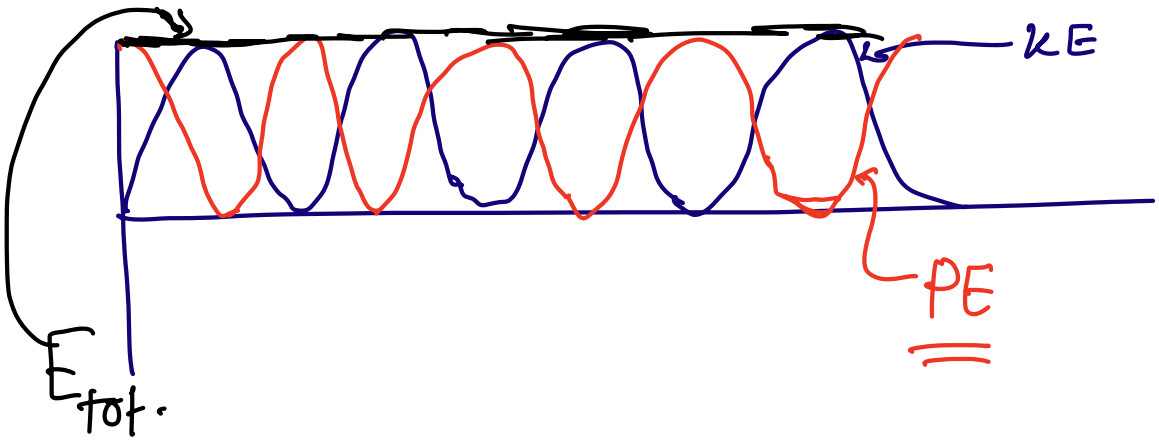
$$\approx \frac{1}{2}mv^2 + mgL\frac{\theta^2}{2} = \frac{1}{4}m\omega_0^2 L^2\theta_{max}^2 = \frac{1}{2}m\omega_0^2 A^2 = \text{constant}$$

Exe

$A \equiv L\theta_{max} = s_{max}$



$h = L - L\cos\theta.$


KE
PE
E tot.

The Simple Pendulum

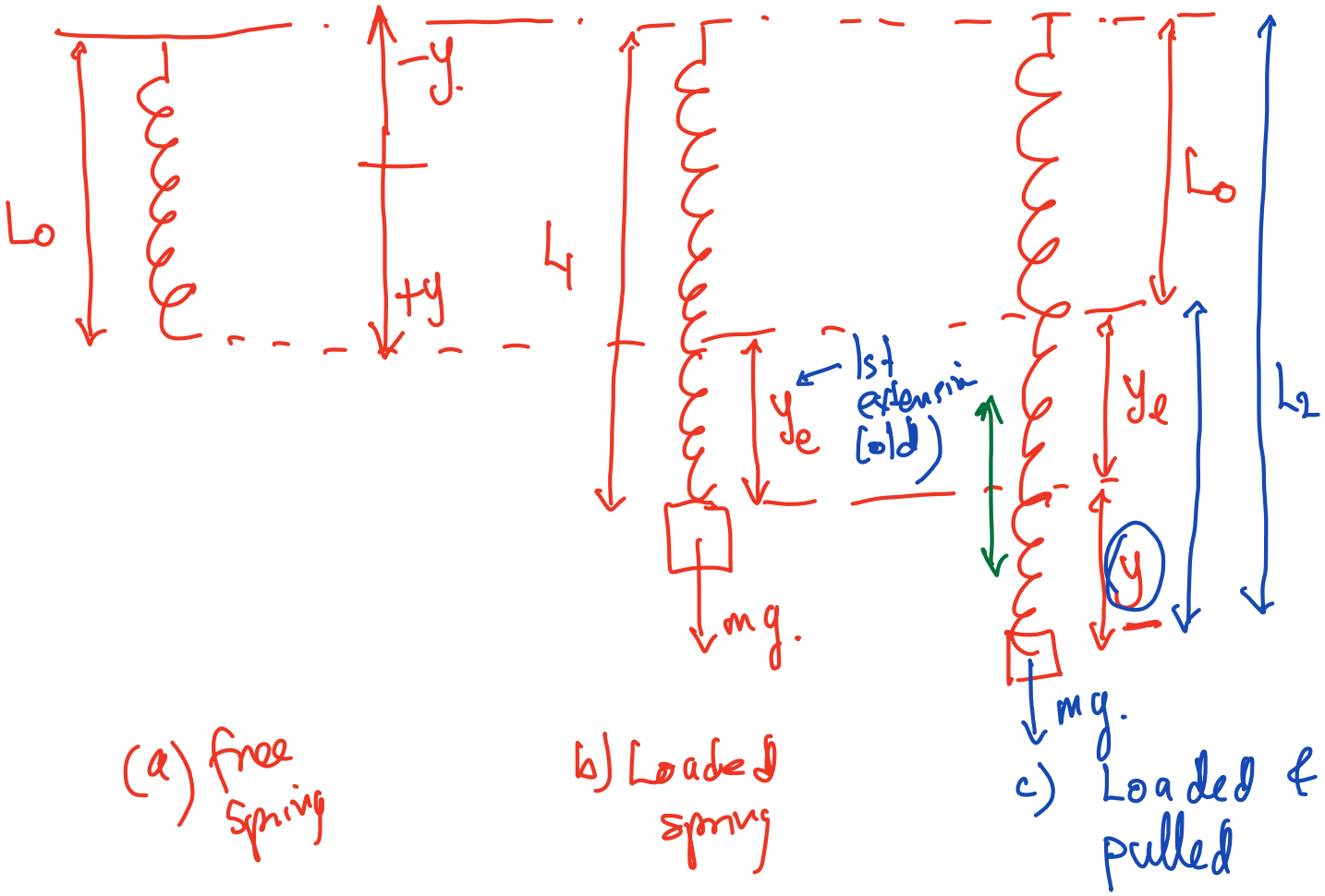
Period of a Simple Pendulum

$$T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{L}{g}}$$

The period depends only on length L and gravity g, not on mass m!

Exe: Determine the length of a simple pendulum whose period is 0.5s on earth. Repeat for the pendulum on a) the moon b) Mars

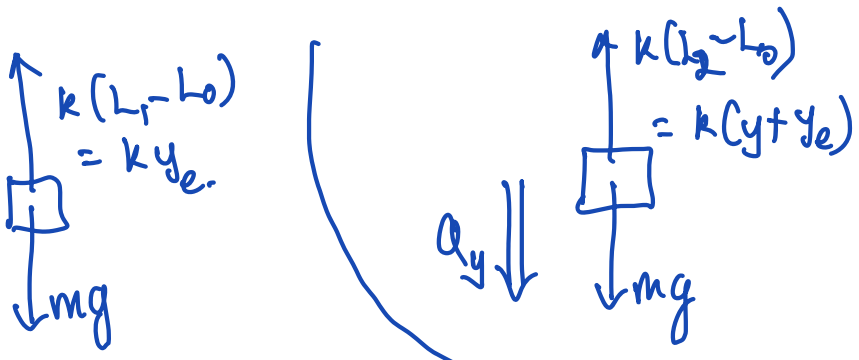
More Examples Of SHM.



(a) free spring

b) Loaded spring

c) Loaded & pulled



$mg - ky_e = ma$ → 0 in equilibrium

$mg - ky_e$ = net force

$mg - ky_e = 0$ ——(1)

$$mg - k(y_e + y) = ma \quad \text{—— (2)}$$

$$(mg - ky_e) - ky = m\ddot{y}$$

$mg - ky_e = 0$

$$\Rightarrow m\ddot{y} = -ky \quad \text{—— (3)}$$

$$\Rightarrow \ddot{y} = -\frac{k}{m} y$$

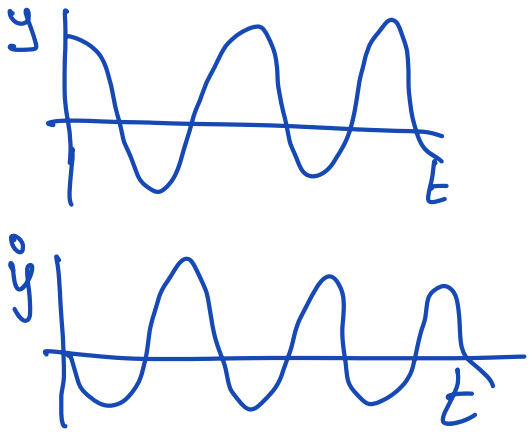
$$= -\omega_0^2 y$$

y → new extension

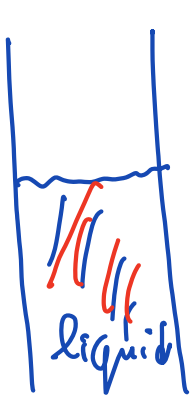
$$\boxed{\ddot{y} = -\omega_0^2 y} \Rightarrow \text{SHM.}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$



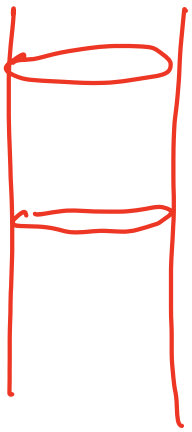
Loaded test tube



(a)

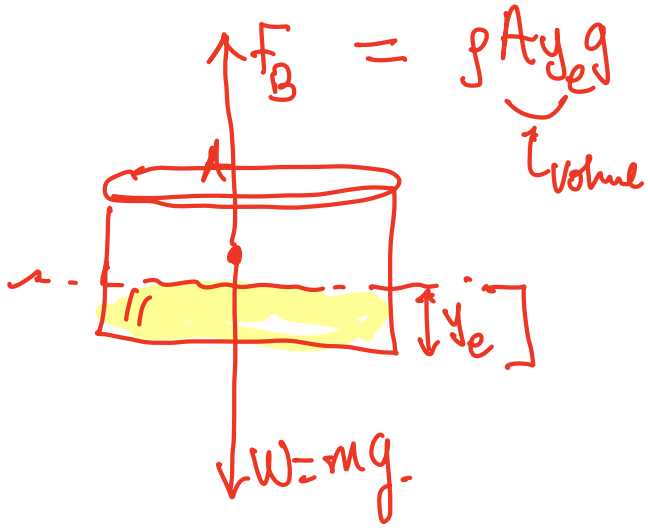


(b)



(c)

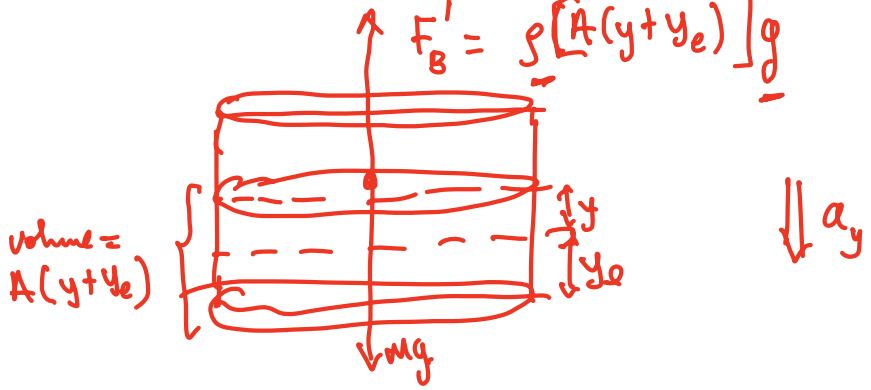
(b)



$$mg - F_B = 0$$

$$mg - \rho A y_e g = 0 \quad \text{——— (1) ✓}$$

(c)



$$mg - F_B' = m a_y \quad \text{——— (2)}$$

$$\Rightarrow mg - \rho A(y+y_e)g = m\ddot{y}$$

$$\Rightarrow (mg - \rho A y_e g) - \rho A y g = m\ddot{y}$$

$(mg - \rho A y_e g) = 0$

$$\Rightarrow m\ddot{y} = -\rho A g y$$

$$\Rightarrow \ddot{y} = -\left(\frac{\rho A g}{m}\right) y \Rightarrow \ddot{y} = -\omega_0^2 y \quad \text{where } \omega_0^2 = \frac{\rho A g}{m}$$

Other Examples of SHM

Mass on a Vertical Spring

$$\frac{2\pi}{T} = \omega_0 = \sqrt{\frac{\rho A g}{m}} \Rightarrow \boxed{T = 2\pi\sqrt{\frac{m}{\rho A g}}}$$

Figure: $T = 2\pi\sqrt{\frac{m}{k}}$ ✓

Floating Body (e.g., in a test tube)

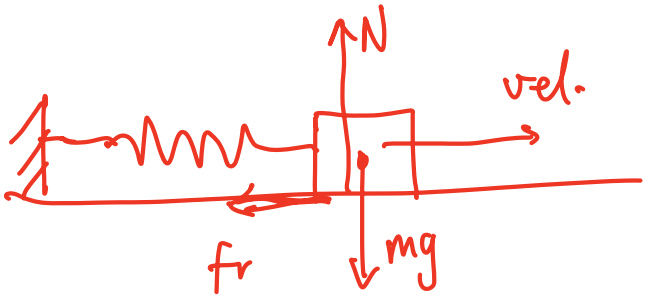
Figure: The buoyant force provides the restoring force. $T = 2\pi\sqrt{\frac{m}{\rho g A}}$ ✓

$f_{\text{air-resistance}} \propto v$

$\uparrow$ −ve. constant.

$f_{\text{air resistance}} = -Bv$

$= -B\dot{x}$

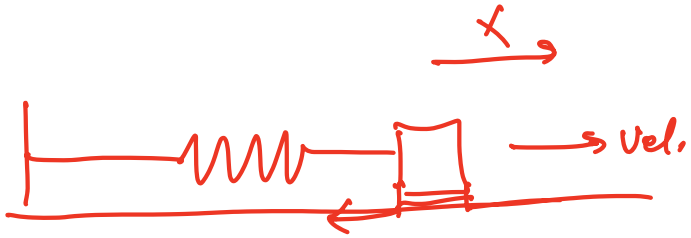


$f_r = \mu N = \mu mg$ ✓ static

$\mu \propto$ velocity or $\mu = -\lambda v$

$f_r = (-\lambda v)\, mg = -Bv$

$= \lambda mg$



$m\ddot{x}$ = net force

$= -kx - Bv$

$\Rightarrow m\ddot{x} = -kx - B\dot{x}$

$\boxed{m\ddot{x} + B\dot{x} + kx = 0}$

$\uparrow$ spring force.

Damping: Real-World Friction

In real systems, energy is lost to friction, air resistance, etc. This is **damping**.

The damping force is often proportional to velocity ($F_d = -bv$).
The new equation of motion becomes:

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + b\frac{\mathrm{d}x}{\mathrm{d}t} + kx = 0$$

- This is a **damped harmonic oscillator**.
- The solution depends on the strength of the damping.

b^2 vs $4mk$

$b^2 < 4mk$

Types of Damping

Underdamped

$b^2 < 4mk$

System oscillates with an exponentially decaying amplitude. Most common.

Critically Damped

$b^2 = 4mk$

System returns to equilibrium as fast as possible **without oscillating**. (e.g., car shock absorbers)

Overdamped

$b^2 > 4mk.$

System returns to equilibrium slowly without oscillating. (e.g., door closer)

Questions?

PHY104:

General Physics IV

(Vibration, Waves and Optics)

2024/25 Session (2025)

tinyurl.com/phy104ui2025



(Discuss the website)

TODAY:

I. Questions from last class?

II. Test Questions

III. Revise Last Class

IV. Waves: Intro and more

II. Test Questions

Q1: Consider the periodic motion of a body. If accn = $-3\ cm/s^2$ when $x=2.5cm$ and $-4.3\ cm/s^2$ when $x=3.0cm$, then the motion is:

a) a Simple Harmonic Motion

b) not a Simple Harmonic Motion ✓

$a \propto x$ ✓ $\Rightarrow \frac{a}{x} = \text{constant}$

$a = -\omega_0^2 x$

$\frac{-3}{2.5} = -1.2$ ✓

$\frac{-4.3}{3} = -1.43$

II. Test Questions

Q2: Consider the periodic motion of a body. If accn = $-3.0\ cm/s^2$ when $x=2.5cm$ and $-3.6\ cm/s^2$ when $x=3.0cm$, then the motion is:

a) a Simple Harmonic Motion

b) not a Simple Harmonic Motion

c) a critically damped harmonic motion

d) none of the other options

$\frac{-3}{2.5} = -1.2$

$\frac{-3.6}{3.0} = -1.2$ ✓

II. Test Questions

Q2B: Consider the periodic motion of a body. If accn = $-3.0\ cm/s^2$ when $x=2.5cm$ and $-3.6\ cm/s^2$ when $x=3.0cm$, then its period of oscillation is:

a) 1.2 s
b) 1.1 s
c) 6.9 s
d) 5.7 s
e) none of the other options

$a = -\omega_0^2 x$ | $\omega_0^2 = 1.2$

$\omega_0 = \frac{2\pi}{T}$ | $T = \frac{2\pi}{\omega_0} = \frac{2\pi}{\sqrt{1.2}} = 5.7s$

II. Test Questions

Q3: Consider an object exhibiting SHM. If it was released (with zero speed) at a point away from equilibrium, at what point in the motion does its total energy become 1/4 of its original energy?

a) 1/2 way through each cycle
b) 1/8 way through each cycle
c) 1/4 way through each cycle
d) none of the above

II. Test Questions

Q4: Consider an object exhibiting SHM. If it was released (with zero speed) at a point away from equilibrium, at what point in the motion does its kinetic energy become 1/4 of its original total energy?

a) 1/2 way through each cycle
b) 1/8 way through each cycle
c) 1/4 way through each cycle
d) none of the above

$E_{tot} = \frac{1}{2} k A^2$



$\frac{1}{4}$ = KE + PE ← $\frac{3}{4}$

$\frac{1}{2} k x^2$

$\frac{1}{2} k x^2 = \frac{3}{4} \frac{1}{2} k A^2$.

$x^2 = \frac{3}{4} A^2$

$x = \frac{\sqrt{3}}{2} A$

$= 0.86A$

$$\frac{1}{2}kx^2 = \frac{3}{4}\cdot\frac{1}{2}kA^2$$

$$x^2 = \frac{3}{4}A^2$$

$$\Rightarrow x = \sqrt{\frac{3}{4}}A$$

$$= 0.866A$$

Exe
Redo for
when $KE = \frac{3}{4}E_{tot}$



① $E_{tot} = \frac{1}{2}kA^2$

② $PE = \frac{1}{2}kx^2$.

③ $E_{tot} = KE + PE$.

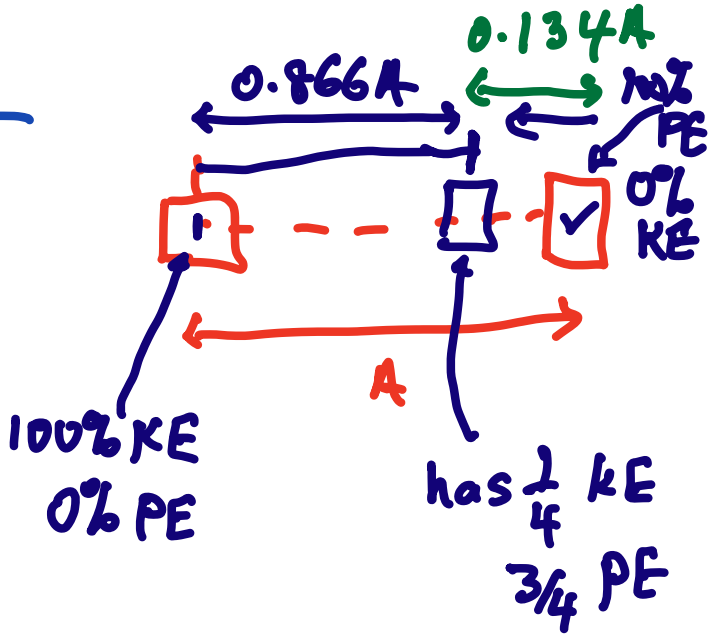
$$KE = \frac{1}{4}E_{tot}.$$

$$\Rightarrow PE = \frac{3}{4}E_{tot}.$$

$$\cancel{\frac{1}{2}}kx^2 = \frac{3}{4}\cdot\cancel{\frac{1}{2}}kA^2.$$

$$x^2 = \frac{3}{4}A^2$$

$$x = \sqrt{\frac{3}{4}}A = 0.866A$$



II. Test Questions

Q5 Consider an object exhibiting SHM. If it was released (with zero speed) at a point away from equilibrium, at what point in the motion does its potential energy become 1/4 of its original total energy?

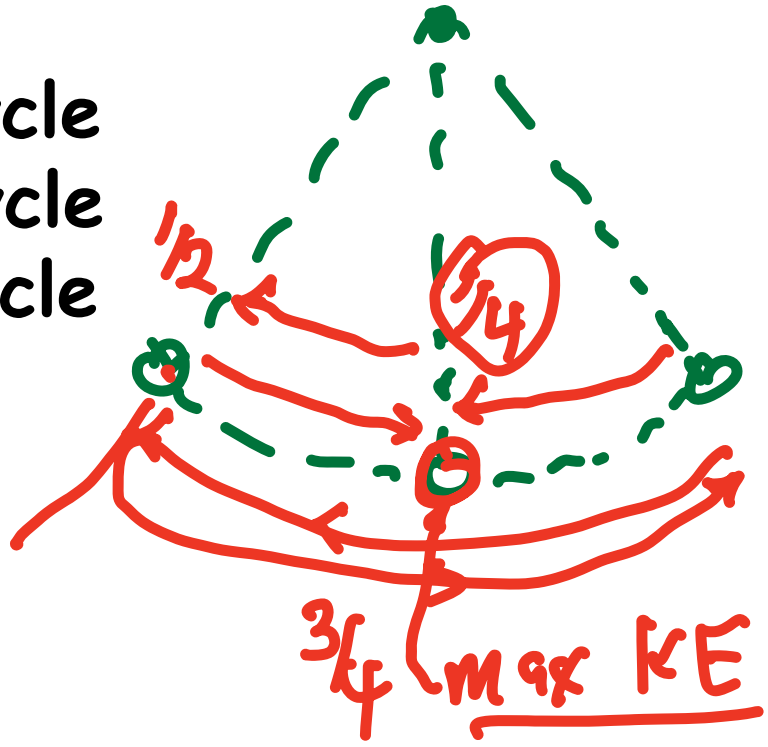
a) 1/2 way through each cycle
b) 1/8 way through each cycle
c) 1/4 way through each cycle
d) none of the above

II. Test Questions

Q6: Consider an object exhibiting SHM. If it was released (with zero speed) at a point away from equilibrium, at what point in the motion is its speed maximum?

a) 1/2 way through each cycle
b) 1/8 way through each cycle
c) 1/4 way through each cycle
d) none of the above

or 3/4 way



II. Test Questions

Q7: Consider a body exhibiting SHM.
If accn = -3.0 cm/s^2 when x=2.5cm. Then when x=3.0cm, its accn should be:

a) -2.5cm/s^2
b) -3.0cm/s^2
c) -3.6cm/s^2
d) -4.3cm/s^2



Imagine the world without waves

Particles: Yes Waves: No

No light (?)
No sound, no music
No stringed instruments: piano, guitars, violins, etc
No cell-phone communications

No Radio waves
No X-rays
No Ultrasound (so, no imaging, no fetus imaging, no sonar,...)
No conduction of heat
No Ocean waves

No life

We can see the world using sound waves

(Movie)

Week3: Propagation of sound in gases, solids and liquids and their properties.

Waves (types and properties of waves as applied to sound).

Transverse and longitudinal waves (superposition, interference, diffraction, dispersion, polarization).

Week4: Waves at interfaces (energy and power of waves).

The wave equation. 2-D and 3-D wave equations.

Wave energy and power.

Phase and group velocities.

Echo and beats.

The Doppler-effect.

We can see the world using sound waves

Species	Approximate Range (Hz)
human	64-23,000
dog	67-45,000
cat	45-64,000
cow	23-35,000
horse	55-33,500
sheep	100-30,000
rabbit	360-42,000
rat	200-76,000
mouse	1,000-91,000
gerbil	100-60,000
guinea pig	54-50,000
hedgehog	250-45,000
raccoon	100-40,000
ferret	16-44,000
opossum	500-64,000
chinchilla	90-22,800
bat	2,000-110,000
beluga whale	1,000-123,000
elephant	16-12,000
porpoise	75-150,000
goldfish	20-3,000
catfish	50-4,000
tuna	50-1,100
bullfrog	100-3,000
tree frog	50-4,000
canary	250-8,000
parakeet	200-8,500
cockatiel	250-8,000
owl	200-12,000
chicken	125-2,000

(Movie)

https://www.lsu.edu/deafness/HearingRange.html

Definitions: What is a wave?

a) Wave your hand?

b) Hair waves?



Definitions: What is a wave?

c) Ocean waves?





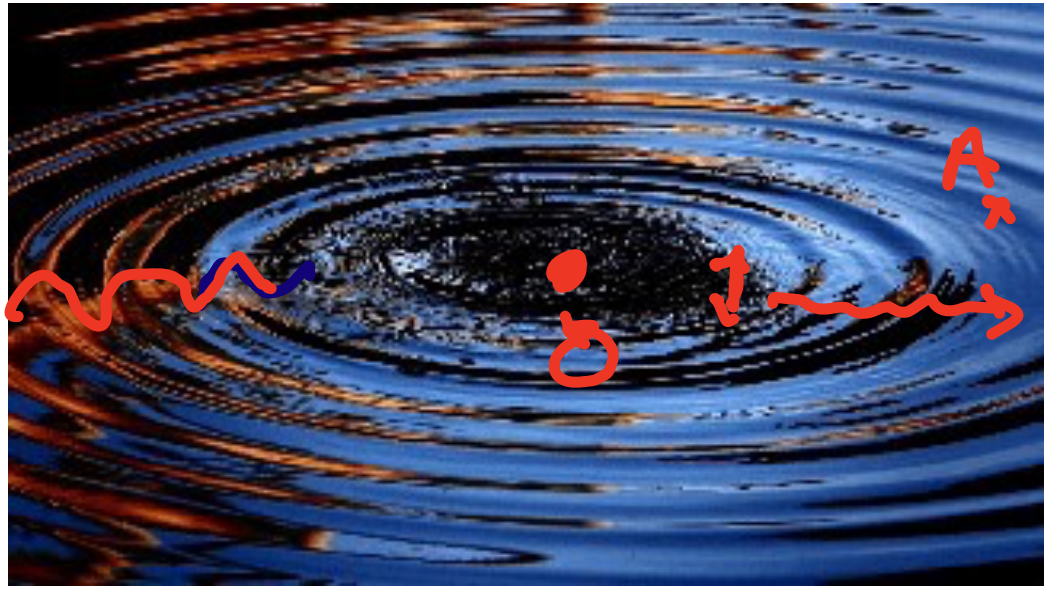
Definitions: What is a wave?

Wave =
Disturbance that propagates energy from a point of creation to another point

-- without the actual movement of the particles of the medium from the point of creation to the other point

What is a wave?: Examples

a) Water ripples in a pond



Observe two motions: (See movie) https://www.vecteezy.com/free-videos/water-drop

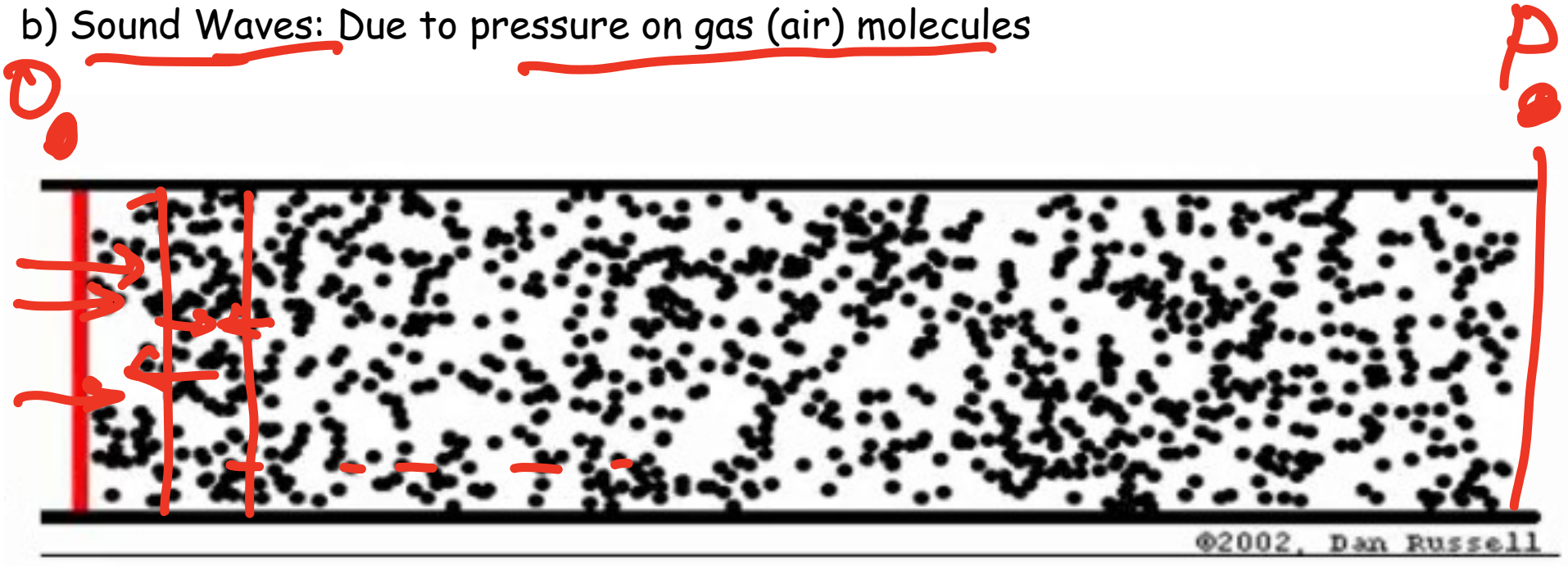
1. Up and down motion ("Oscillatory motion")

2. Outward motion

→ Transmits energy

What is a wave?: Examples

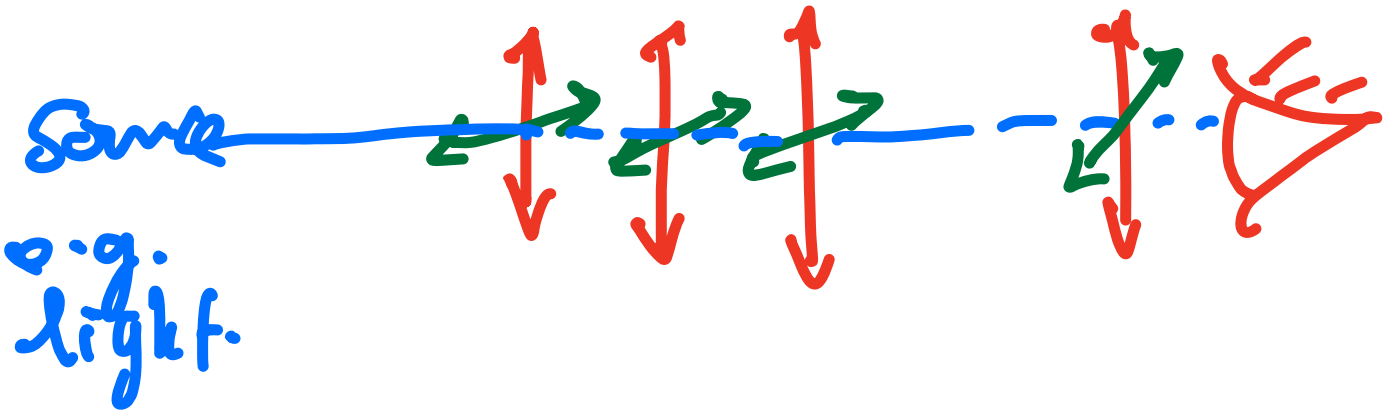
b) Sound Waves: Due to pressure on gas (air) molecules



The air particles at my mouth do not actually get to you. So, how do you hear my voice?
The air particles at the speakers do not actually get to you, either. :-)

What is a wave?: Examples

c) Electromagnetic Waves: Due to Electric and Magnetic Fields



Types of Waves

* First Classification: Mechanical vs. Non-Mechanical waves

a) Mechanical waves: require a material medium i.e., non-vacuum
Examples: Sound waves, water waves, seismic waves
Question: 1) Can sound travel in a vacuum?
2) If you shout in a vacuum, can we hear you?
3) [If a tree falls in the forest with nobody around, does it make a sound?]

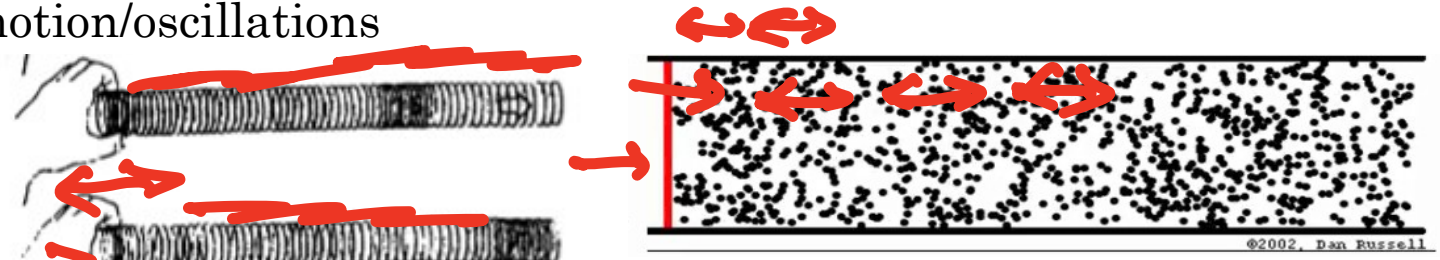
b)Non-Mechanical waves: can travel without a material medium. i.e., can travel in vacuum.
Examples:
* Electromagnetic waves [Radiowaves, microwaves, infrared waves (heat waves),
Visible light, ultraviolet waves, x-rays, gamma rays]
* Gravitational waves (not Gravity waves!)

Types of Waves

* Second Classification: Longitudinal vs. Transverse waves

Longitudinal: Direction of propagation is same as direction of wave (or medium) motion/oscillations

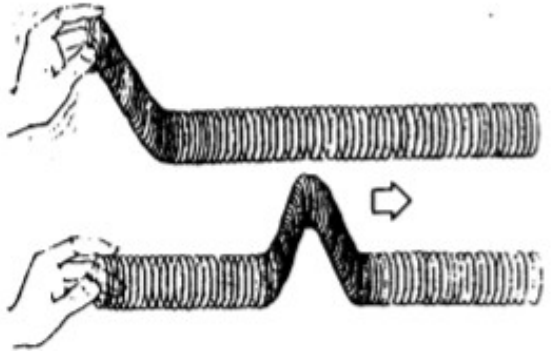
Examples: Sound waves
Seismic S waves



Sound waves in air/fluid.

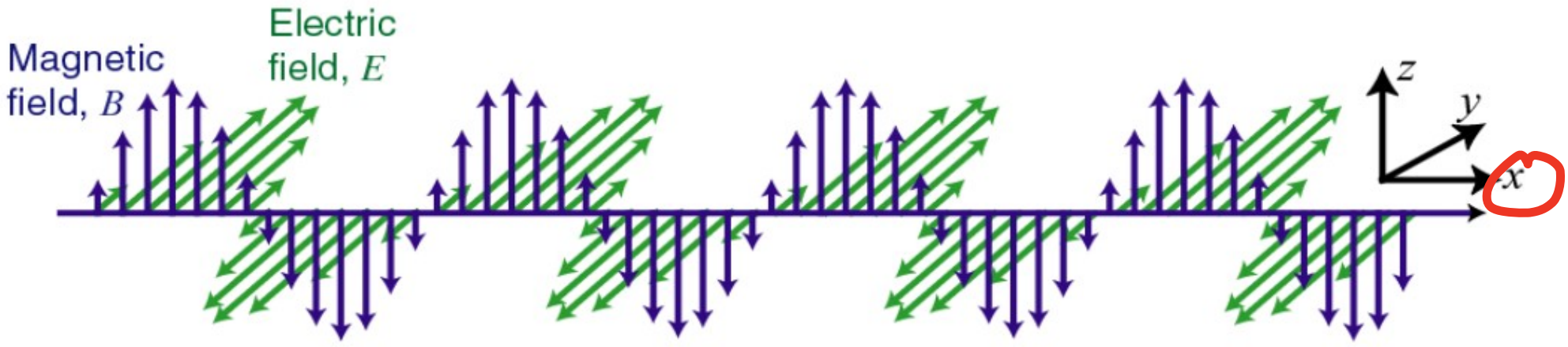
Transverse: Direction of propagation is perpendicular to direction of wave (or medium) motion/oscillations

Examples: Ripple Waves in water ✓
Electromagnetic waves ✓
Seismic P waves
Gravitational waves ✓



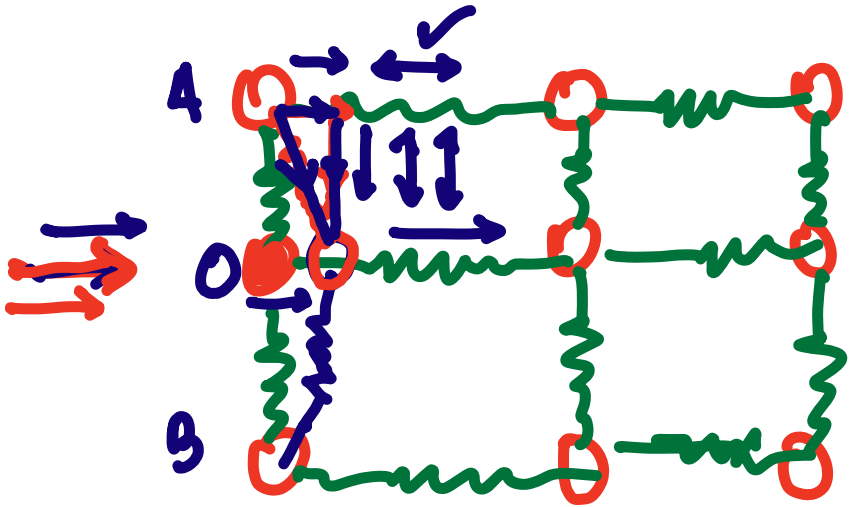
EM Waves are transverse

Sound waves in solids have two possible components — longitudinal wave — transverse wave.



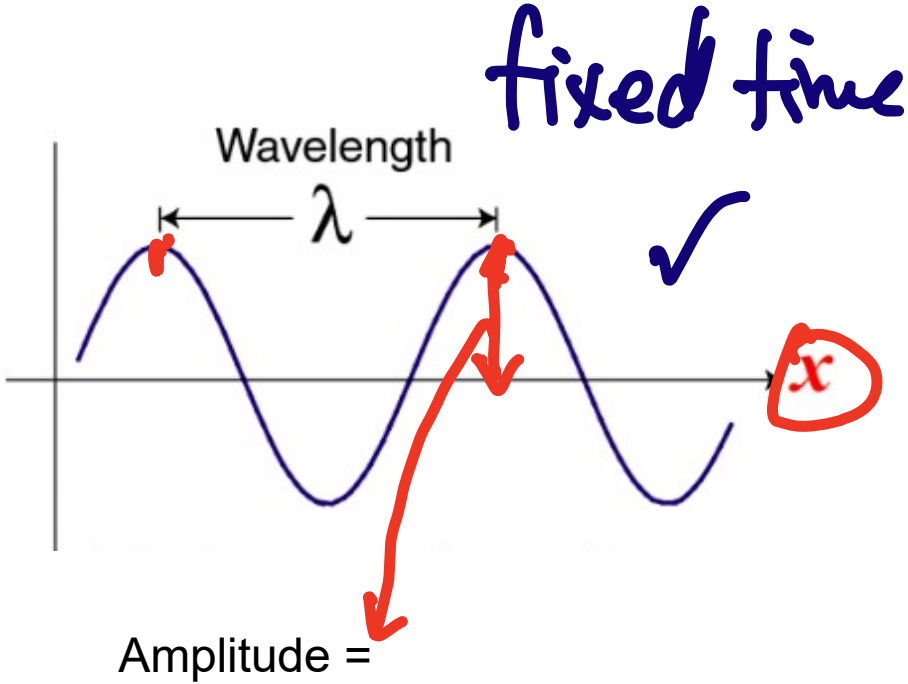
Types of Waves

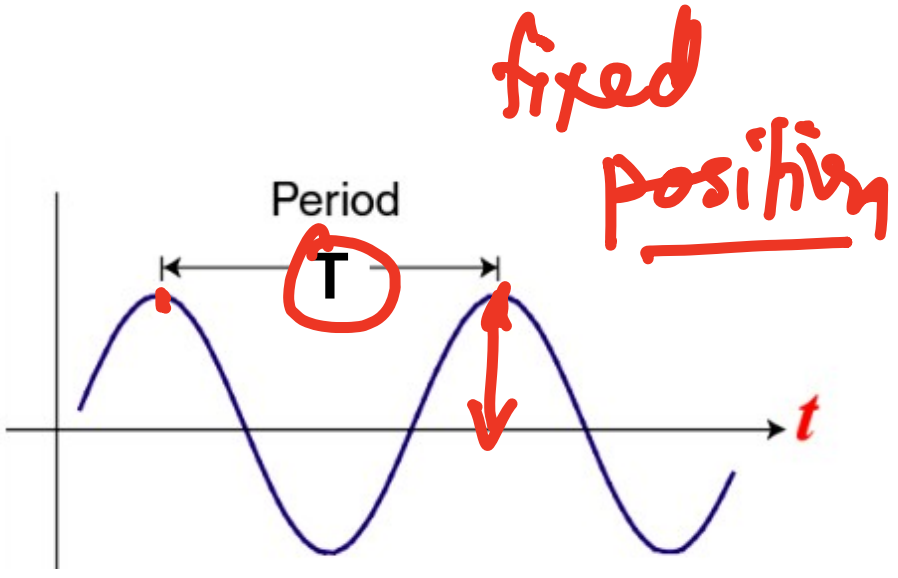
Questions: 1) Can a mechanical wave be transverse?
2) Can a non-mechanical wave be longitudinal?
3) Can EM waves become longitudinal?



Waves: More Definitions

Questions: 1. What are these pictures?
2. How are they related to waves?





Note: Not all waves are periodic
Not all waves are simple/harmonic

Waves: More Definitions

$$v = \frac{\lambda}{T}$$

$$f = \frac{1}{T}$$

Question: How?

$$v = f\lambda$$

$$\omega = \frac{2\pi}{T} = 2\pi f \qquad k \equiv \frac{2\pi}{\lambda}$$

Wave Equation

Like particles obey F=ma, waves do not have masses, so which equations do they obey?

$$\frac{\partial^2 J}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}$$

Here, J could be:

1. ~~Magnitude of the~~ Electric or Magnetic fields
2. Pressure of air (sound waves)
3. Height above the horizontal (water ripples)

Wave Equation

$$\frac{\partial^2 J}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}$$

What are these funny-looking d's?

$\frac{dJ}{dx}$ $\frac{\partial J}{\partial x}$

$J = J(x, t)$

$\frac{\partial J}{\partial x} = \left(\frac{\partial J}{\partial x}\right)_t$ ← time is constant

$\varphi = 3x^2 t$

$\frac{\partial \varphi}{\partial x} = 6xt$

$\varphi = 3x^2 t + t^7$

$\frac{\partial \varphi}{\partial x} = 6xt$

$\frac{\partial^2 \varphi}{\partial x^2} = 6t$ ✓

Wave Equation

Solution of the wave equation

Partial Differential Eq.

$$\frac{\partial^2 J}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}$$

v = phase velocity

is any function of : $(x - v\,t)$ or of $(x + v\,t)$

e.g. $J = A(x - vt)^2$ is a solution

$J = A(x - vt)^4$ is a solution

$J = e^{-\alpha(x-vt)}$

$J = e^{-\beta(x+vt)}$

$J = 6x^2t$

Wave? How to know

$$\frac{\partial^2 J}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}$$

Best way: Check if LHS = RHS

Examples:

Is $J = (x^2 - v^2t^2)$ a wave? No!

$= (x - vt)(x + vt)$

$\frac{\partial J}{\partial x} = 2x \rightarrow \frac{\partial^2 J}{\partial x^2} = 2$

LHS

$\frac{\partial J}{\partial t} = -2v^2 t$

$\frac{\partial^2 J}{\partial t^2} = -2v^2$

$\frac{1}{v^2}\frac{\partial^2 J}{\partial t^2} = -2$

= RHS

Is $J = (x^2 - v^2t^2)$ a wave? NO!

LHS:

$$\frac{\partial J}{\partial x} = 2x$$

$$\frac{\partial^2 J}{\partial x^2} = 2$$

RHS

$$\frac{\partial J}{\partial t} = -2v^2t$$

$$\frac{\partial^2 J}{\partial t^2} = -2v^2$$

$$\frac{1}{v^2}\frac{\partial^2 J}{\partial t^2} = -2$$

Exe: Check if $J = A\sin[k(x-vt)]$ is a wave

$$\frac{\partial J}{\partial x} = kA\cos k(x-vt)$$

$$\frac{\partial^2 J}{\partial x^2} = -k^2 A\sin k(x-vt)$$

$$\frac{\partial J}{\partial t} = -kvA\cos k(x-vt)$$

$$\frac{\partial^2 J}{\partial t^2} = -kv(-kv)A\sin[\,] = -k^2v^2A\sin k(x-vt)$$

$$\frac{1}{v^2}\frac{\partial^2 J}{\partial t^2} = -k^2A\sin k(x-vt)$$

same ✓

Exe: $J = A\sin k(x+vt)$

Wave? How to know

$$\frac{\partial^2 J}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}$$

Best way: Check if LHS = RHS

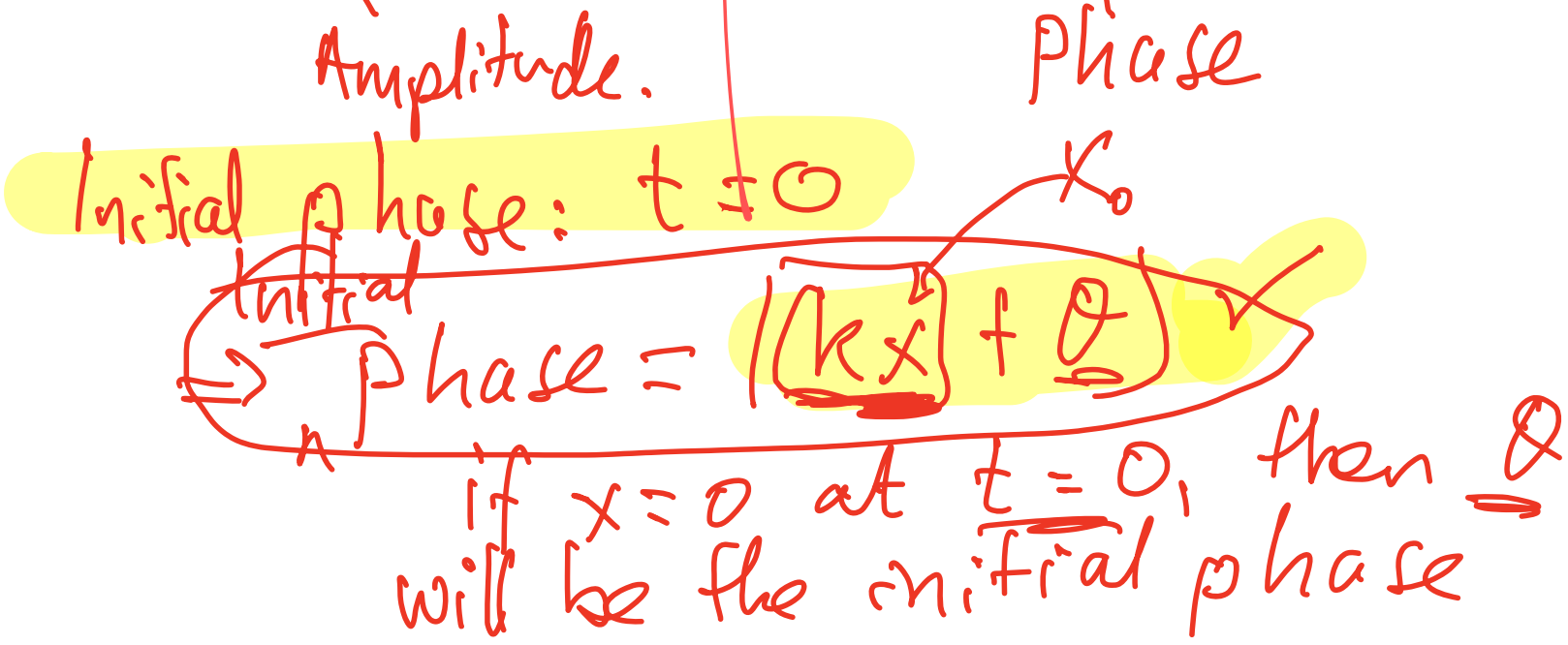
Examples:

LHS

RHS

$$\mathbf{J(x,t) = A \sin [k(x - vt) + \theta]}$$

Amplitude, Phase, Initial Phase, Wave velocity = Phase velocity



Exe: Progressive wave – Is this wave moving left or right?

$$J(x,t) = A \sin[k(x - vt) + \theta]$$

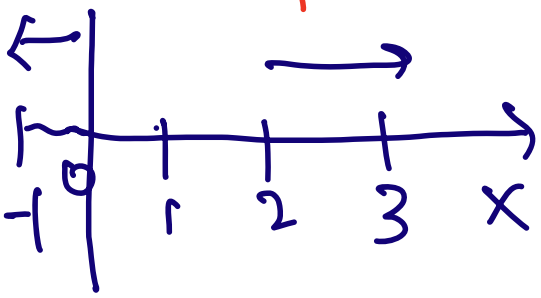
Amplitude, Phase, Initial Phase, Wave velocity = Phase velocity



+ve

+ve

$t \uparrow$



we want to keep $k(x - vt)$ constant

$\Rightarrow x \uparrow$

$t \uparrow$

Exe: Progressive wave – Is this wave moving left or right?

$B = Q \cos[-k(x + vt)]$ — Left

$x \downarrow \Leftarrow t \uparrow$

$1 \hookleftarrow 2 \hookleftarrow 3$

$x + vt = 3$

$t: 0 \to 1 \to 2$

$x \downarrow$

$C = Q \cos[k(x + vt)]$ — Left

$D = W \sin[k(vt - x)]$ — Right

e.g. 3 3 phase = 0

4 4

200 200 3 4 ... 2a

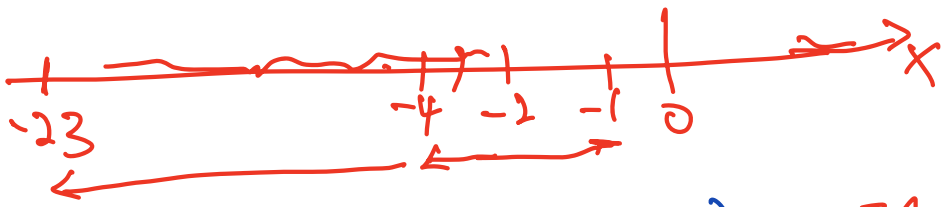
$v = |v|$

$C = Q \cos[-k(x + vt)]$ — left.

e.g. $-1 + 3$ Init phase = ?

$-2 + 4$

$-23 + 25$



$W = C_0 \sin(-kx - \omega t) = \sin[-(kx + \omega t)]$

+ve +ve

$-0 - 3 = -3$

$\pm 1 \quad -4$

-1

$$S(x,t) = 0.1 \sin[0.4t - 0.1x + 2] \text{ m}$$

Exe: Determine

a) Whether to the left or to the right

b) Maximum magnitude of S : 0.1

c) Amplitude of S : 0.1

d) Initial phase ; $-0.1x_0 + 2$

e) Phase velocity

f) Wave vector

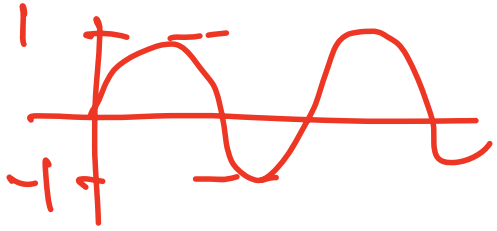
g) Wave length

h) Period ✓

i) Frequency ✓

h) Wave equation

-1 to 1



initial position

$k = \frac{2\pi}{\lambda}$

If $x_0 = 0$ (i.e. at $t = 0$, $x = 0$)

then init phase : $-0.1 \cdot 0 + 2$

$\sin[0.4t - 0.1x + 2]$

$\frac{+2}{2}$

$\sin[0.4t - 0.1x + 2]$

$\sin[\pm k(x \pm vt)]$ — wave vector.

$x - vt$

$x + vt$

$\sin\left[-0.1\left[\frac{0.4t - 0.1x}{-0.1}\right] + 2\right]$

$\sin[-0.1(-4t + x) + 2]$

wave vector: 0.1

$k = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{2\pi}{k}$

phase velocity = 4 (magnitude) & moving to right.

$\lambda = \frac{2\pi}{0.1} = 20\pi$

period = ?

$v = 4$, $\lambda = 20\pi$

$v = f\lambda$

$\therefore f = \frac{v}{\lambda} = \frac{4}{20\pi}$

$\Rightarrow T = \frac{1}{f} = \frac{20\pi}{4} = 5\pi$

$\frac{\partial^2 \zeta}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 \zeta}{\partial t^2}$ ✓ correct = Wave equation in 1D (in 1 Dimension)

$\Rightarrow \frac{\partial^2 \zeta}{\partial x^2} = \frac{1}{16}\frac{\partial^2 \zeta}{\partial t^2}$ ✓✓ more correct

1D: (x)

$$\frac{\partial^2 J}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}.$$

2D: $(x, \& y)$

$$\frac{\partial^2 J}{\partial y^2} + \frac{\partial^2 J}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}$$

3D: (x, y, z)

$$\frac{\partial^2 J}{\partial x^2} + \frac{\partial^2 J}{\partial y^2} + \frac{\partial^2 J}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 J}{\partial t^2}$$

Revision: Light and Gravity as waves

What is the color of the Sun now?

Revision: Light and Gravity as waves

If the Sun disappears, how long will it take for us on Earth to realize it?

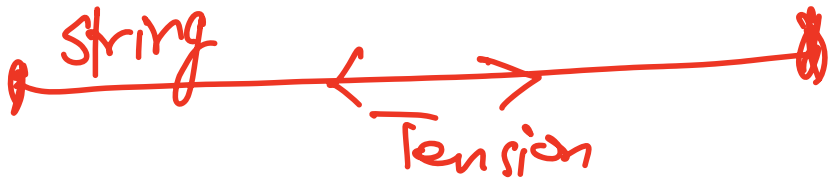
-- c=3X10^{8}m/s and Earth-Sun distance is 148.29 million kilometres
(Give your answer in minutes and seconds)

Speed of Sound waves

$$v_{\text{mech}} = \sqrt{\frac{\text{Stiffness Factor}}{\text{Mass(Density) Factor}}}$$

$$v = \sqrt{\frac{\text{Tension (Force)}}{\mu}}$$

(μ = mass per unit length)



$$v = \sqrt{\frac{\gamma P}{\rho}} \Rightarrow v_{\text{air(sea level)}} = 331\sqrt{\frac{T}{273}}\ \text{m/s}$$

($\gamma = C_p/C_v$)

T in Kelvin

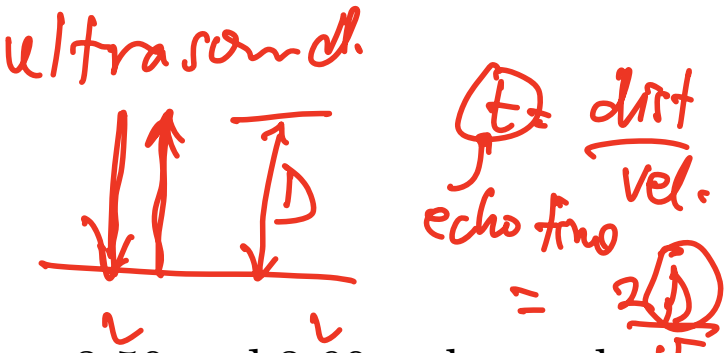
$$= 331\sqrt{\frac{t+273}{273}}\ \text{m/s}$$

Speed of Sound waves

At 0°C
Air ~ 331 m/s Oxygen ~ 316 m/s Hydrogen ~ 1290 m/s

At 20°C
Sea water ~ 1540 m/s Human Tissue ~ 1540 m/s

Solids (longitudinal)
Steel ~ 5960 m/s Bones ~ 3,000 – 4,000 m/s

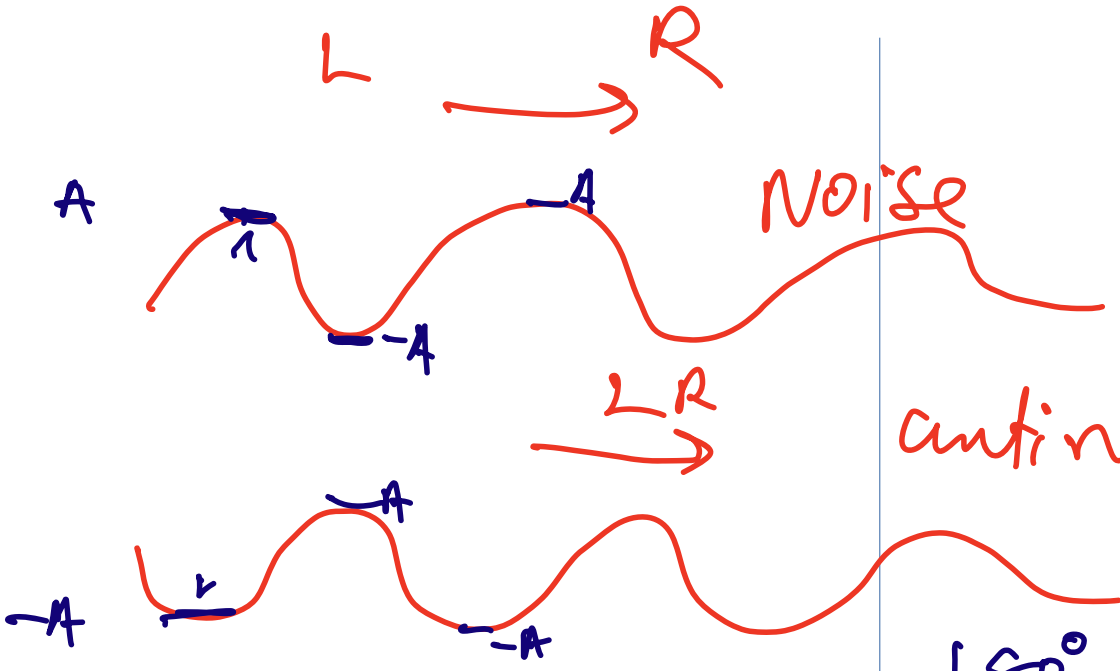


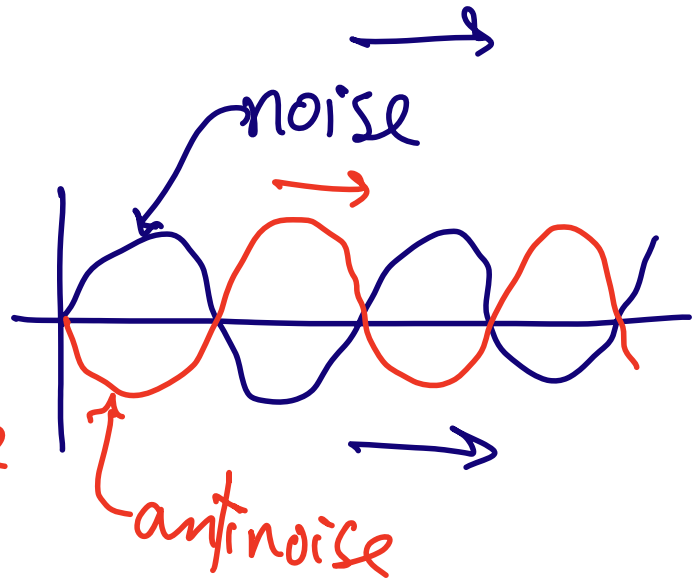
Exe (Copied):

a) What is the difference in echo times for tissues that are 3.50 and 3.60cm beneath the surface?

b) How far apart are two layers of tissue that produce echoes having round-trip times (used to measure distances) that differ by 0.750 μs?

Adding Waves: x → kx, t → ωt





180° Out of phase.

Adding Waves

$$\sin(a) + \sin(b) = 2\sin\left[\frac{a+b}{2}\right]\cos\left[\frac{a-b}{2}\right]$$

$\sin(A+B) = \sin A\cos B + \cos A\sin B$

$\cos(A-B) = \cos A\cos B + \sin A\sin B$

Adding Waves: Simple Noise Canceling

$$\sin(a) + \sin(b) = 2\sin\left[\frac{a+b}{2}\right]\cos\left[\frac{a-b}{2}\right]$$

$Y_1 = A\sin(kx - \omega t)$ = $A\sin\left[k\left(x - \frac{\omega}{k}t\right)\right] = A\sin k(x - vt)$

$\Rightarrow \frac{\omega}{k} = v$

$Y_2 = A\sin(kx - \omega t + 180^\circ)$

$a = (kx - \omega t)$

$b = (kx - \omega t + 180^\circ)$

$a + b = ?$ $2kx - 2\omega t + 180^\circ$

$a - b = ?$ -180°

Adding Waves: Simple Noise Canceling

$$\sin(a) + \sin(b) = 2\sin\left[\frac{a+b}{2}\right]\cos\left[\frac{a-b}{2}\right]$$

$$Y_1 = A\sin(kx - \omega t)$$

$$Y_2 = A\sin(kx - \omega t + \pi)$$

$$a = (kx - \omega t)$$

$$b = (kx - \omega t + \pi)$$

$a + b = ?$ $2kx - 2\omega t + \pi$

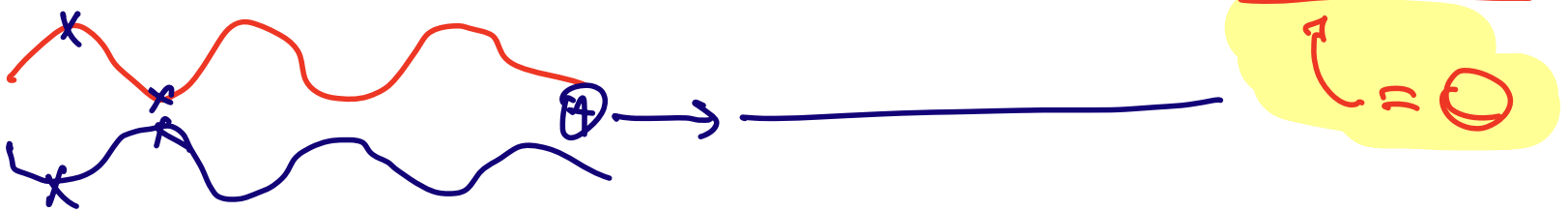
$a - b = ?$ $-\pi$ rad

$\Rightarrow \frac{a+b}{2} = kx - \omega t + \frac{\pi}{2}$

Adding Waves: Simple Noise Canceling

$$\sin(a) + \sin(b) = 2\sin\left[\frac{a+b}{2}\right]\cos\left[\frac{a-b}{2}\right]$$
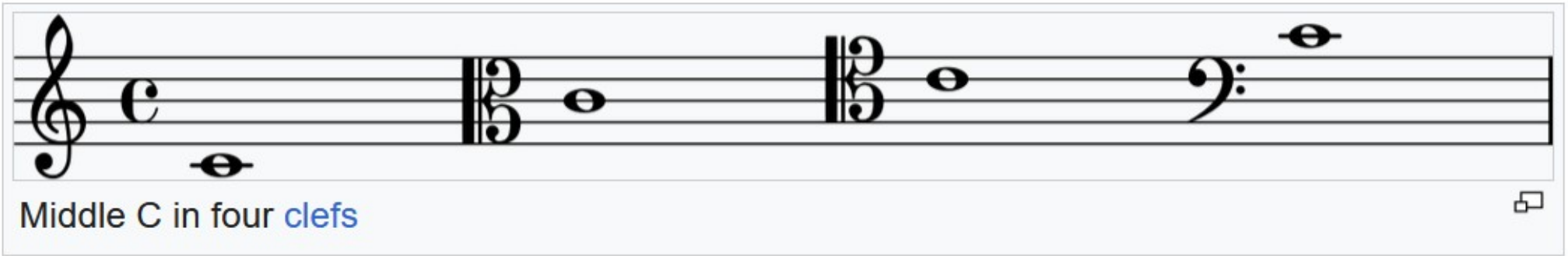
$$Y = Y_1 + Y_2 = 2A\sin(kx - \omega t + \pi/2)\cos(-\pi/2)$$



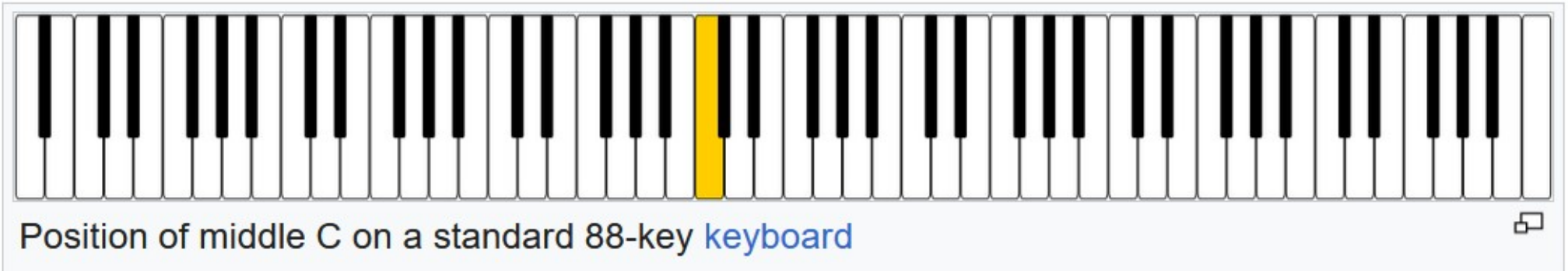
Note: Both waves are coming <u>TOWARDS</u> you!

Yet, they cancel!!

Adding Waves: Music Overtones



Middle C in four clefs



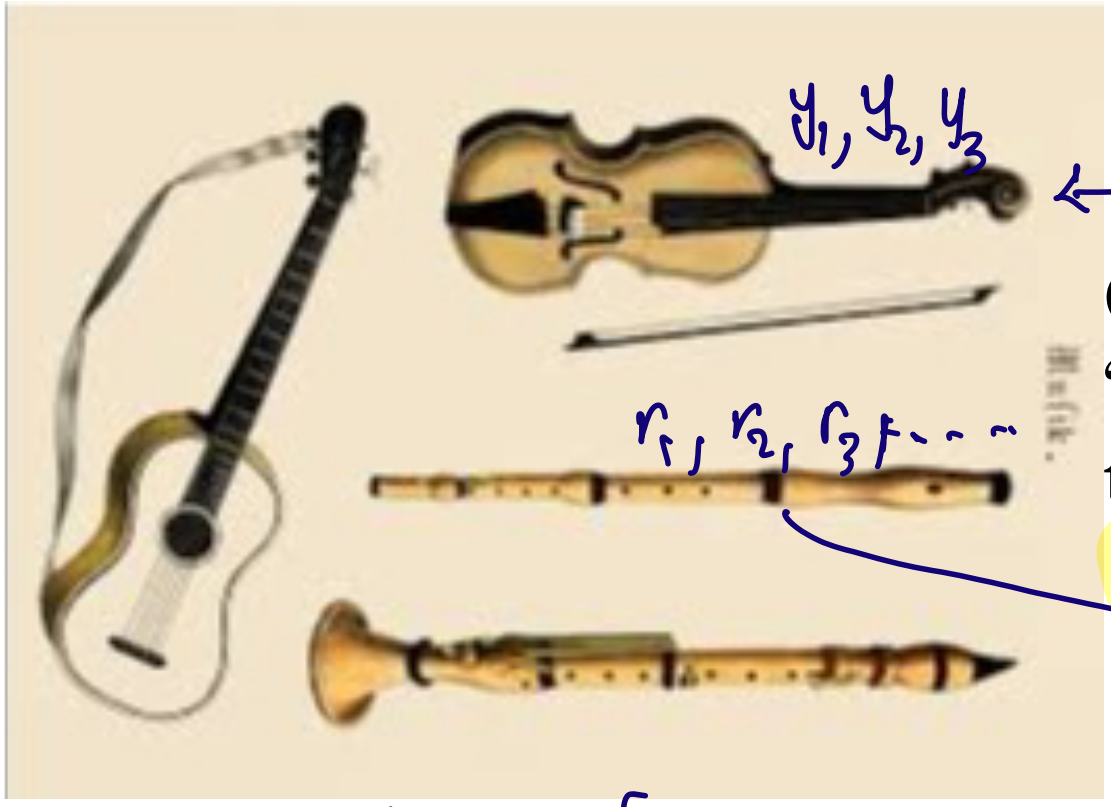
Position of middle C on a standard 88-key keyboard

Middle C (= C_4) → 256 Hz (Science)

→ 261.63 Hz (A440 Standard)

(Relative PITCH vs. Absolute PITCH)

Adding Waves: Music Overtones



$Y \rightarrow$ middle C wave

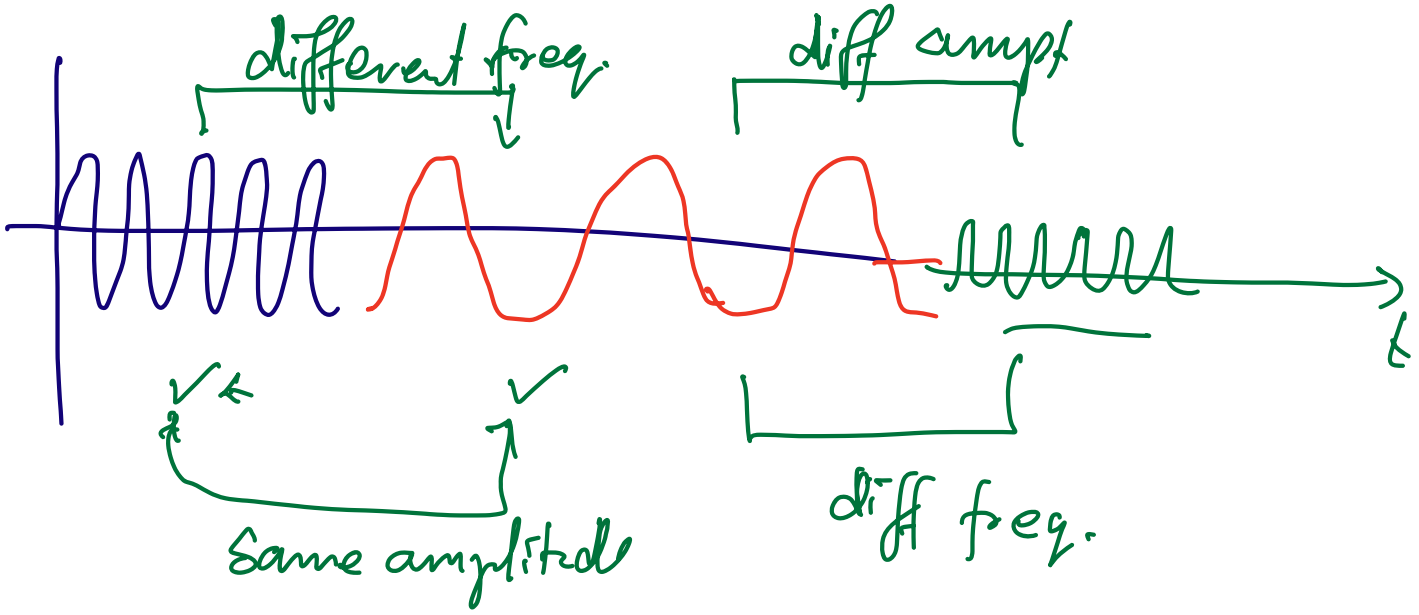
$Y + y_1 + y_2 + y_3 + \ldots$

Q: Why do they sound 'different' if they all have the same middle C (256 Hz)

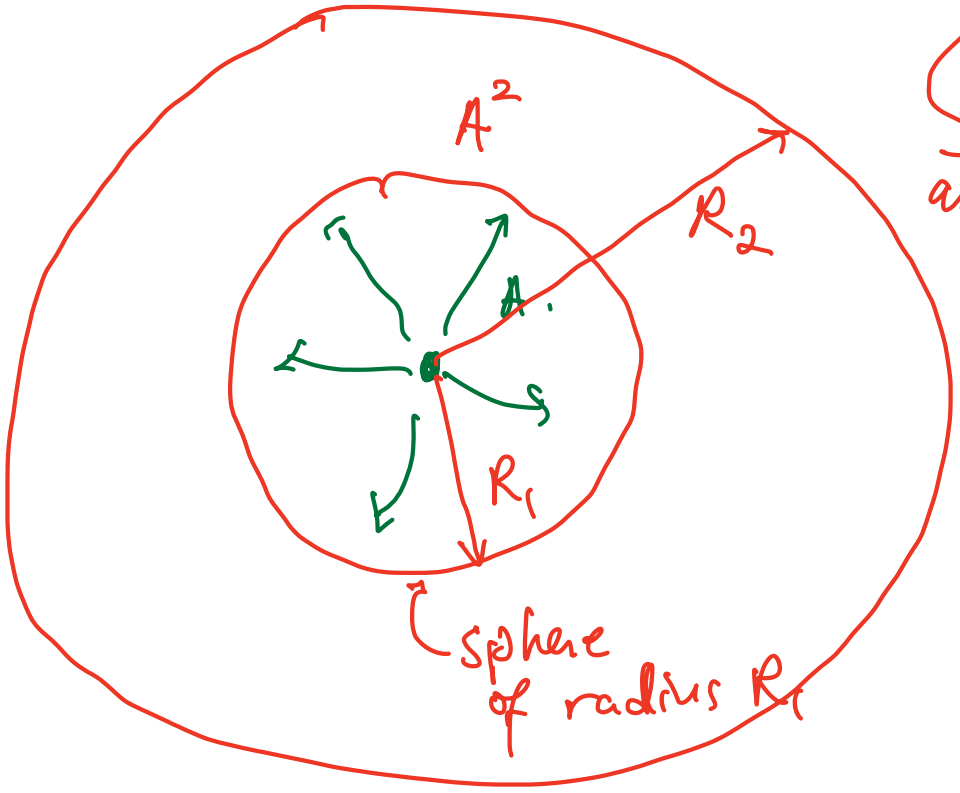
$Y + r_1 + r_2 + r_3 + \ldots$

Pitch ~ frequency (Sound)

Colour ~ frequency (Light)



Loudness $\propto |A|^2$



$P \propto A^2$

area area

$$\text{Intensity} = \frac{\text{Power}}{\text{area}}$$

Sound Intensity: Decibels

$$I = \frac{Average\ Power}{Area} = \left(\frac{P_{ave}}{4\pi R^2}\right)$$

$$\beta = 10 \log_{10}\left[\frac{I}{I_0}\right] \qquad \mathrm{dB} = \text{decibels}$$

Bell & (Bell)

I_0 =0.98 pW/m^2 at 1 atmosphere and 25 °C

~ 1 X 10^{-12} W/m^2 (Usually)

Sound Intensity: Decibels

$$\beta = 10 \log_{10}\left[\frac{I}{I_0}\right]$$

$10 \log(1/2) \approx -3\,dB$

80 dB: 10^{-4} / 40 dB: 10^{-8} $= 10^{4}$

60 dB (normal speech) is 10^{6} more intense than 1 dB (faint whisper)

decibels	intensity*	type of sound
130	10	artillery fire at close proximity (threshold of pain)
120	1	amplified rock music; near jet engine
110	10^{-1}	loud orchestral music, in audience
100	10^{-2}	electric saw
90	10^{-3}	bus or truck interior
80	10^{-4}	automobile interior
70	10^{-5}	average street noise; loud telephone bell
60	10^{-6}	normal conversation; business office
50	10^{-7}	restaurant; private office
40	10^{-8}	quiet room in home
30	10^{-9}	quiet lecture hall; bedroom
20	10^{-10}	radio, television, or recording studio
10	10^{-11}	soundproof room

https://www.britannica.com/science/sound-physics/The-decibel-scale

Adding Waves: Standing Waves

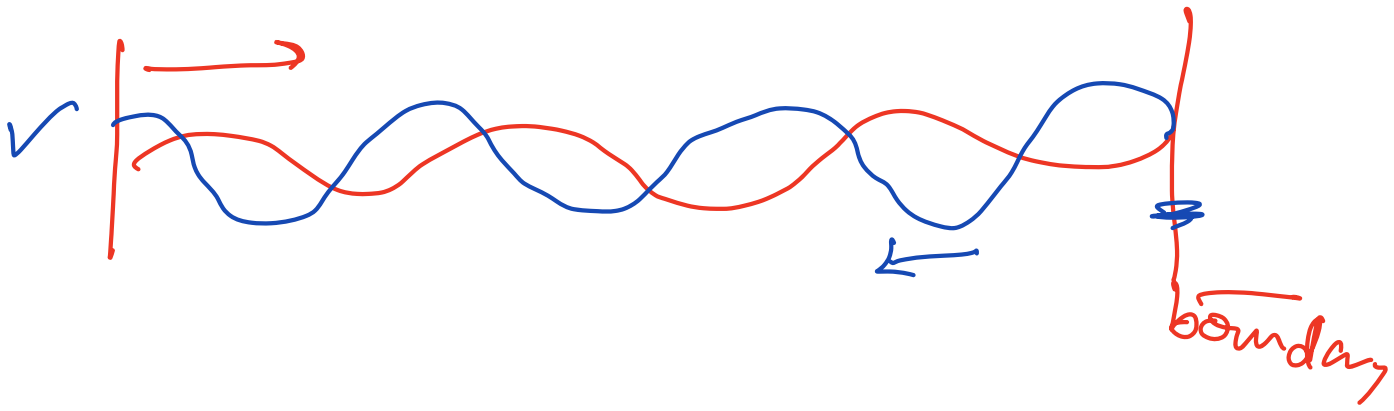
$$\sin(a) + \sin(b) = 2\sin\left[\frac{a+b}{2}\right]\cos\left[\frac{a-b}{2}\right]$$

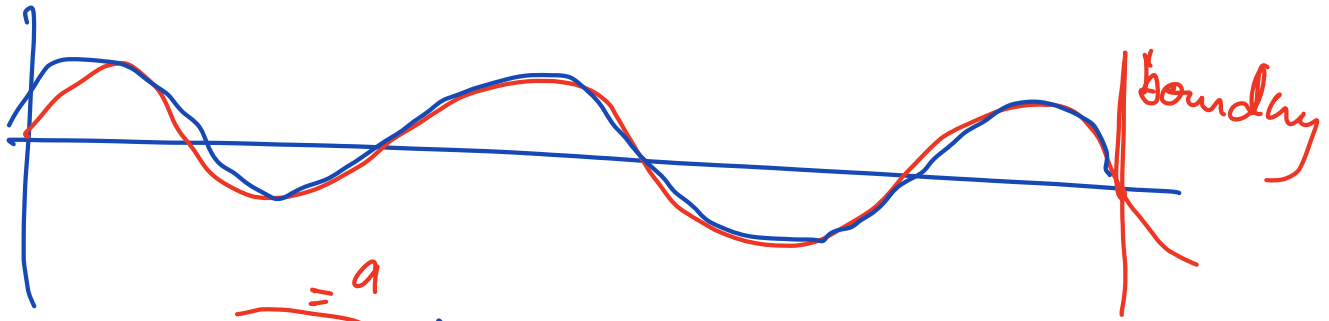
$$Y_1 = A\sin(kx - \omega t)$$

$$Y_2 = A\sin(kx + \omega t)$$

$$Y = Y_1 + Y_2 = ?$$

https://www.compadre.org/osp/EJSS/4035/143.htm





$$I = A\sin(\underbrace{kx - \omega t}_{=a}) \qquad \rightarrow \frac{a+b}{2} = kx$$

$$R = A\sin(\underbrace{kx + \omega t}_{=b}) \qquad \rightarrow \frac{a-b}{2} = -\omega t$$

$$I + R = 2A\sin kx\cos(-\omega t)$$

$$kx = \frac{2\pi}{\lambda}x$$

$$= 2A\sin\left(\frac{2\pi x}{\lambda}\right)\cos(-\omega t)$$

Exe:

$I = A\sin(kx - \omega t)$

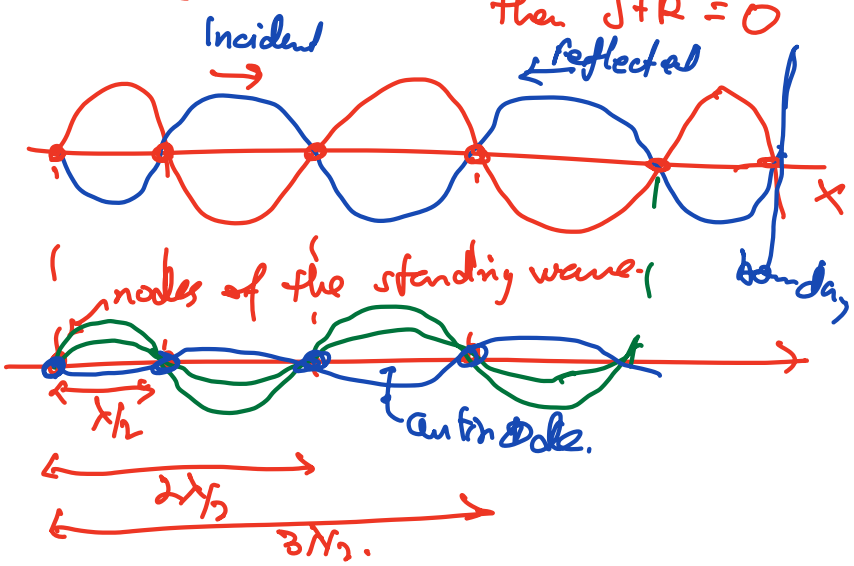
$R = A\sin(kx + \omega t + \pi)$

$I + R = ?$

$\sin(2\pi) = ? \; 0$ when $x = \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \frac{4\lambda}{3}$

$\sin(\pi) = 0$

then $I + R = 0$

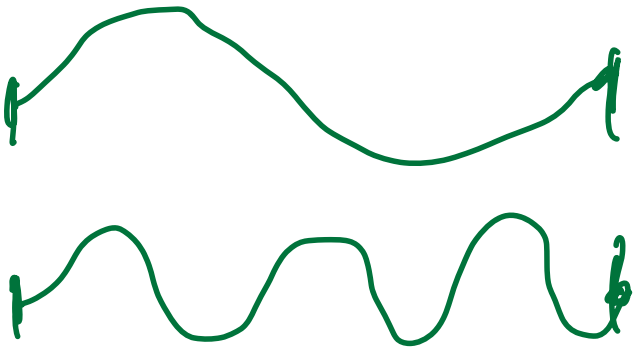


Standing Waves: Strings and Pipes

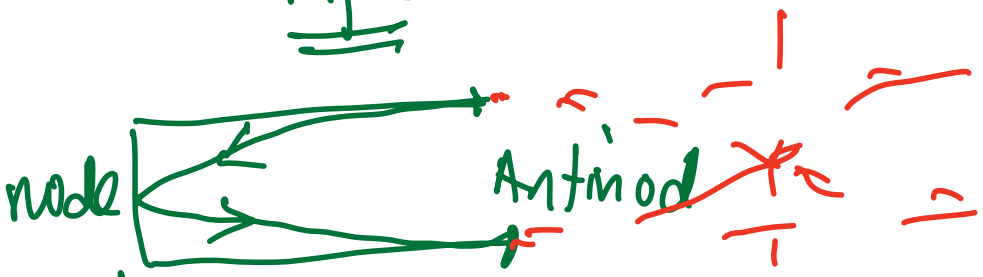
$$\sin(a) + \sin(b) = 2\sin\left[\frac{a+b}{2}\right]\cos\left[\frac{a-b}{2}\right]$$

Guitar string





Pipe

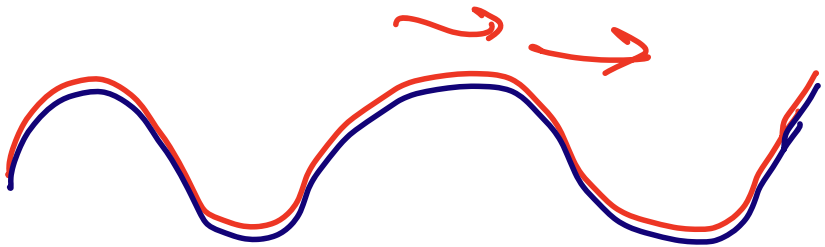


$L = \lambda/4$

Adding Waves: Path Difference

$$\sin(a) + \sin(b) = 2\sin\left(a + \frac{b}{2}\right)\cos\left(a - \frac{b}{2}\right)$$

A)

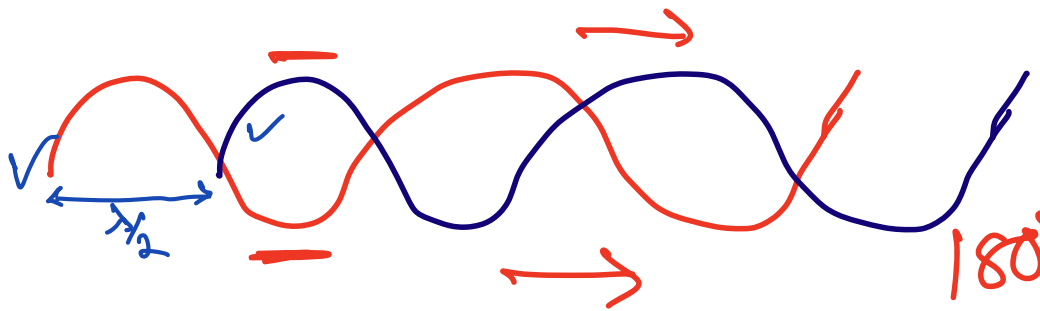


$\psi = \psi_1 + \psi_2$ in phase. ⇒ reinforce. ⇒ Constructive Interference

Path difference = 0

or integer Multiple of λ

B)



$\psi = \psi_1 + \psi_2$.

180° out of phase ⇒ They cancel.

Destructive interference

Path difference = $\lambda/2$ or an odd integer of $\lambda/2$

Superposition = adding together.

Superposition of Waves = Adding Waves

$$\sin(a) + \sin(b) = 2\sin\left[\frac{a+b}{2}\right]\cos\left[\frac{a-b}{2}\right]$$

a) Noise Cancellation
b) Standing Waves
c) Interference
d) Diffraction
* Resonance,
* Refraction,
* Doppler effect
* Diffraction (and Huygen's principle),
* Polarization (as a property of transverse waves only).

Standing Waves: Strings and Pipes

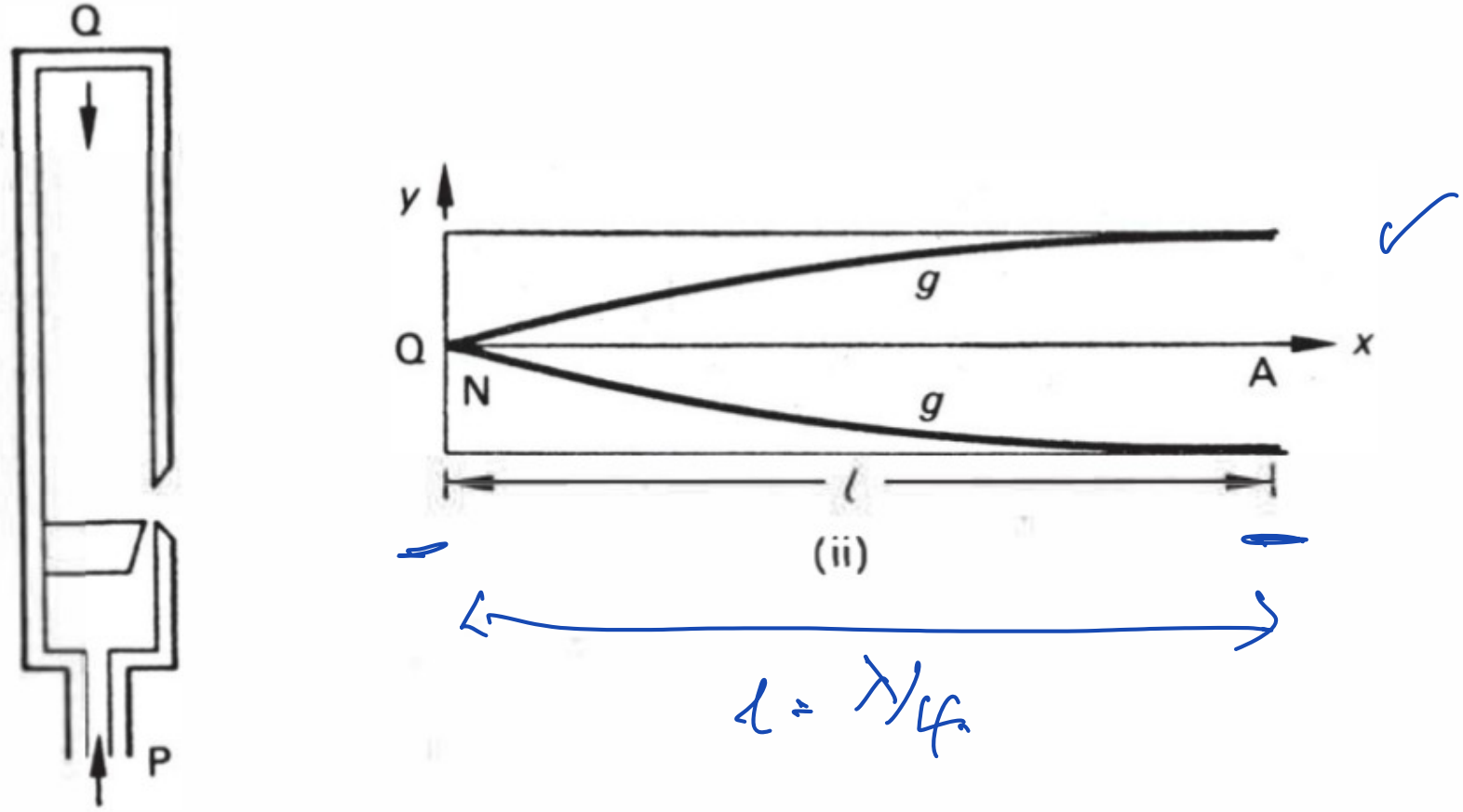
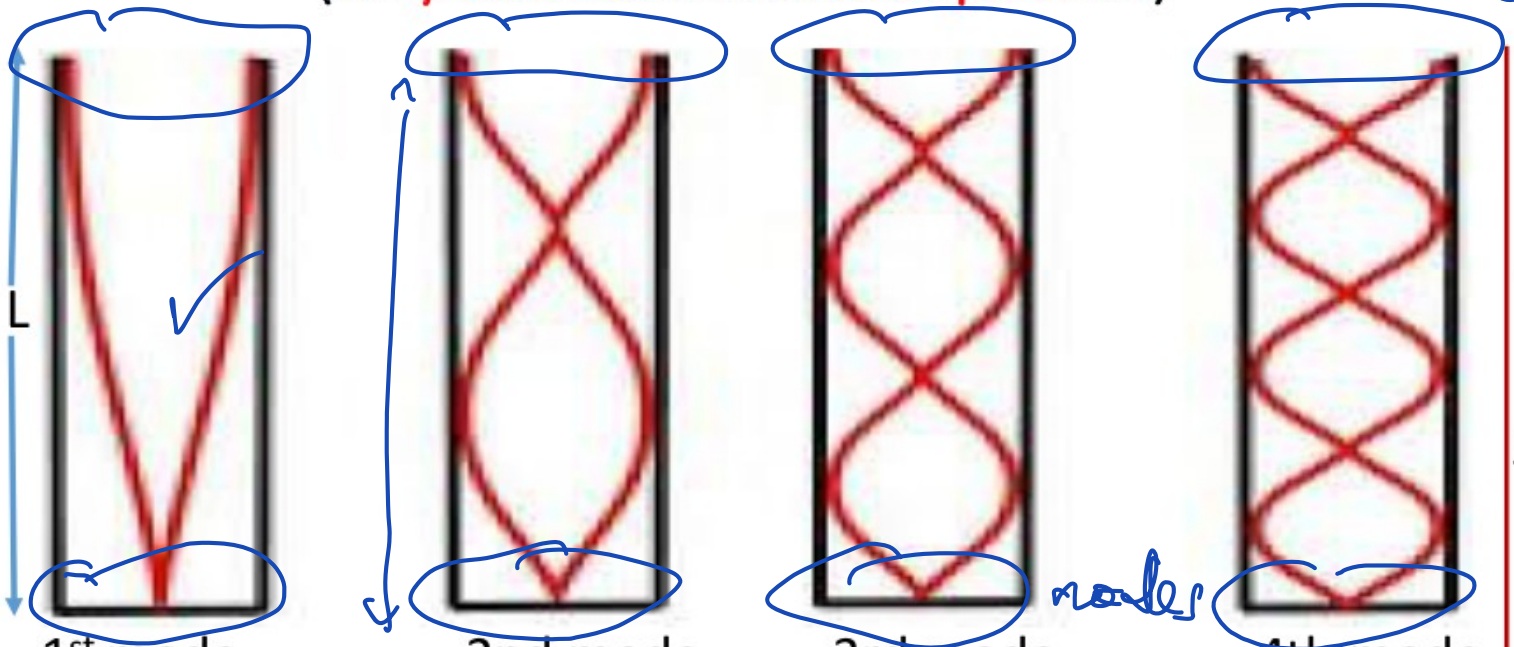


Figure 23.8 *(i) Closed (stopped) (ii) Fundamental of closed (stopped) pipe*

Standing Waves: Strings and Pipes

A cylindrical pipe open at one end and closed at the other end.

(only odd harmonics are present)



- $L = \lambda_1/4$
- $\lambda_1 = 4L$

$L = 3\lambda_2/4$ $\quad \lambda_2 = 4L/3$

$L = 5\lambda_3/4$ $\quad \lambda_3 = 4L/5$

$L = 7\lambda_4/4$ $\quad \lambda_4 = 4L/7$

overtones

$$\lambda_1 = 4L$$

$$\lambda_n = \frac{4L}{2n-1}$$

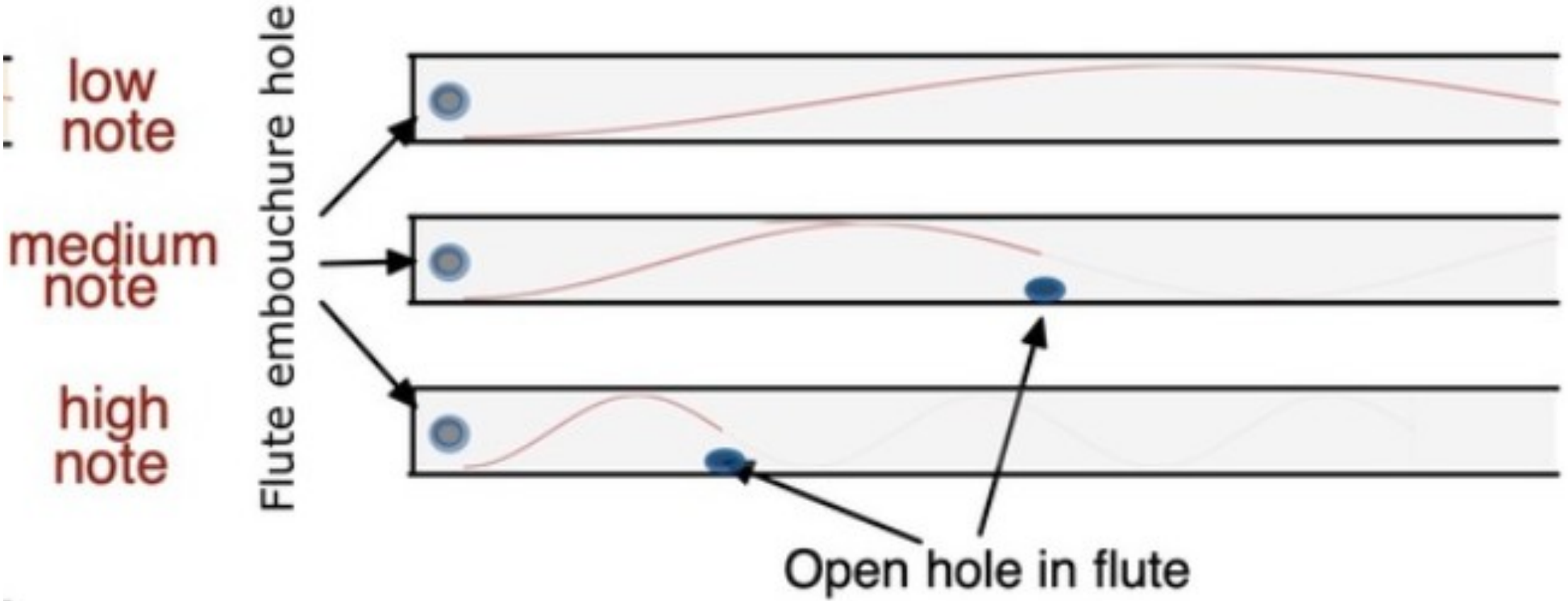
$$f_n = \left(\frac{2n-1}{4L}\right)V$$

$$f_1 = \frac{V}{4L}$$

$$f_n = (2n-1)f_1$$

n = 1, 2, 3,.....

Standing Waves: Flutes (Recorder)



PITCH ~ FREQUENCY & $f \sim 1/L$

Waves: Resonance

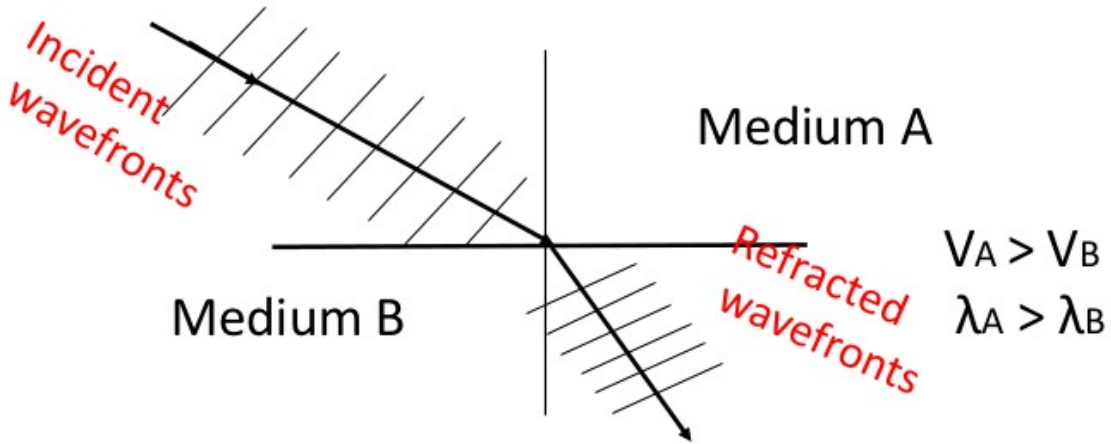


Figure 23.16 *Resonance in closed pipe*

& Beyond Waves

Propagation of Waves: Refraction

- **Refraction:** This is the change in the direction of a wave that results from the change in speed as a wave passes from one medium to another.
- Increase in speed results into increase in the wavelength and vice versa. The frequency of the wave remains unchanged when refracted.

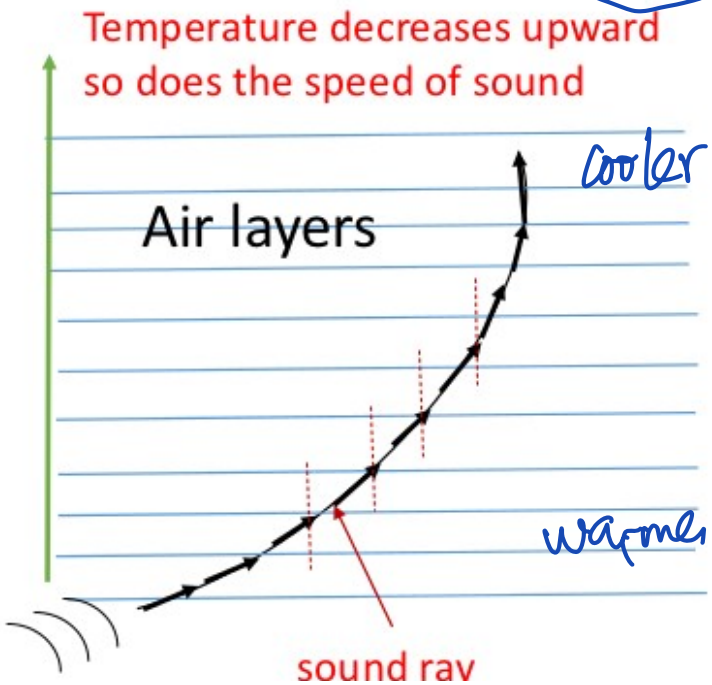


Propagation of Waves: Refraction

- DAY TIME

Refracted ray bends towards the normal all through.

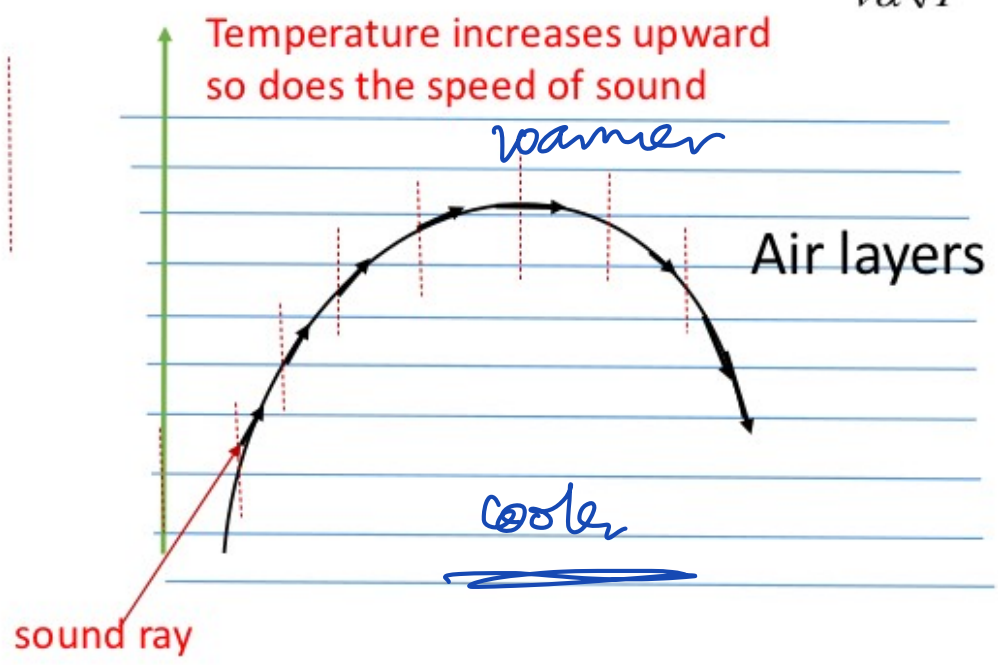
$v \alpha \sqrt{T}$



NIGHT TIME

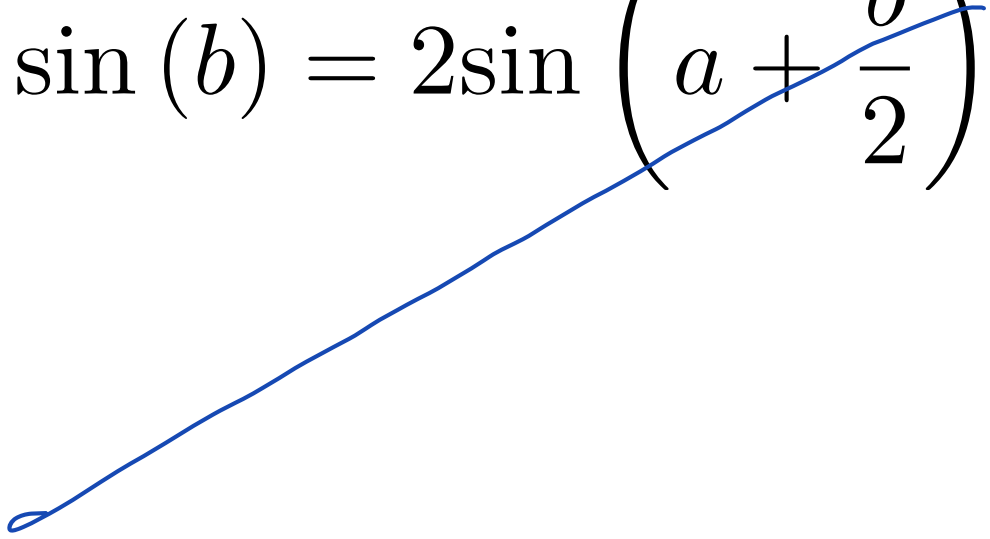
Refracted ray bends away from The normal on its way up

$v \alpha \sqrt{T}$



Adding Waves: Path Difference

$$\sin(a) + \sin(b) = 2\sin\left(a + \frac{b}{2}\right)\cos\left(a - \frac{b}{2}\right)$$



Propagation of Waves: Diffraction

Diffraction

- Diffraction is the spreading out of a wave after passing through an opening or bending around an obstacle.
- As a result of diffraction, sound waves can enter a room through a narrow opening and the whole intensity of the sound can be perceived.

Propagation of Waves: Diffraction

Bending/scattering of waves when they encounter an obstacle.

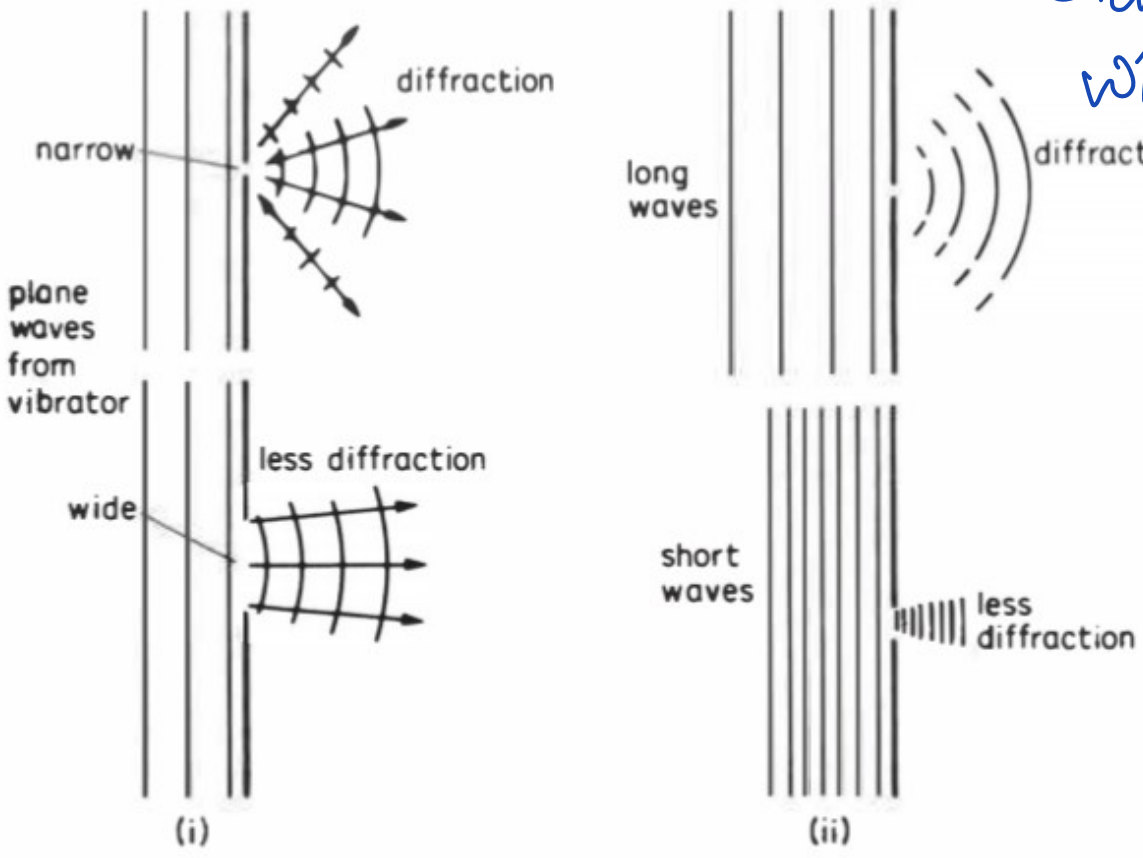


Figure 17.24 *Diffraction of waves*

Waves: Doppler Effect

(Movie)

Change in frequency observed due to relative motion of observer and source

Applications:

* (Sirens) ✓

* Astronomy ✓

* Radar

* Doppler Ultrasonography (Exe: Read up)

Doppler Effect: No relative motion

* f waves sent out in 1 second occupy a distance c. So,

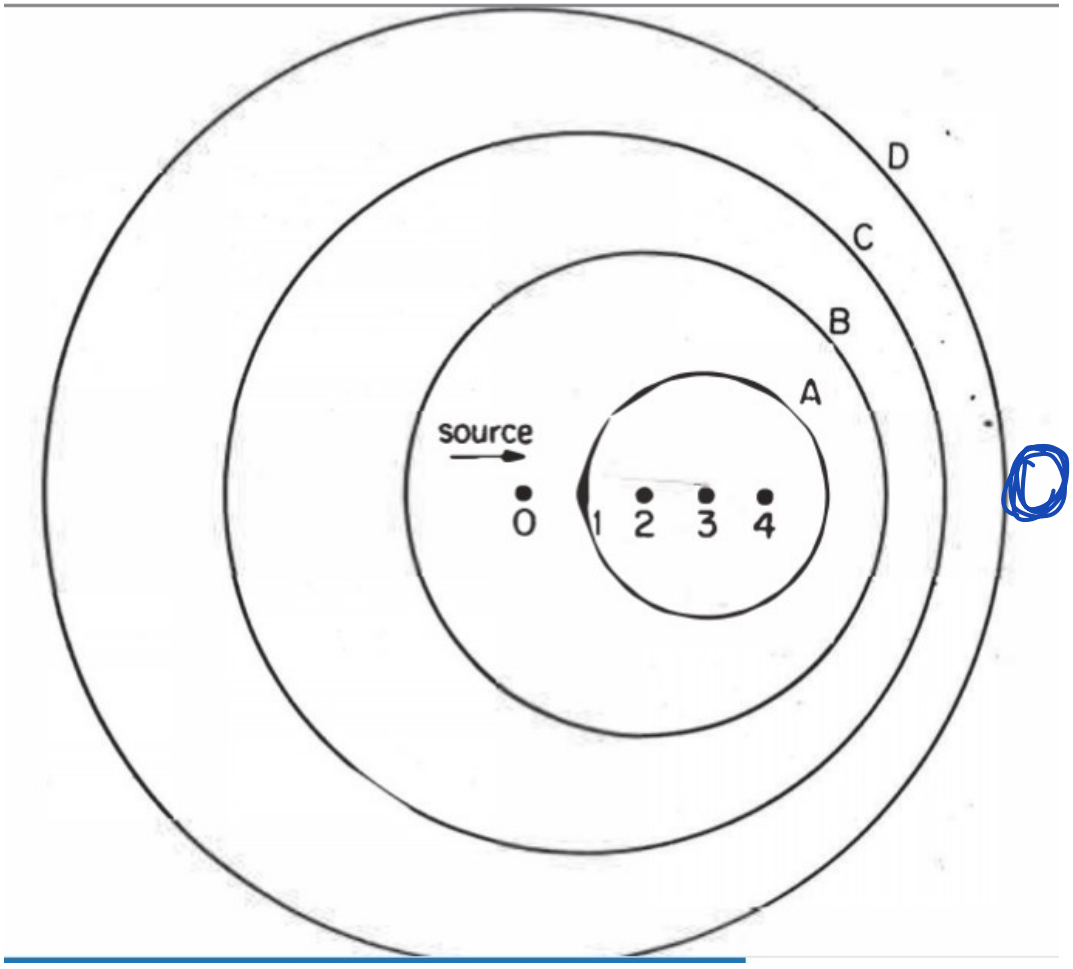
f waves

$$\lambda = c/f$$ ✓

Speed of Sound

$\lambda =$ dist / # of waves

Doppler Effect: S moving towards O



* Now Source moves and observer is on the right. f waves sent out in 1 second occupy a distance $(c - u_s)$. So:

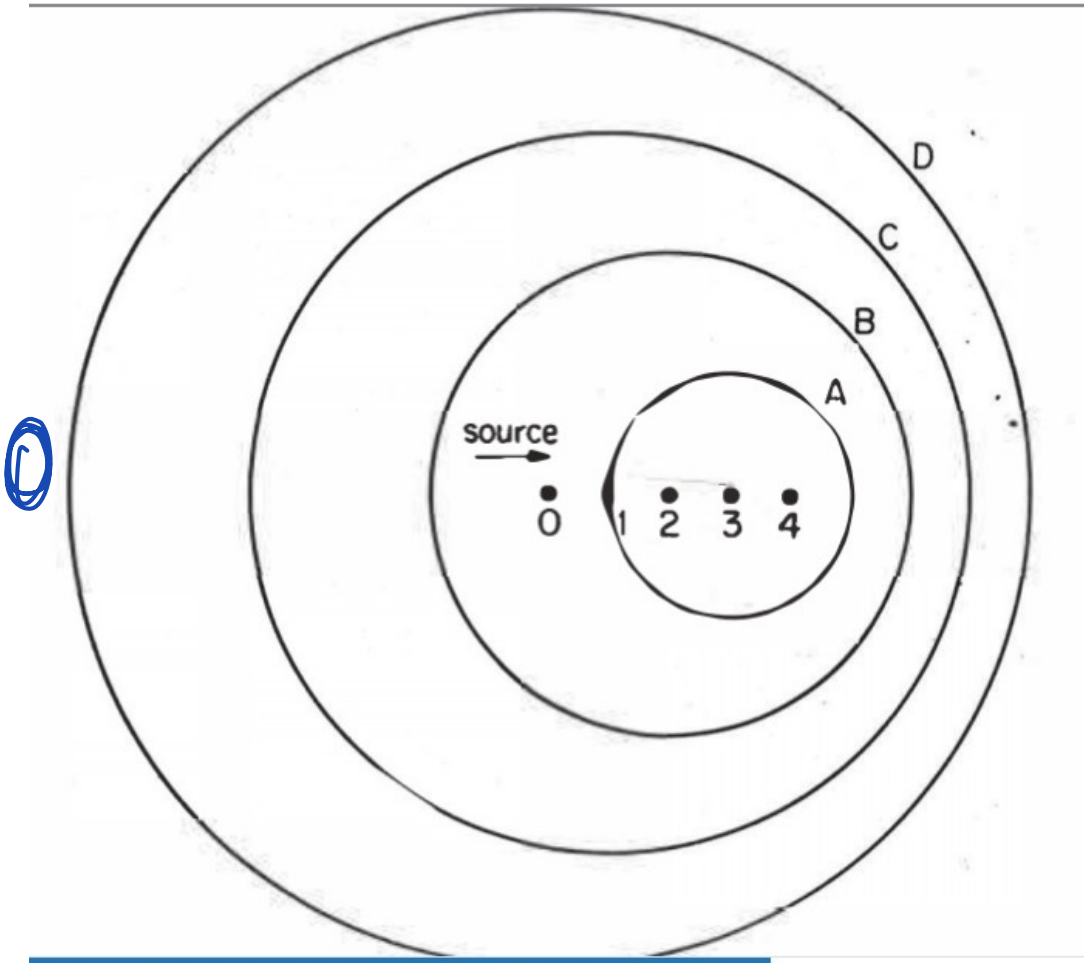
$$\lambda' = (c - u_s)/f$$

$$\text{App. freq} = \frac{\text{speed of sound}}{\text{apparent wavelength}}$$

$$\Rightarrow f' = \frac{c}{(c - u_s)/f}$$

$$f' = \frac{c}{c - u_s} f$$

Doppler Effect: S moving away from O



* Now Source moves and observer is on the left. f waves sent out in 1 second occupy a distance $(c + u_s)$. So:

$$\lambda' = (c + u_s)/f$$

$$\text{App. freq} = \frac{\text{speed of sound}}{\text{apparent wavelength}}$$

$$\Rightarrow f' = \frac{c}{(c + u_s)/f}$$

$$f' = \frac{c}{(c + u_s)} f$$

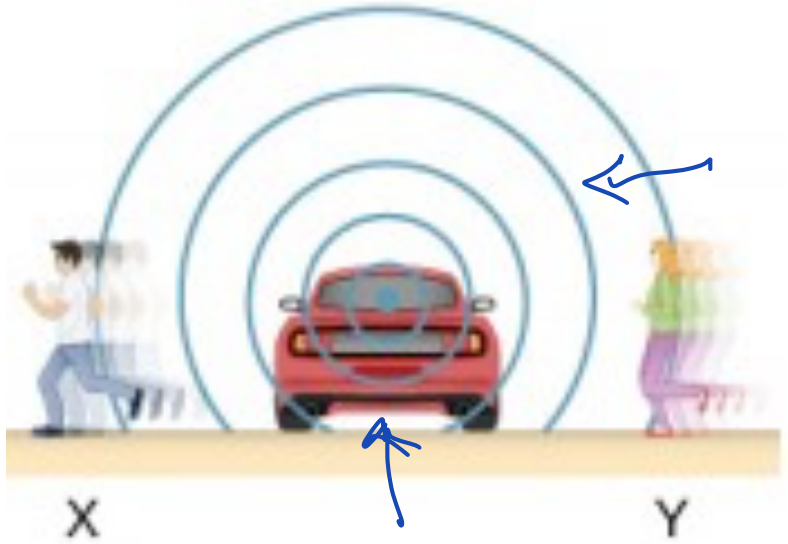
Doppler Effect: O moving towards S

* Speed of sound relative to Observer: $c + u_0$

* Wavelength = c/f

$$f' = \frac{(c + u_o)}{c/f}$$

$$f' = \frac{c + u_o}{c} f$$

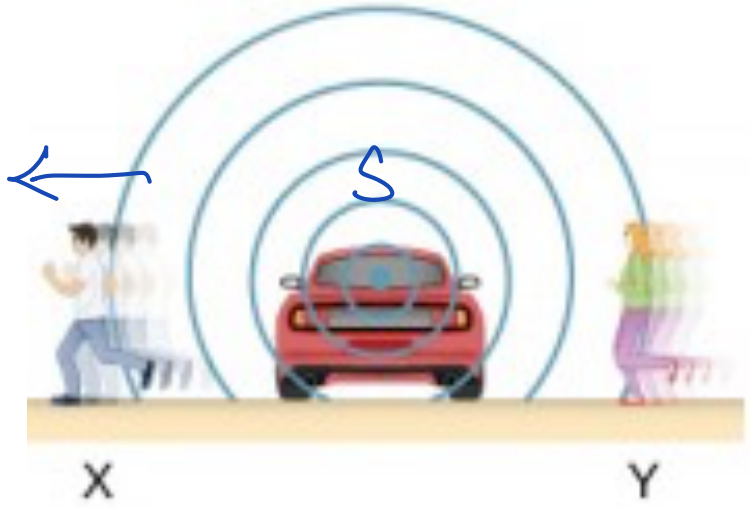


(c) Stationary source, moving observers

Doppler Effect: O moving away from S

* Speed of sound relative to Observer: $c - u_0$

* Wavelength = c/f



(c) Stationary source, moving observers

$$f' = \frac{c - u_o}{c} f$$

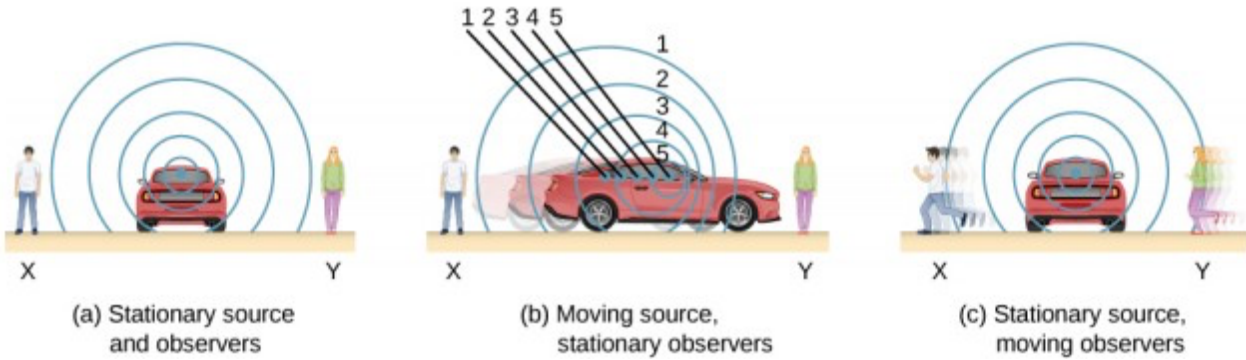
Doppler: Summary

$$f' = \frac{c}{c \mp u_s} f$$

$$f' = \frac{c \pm u_o}{c} f$$

$$f' = \frac{c - u_o}{c - u_s} f$$

$$f' = \frac{c + u_o}{c - u_s} f$$



(a) Stationary source and observers

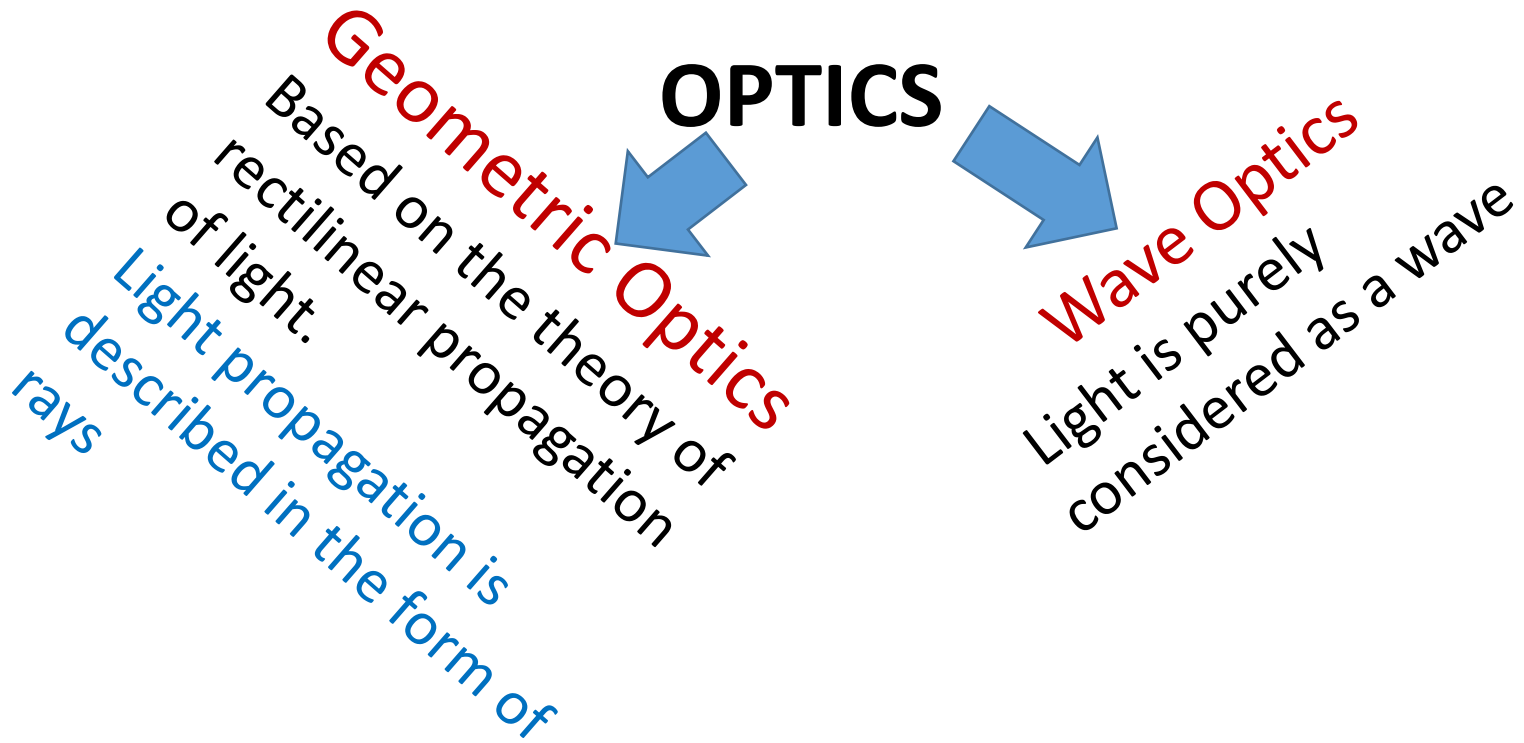
(b) Moving source, stationary observers

(c) Stationary source, moving observers

1. For wind effect, add or subtract u_w from c
2. Exe: What happens at supersonic speeds $u_s > c$?

OPTICS

- Optics is the study of Light. By light we mean the visible part of the electromagnetic spectrum.
- Early scientists were curious to study light because without it human sense of vision cannot be stimulated.



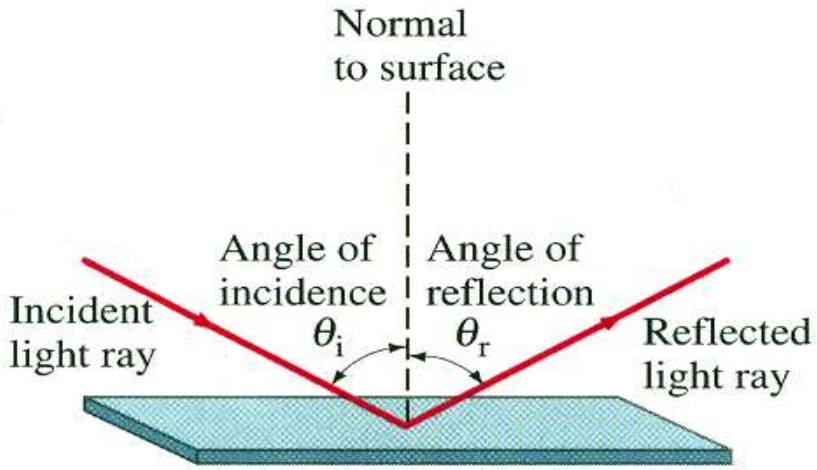
Vocabulary

- Image: That which is formed when light is reflected or refracted.
- Real Image: –
  - It is made from “real” light rays that converge at a point.
  - Can be projected unto a screen because light actually passes through the point where the image appears.
  - Always inverted.

Vocabulary

- Virtual Image–
  - “Not Real” because it cannot be projected on a screen
  - It is formed from virtual rays.
  - Normal: An imaginary line that is perpendicular to the reflecting surface.

Reflection of light from a smooth plane surface



- Laws of Reflection
- The incident and reflected rays lie in the same plane with the normal.
- The angle of incidence equals the angle of reflection.
- $\theta_i = \theta_r$

Image formation.

- When scattered light from an object falls on a plane smooth surface such as a mirror, the light is reflected, resulting in image formation
- The image is located behind the mirror.
- Image properties:

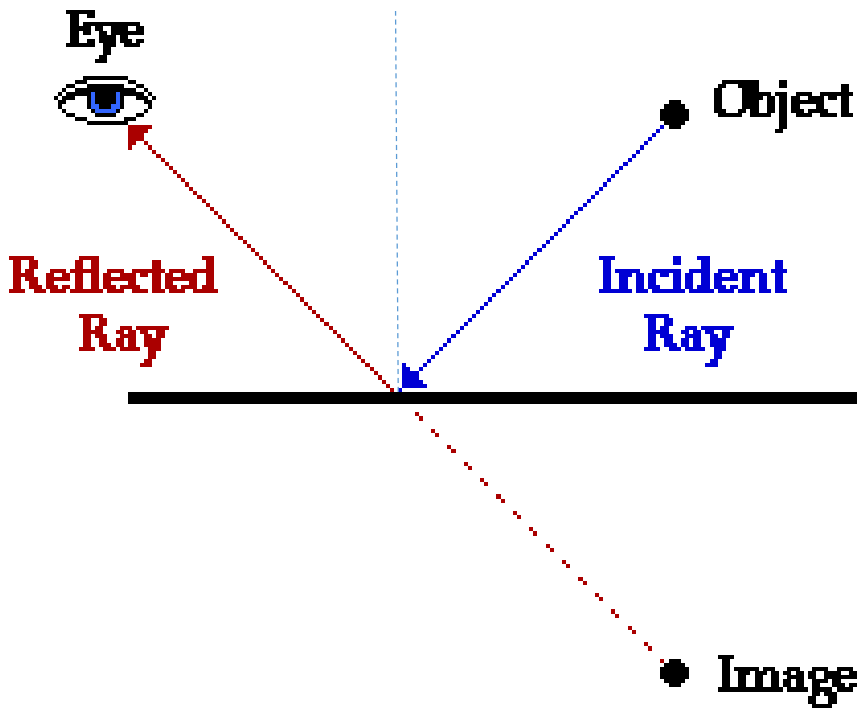
i) virtual

ii) Erect

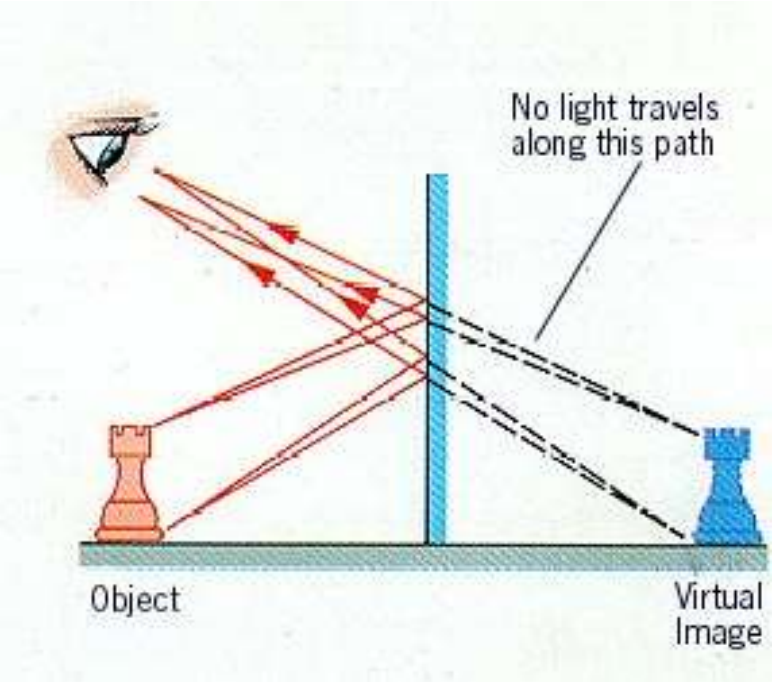
iii) Same size as the object

iv) Same distance from the reflecting surface as the object.

v) Laterally inverted

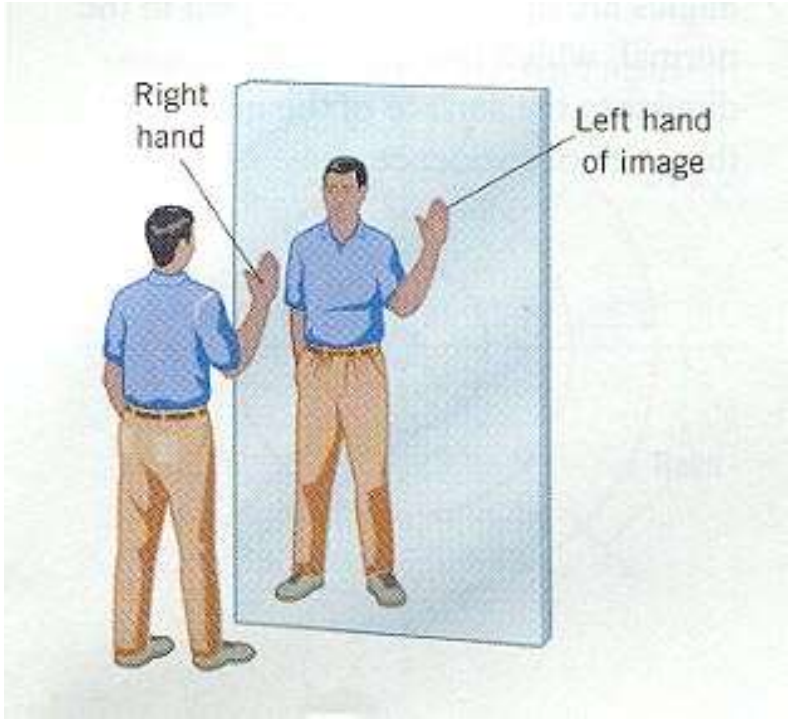


Virtual Images in Plane Mirrors



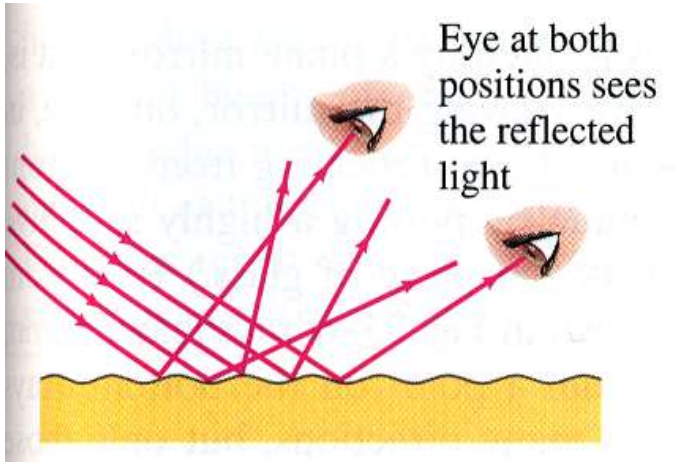
Since light energy doesn't flow from the image, the image is "virtual".

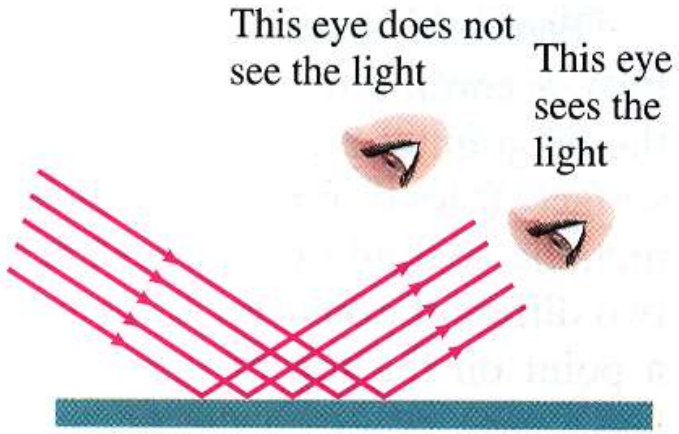
LATERAL INVERSION



Diffuse vs Specular Reflection

- Diffuse Reflection
  - Light incident upon a rough surface
  - Laws of reflection still hold; Normals are not parallel.
- Specular Reflection
  - Mirror like reflection
  - All Normals are parallel





Deviation of Light by a Plane-Mirror

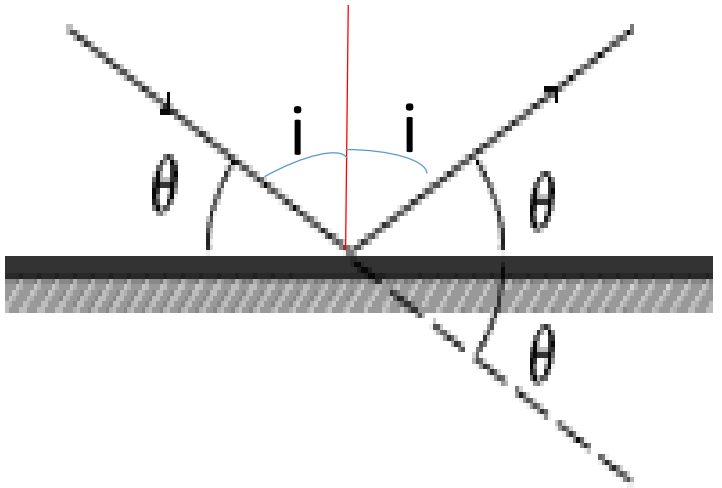
The total angle between the straight-line path of the incident ray and the reflected ray is twice the glancing angle

This is called the *deviation* of the light. It is a measure of the angle through which the light has strayed from its initial straight-line path.

$D = 2\theta$

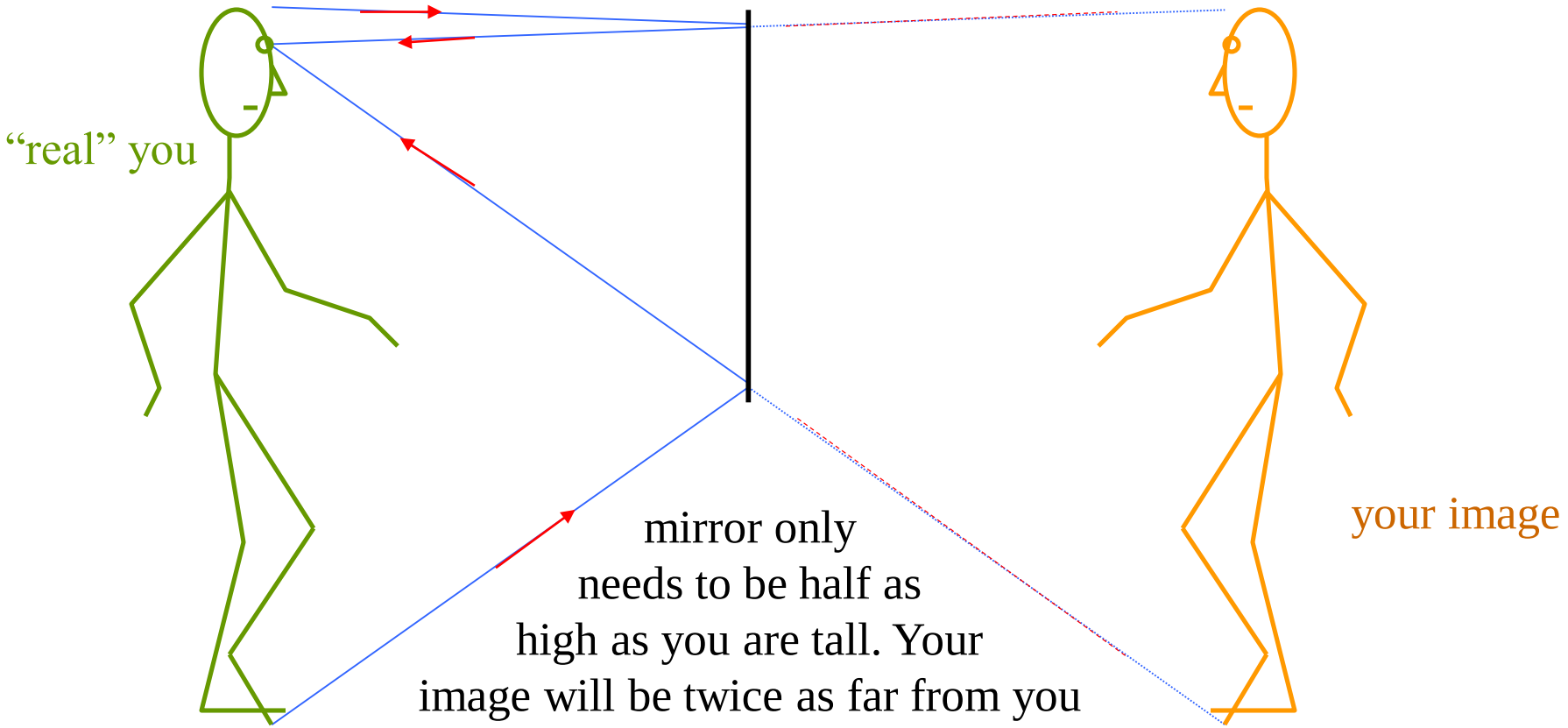
$\theta = 90 - i$

$D = 180 - 2i$



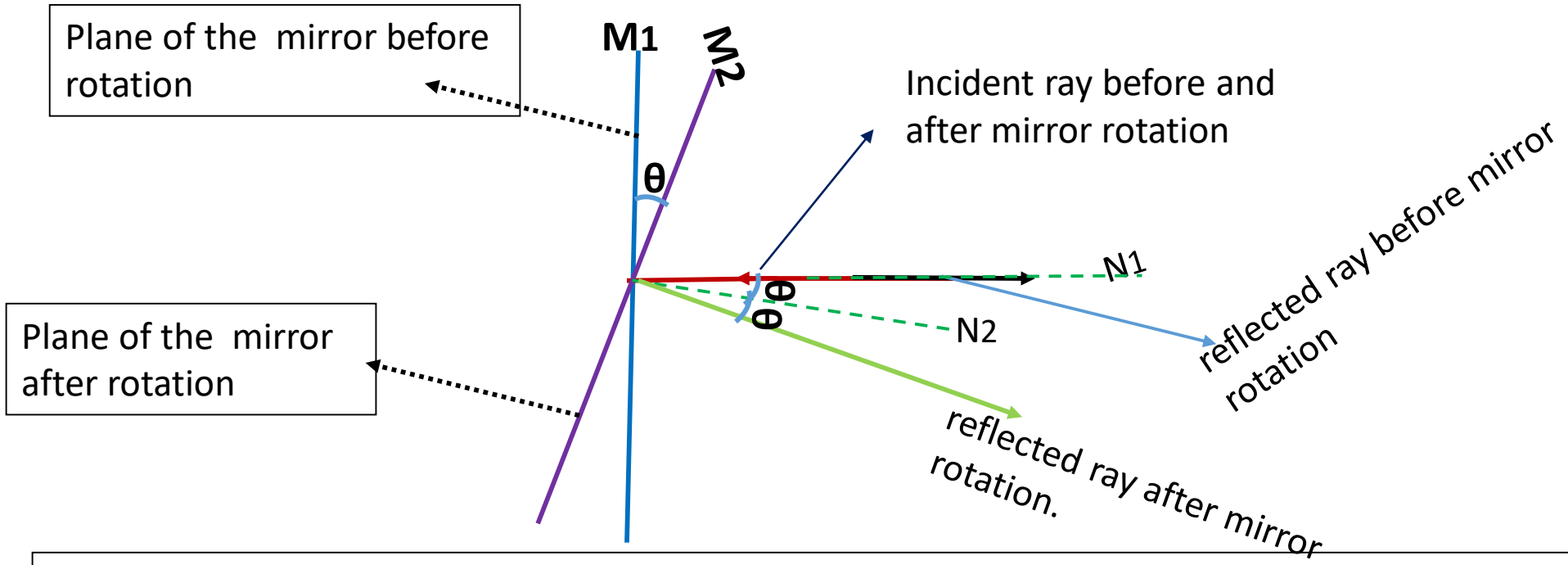
Effects of Reflection From Plane surfaces

- *To view your full image in a mirror, the mirror length needs to be only half your height.*



Effect of mirror rotation

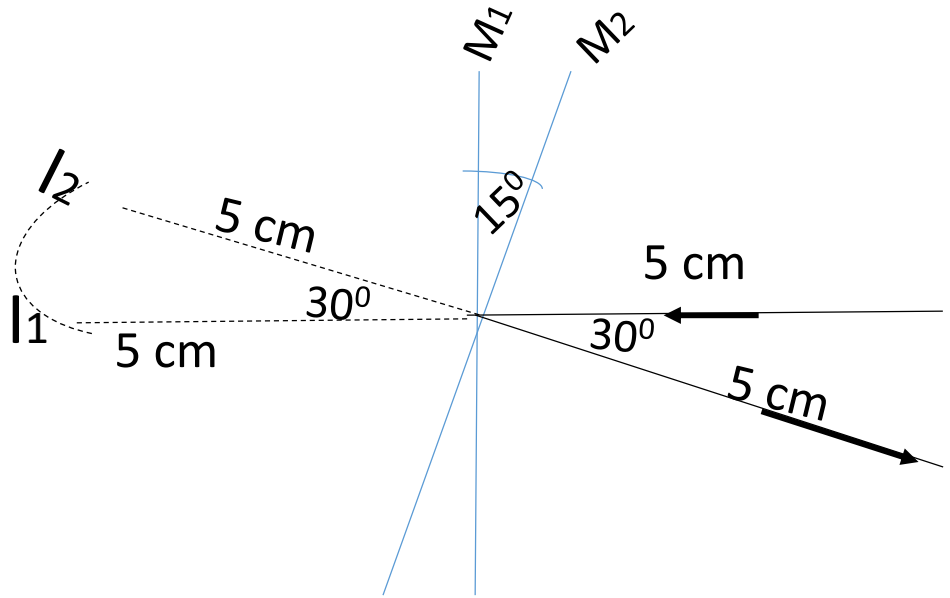
- When a mirror is rotated through angle θ about the point of incidence, the reflected ray rotates through angle 2θ



We can see that the reflected ray has rotated through angle 2θ. Because the reflected ray ends virtually on the image, the image moves on an arc which also substends angle 2θ at the point of incidence.

Worked Example

- A point object is placed 5 cm away from a plane mirror. Calculate the length of the arc that is traced by the image of the object as the mirror is rotated through angle 15^0

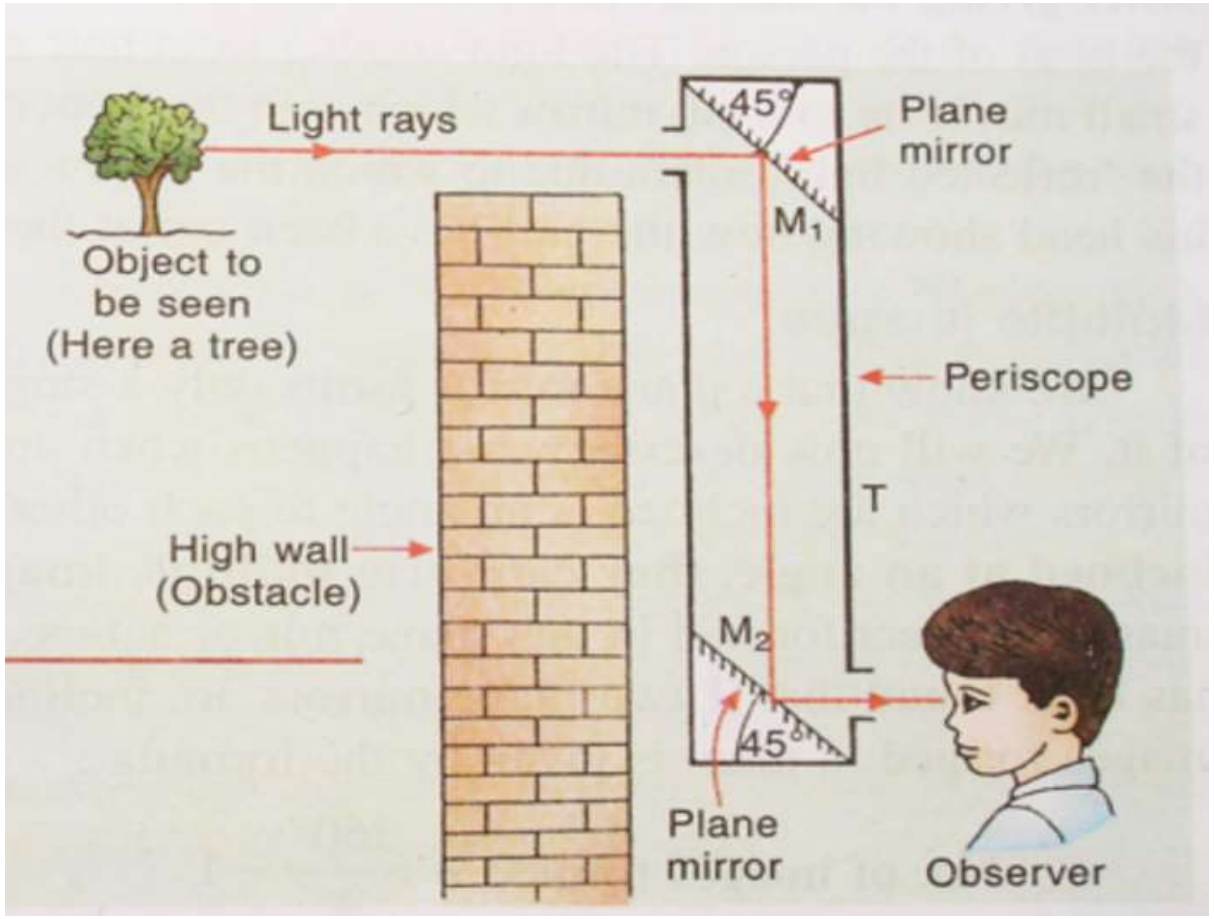


- Length of arc $I_1 I_2 = (15/360)(2\pi x30) \approx 7.9$ cm

Applications of Reflection From Smooth Plane Surfaces

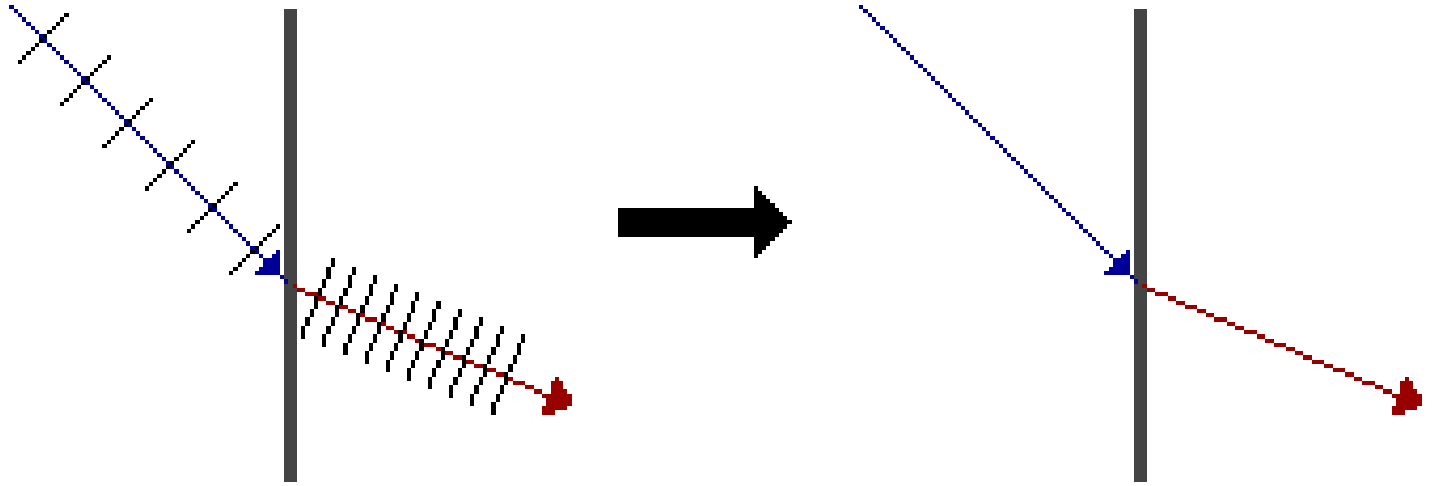
- Plane mirrors are used as looking glasses (dressing mirrors)
- In periscopes
- In solar cookers
- In kaleidoscopes

Mirror Periscope



Refraction

- Refraction: Sudden change in the direction of a wave as it goes from one medium to another.
- For refraction to occur, light ray must enter the medium obliquely
- It is always accompanied by change in speed



A ray will be used to depict the direction which a wavefront travels.

Refraction at Plane Surfaces

Laws of Refraction

1. The incident ray, the normal and the refracted ray all lie in the same plane.
2. For two particular media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant, i.e.

$$\frac{\sin i}{\sin r} = \text{constant} = n_2/n_1$$

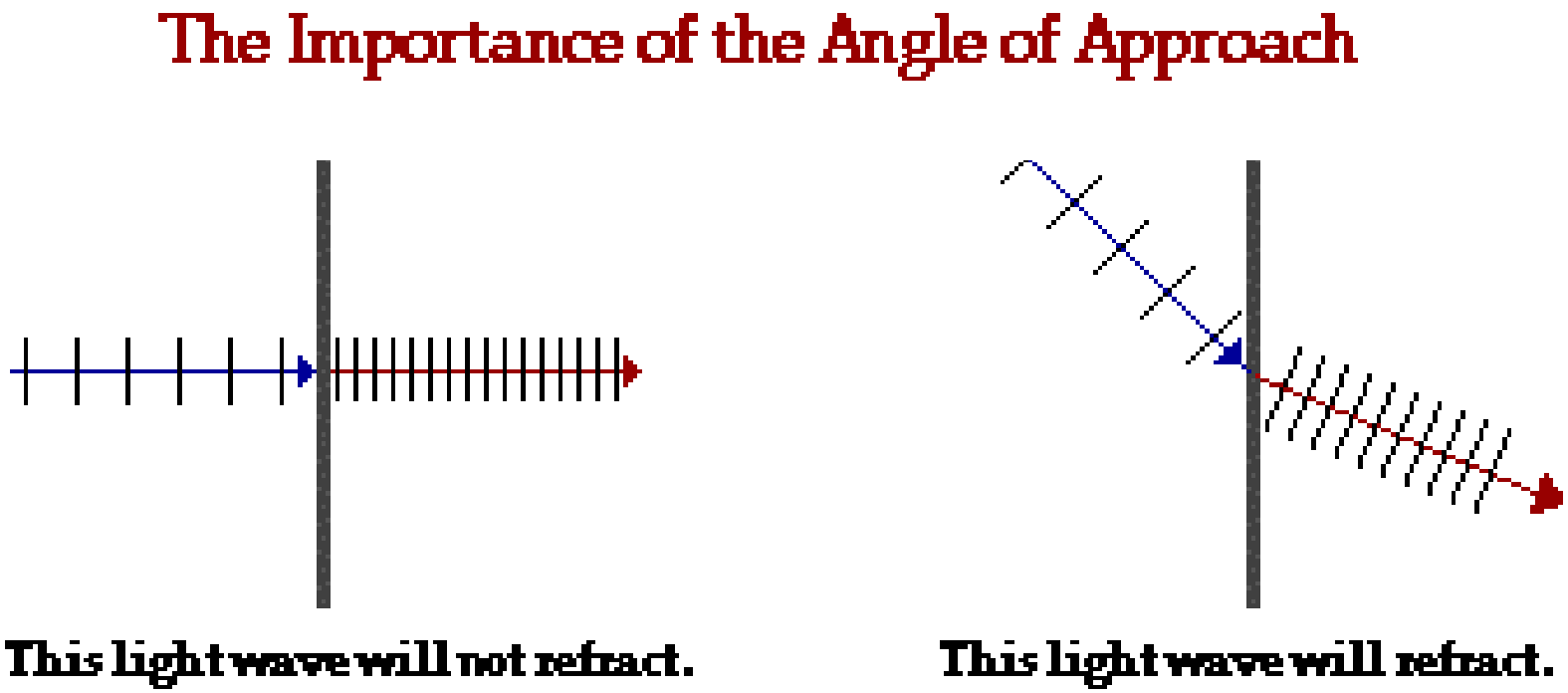
n_1 = Refractive index in the first medium

n_2 = Refractive index in the second medium

The second law of refraction is also known as Snell's Law.

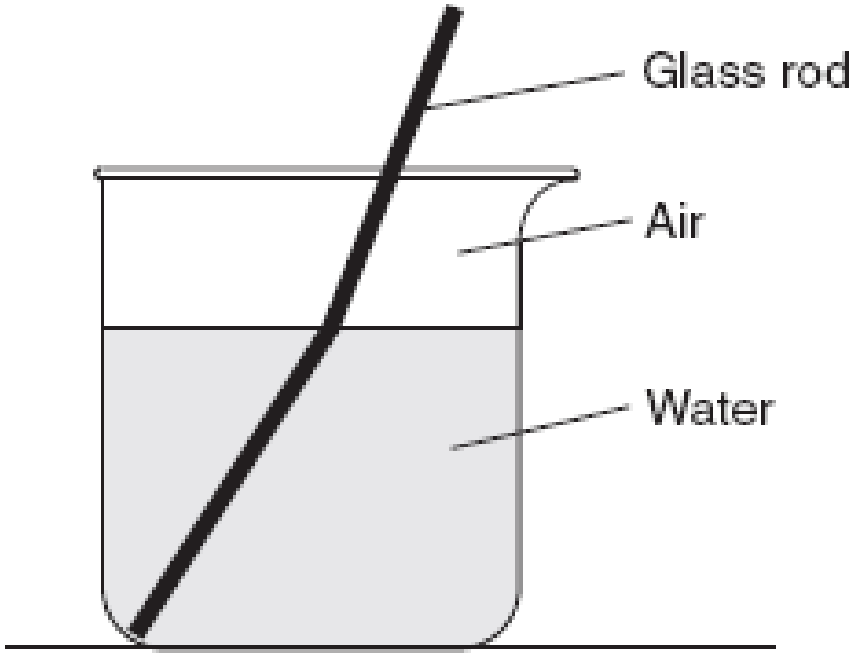
Refraction from plane surfaces

- In both cases the speed of the wave has decreased. This is indicated by the decrease in wavelength!



Refraction

- In which medium does light travel faster? (glass rod appears bent)



Absolute Refractive Index of a medium

$$n = \frac{c}{v}$$

- v is the speed of light in the new medium.

n is the absolute index of refraction. This is a measure of optical density. n is defined as the ratio of the speed of light in air (and vacuum) to the speed of light in a new medium.

- $c = 3.0 \times 10^8$ m/s
- As the index increases the speed decreases.
- Note: $n_{(air)} = 1.0$

Relative Refractive Index

- When light travels from medium 1 of absolute refractive index n_1 to medium 2 of absolute refractive index n_2, then relative refractive index from medium 1 to medium2 is given by:

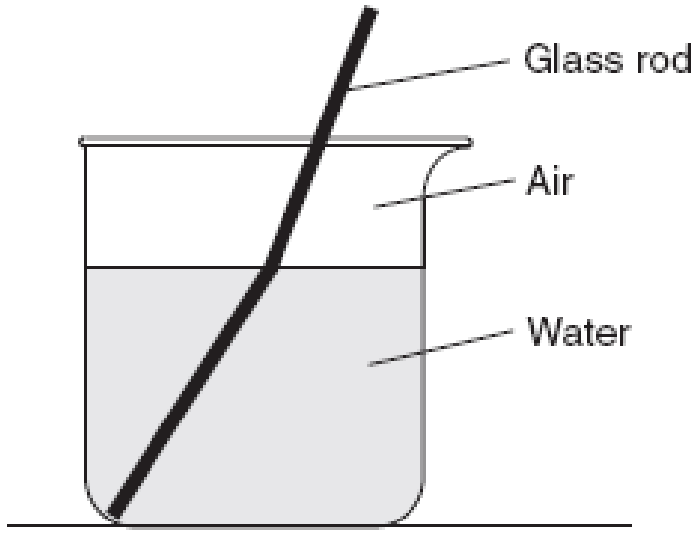
$$n_{rel} = \frac{n_2}{n_1} = \frac{c}{v_2} \div \frac{c}{v_1}$$

$$n_{rel} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

- If $n_{rel} < 1$; speeds up
- If $n_{rel} > 1$; slows down

Refraction

$n_{(water)}=1.33$; $n_{(glass)}=1.50$; $n_{(air)}=1.00$

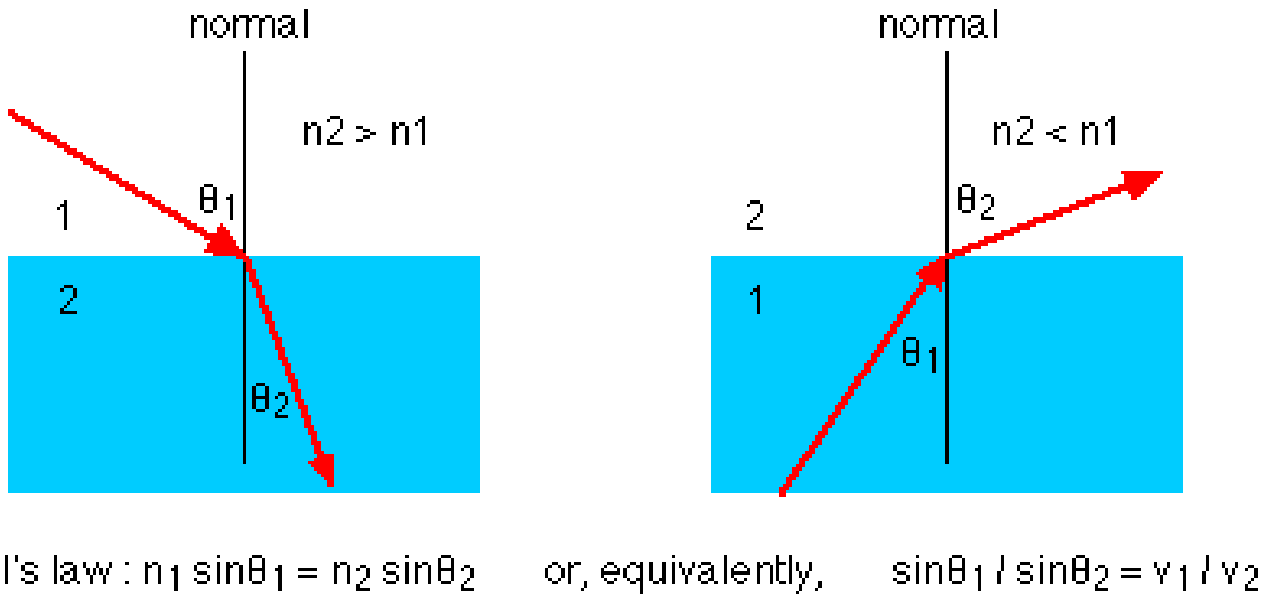


Calculate the speed of light in water and glass.

$V_w = 2.26 \times 10^8 m/s$

$V_g = 2.00 \times 10^8 m/s$

Snell's Law of Refraction ($n_k \sin\theta_k$ = constant)

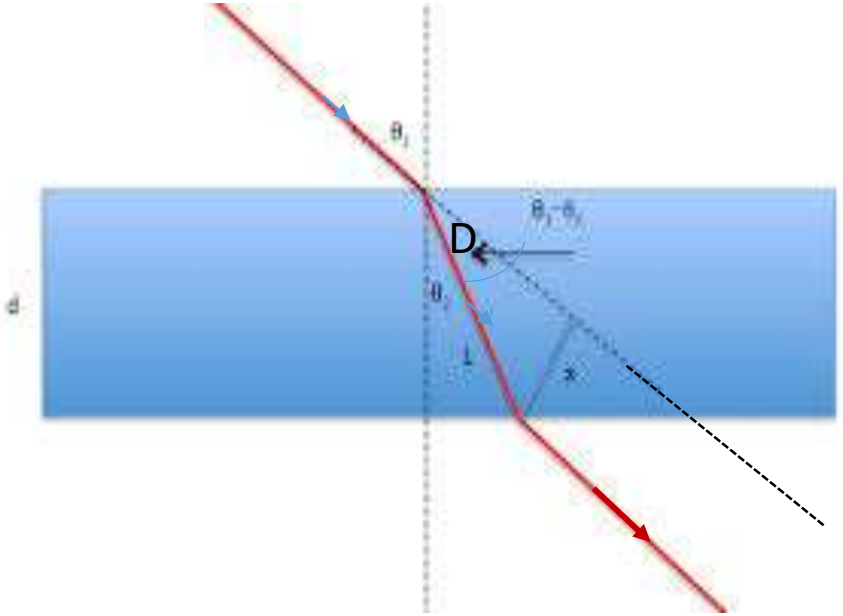


Snell's law : $n_1 \sin\theta_1 = n_2 \sin\theta_2$ or, equivalently, $\sin\theta_1 / \sin\theta_2 = v_1 / v_2$

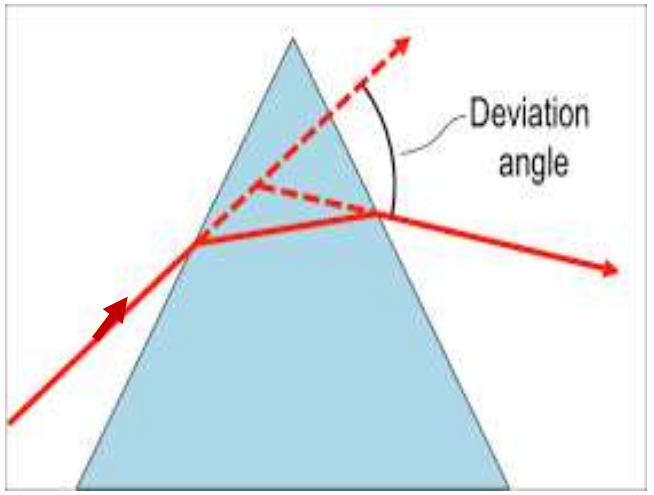
- When a wave slows down it bends closer to the normal.{n2>n1}
  When a wave speeds up it bends away from the normal. {n2<n1}

Deviation Due to Refraction of Light

Glass Slab



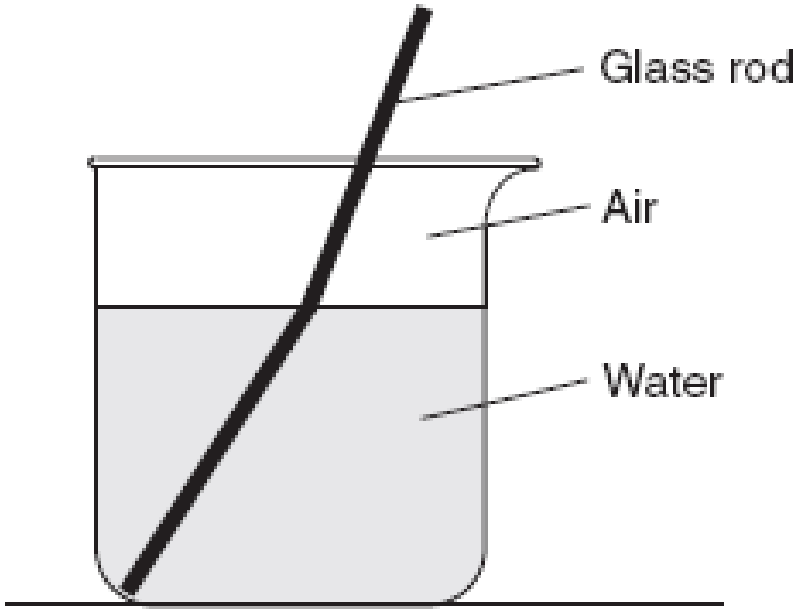
Glass Prism



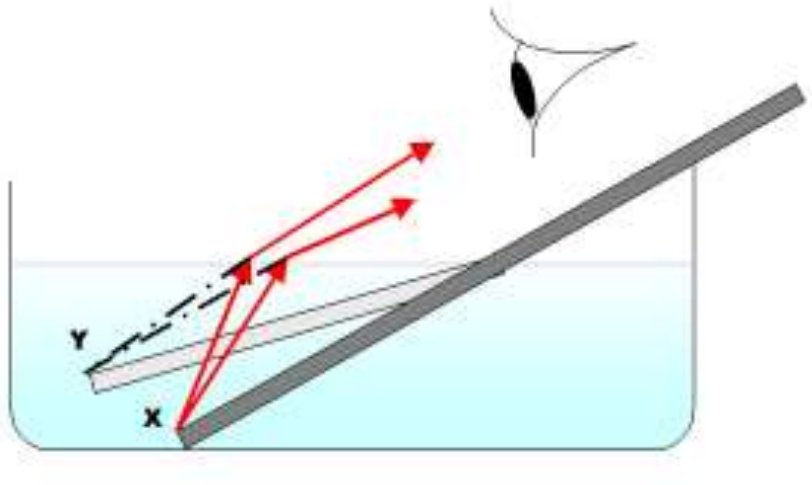
Deviation inside the
slab = angle of incidence – angle of refraction
Final deviation outside the slab = 0
Because emergent ray is parallel to
The direction of incident ray

Refraction

- If light rays bend closer to the normal when slowing down, why does the glass rod seem to bend away form the normal?



Refractive index in terms of Apparent and Real Depth



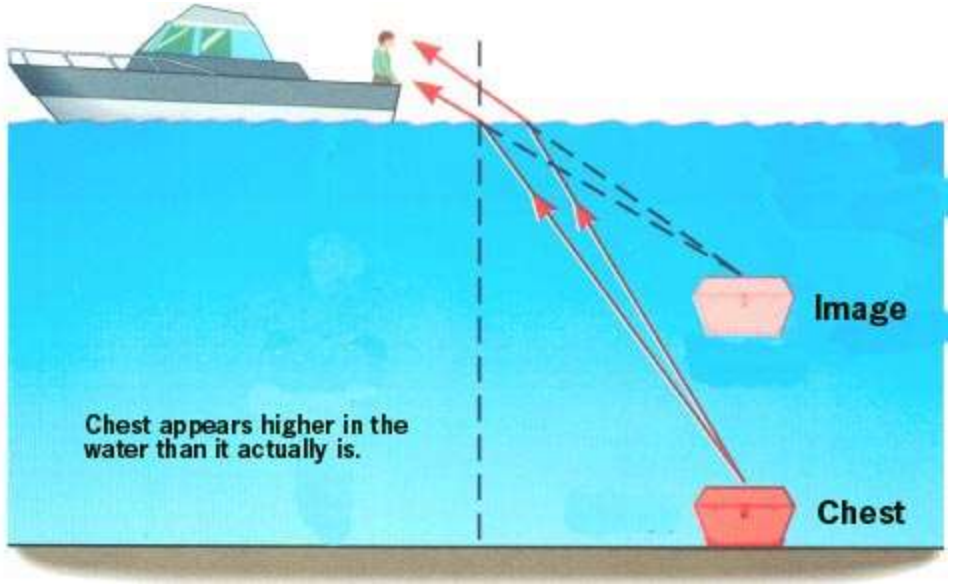
- It is the rays from the rod inside the water that enter your eyes.
- A virtual image appears to come from point **y**

$$n = \frac{R}{A}$$

R – Real Depth

A – Apparent Depth

Apparent Depth



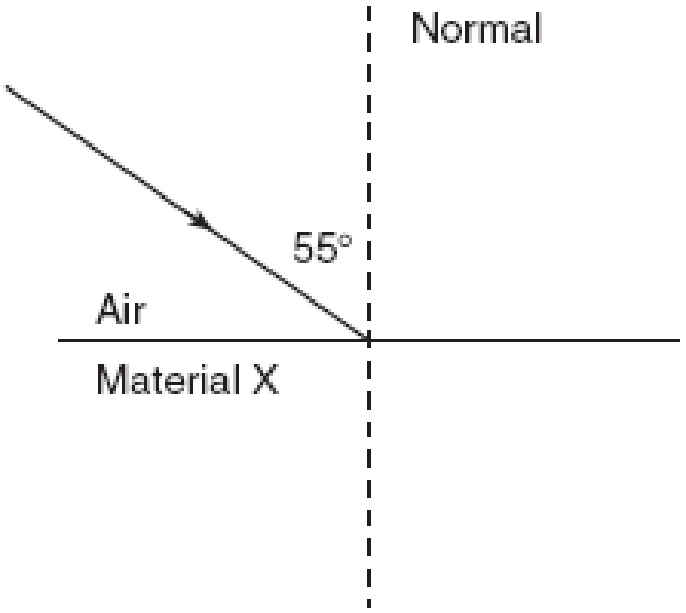
- If the chest is 20 m below the surface at what depth will the image appear? Assume $n_{\text{sea water}} = 1.34$

Snell's Law

- $n_1 \sin\theta_1 = n_2 \sin\theta_2$
- $v_1 / v_2 = \lambda_1 / \lambda_2$

Exercises

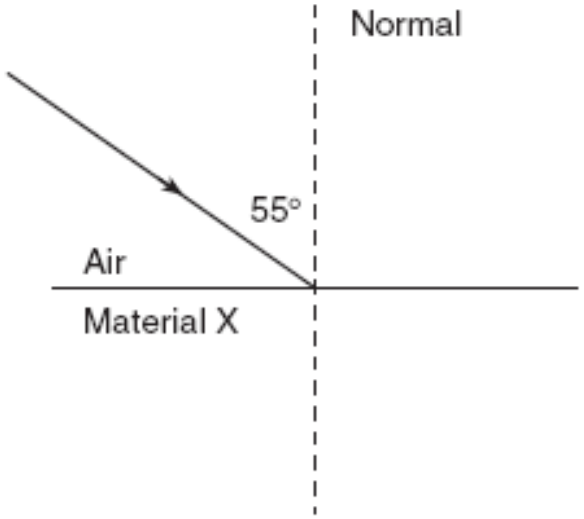
1. A monochromatic light ray f= 5.09 x 10^{14} Hz is incident on medium X at 55˚. The absolute index of refraction for material X is 1.66



a) Determine the angle of refraction.

b) Determine the speed of light in medium X.

Hint:



1. use Snell's Law to obtain $\theta_2 = 30^\circ$

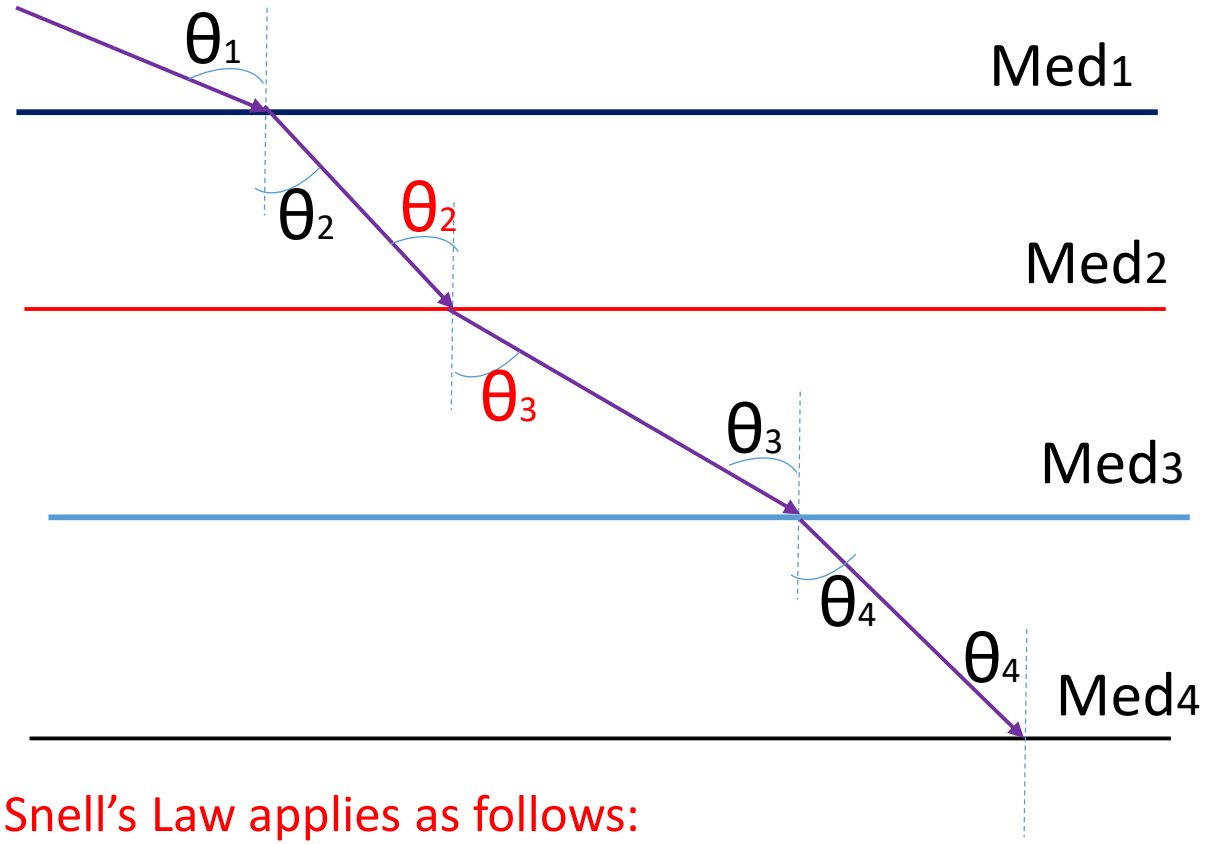
2. use $n=c/v$ to obtain $v = 1.8 \times 10^8$ m/s

2. A glass block (n=1.52) is placed inside water contained in a basin. A layer of oil (n=1.44) is poured on the water surface.. A ray of light strikes the oil surface at an angle of incidence of 38°

Calculate the angle of refraction at the water glass interface ($n_{(air)}$ = 1.00)

- Hint. Use Snell's law to obtain $\theta_{glass} = 23.9^{o}$

Refraction Through Multiple Layers



Snell's Law applies as follows:

$n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3 = n_4 \sin\theta_4$ ------

$n_i \sin\theta_i = \text{constant}$

- Snell's law: $n_k \sin\theta_k$ = constant , i.e.
- $n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3 = n_4 \sin\theta_4$ …….
- It implies that if light travels from medium 1 to 4 medium 4 through 2 and 3, refraction through 4 occurs as if light travels directly from Medium 1 to 4.

Exercise

A glass block (n=1.52) is placed inside water contained in a basin. A layer of oil (n=1.44) is poured on the water surface.. A ray of light strikes the oil surface at an angle of incidence of 38°

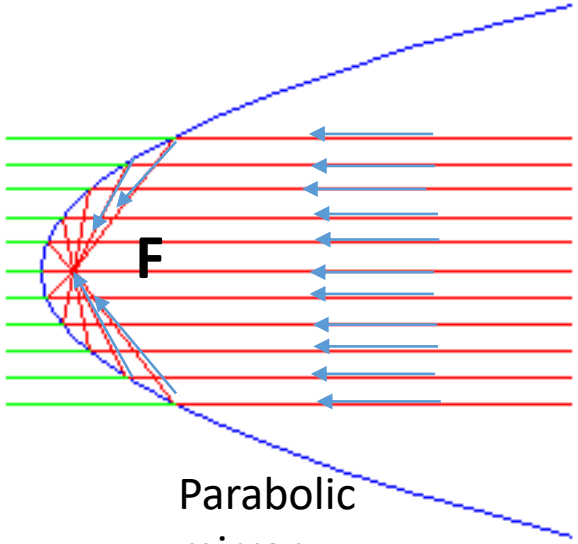
Calculate the angle of refraction at the water glass interface ($n_{(air)}$ = 1.00)

- Hint. Use Snell's law to obtain $\theta_{glass} = 23.9^{o}$

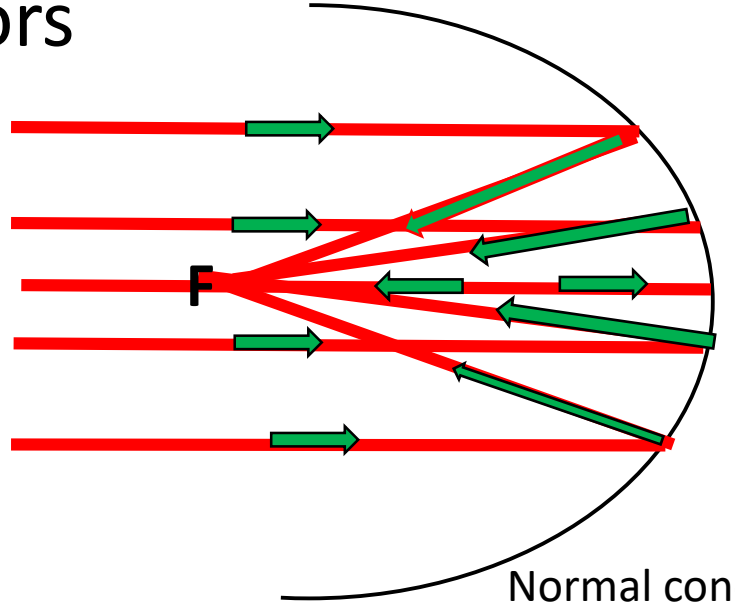
REFLECTION FROM CURVED SURFACES

Concave Mirrors

- Note: Concave mirrors are also known as converging mirrors because parallel rays that fall on their surfaces are reflected to meet at its focal point.
- Used as a magnifying mirrors



Parabolic mirror



Normal concave mirror

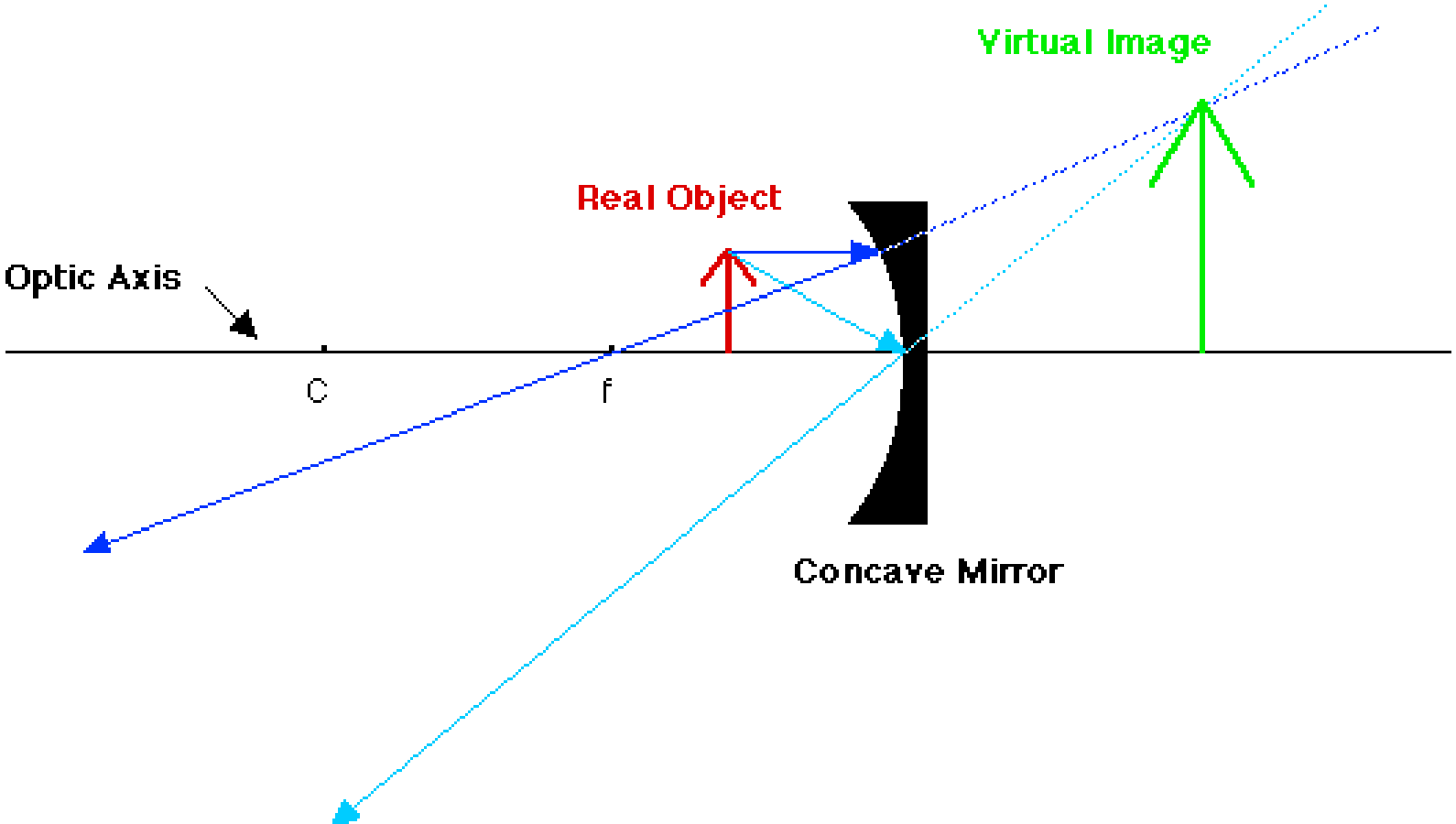
Concave Mirrors

- Curves inward
- May be real or virtual image

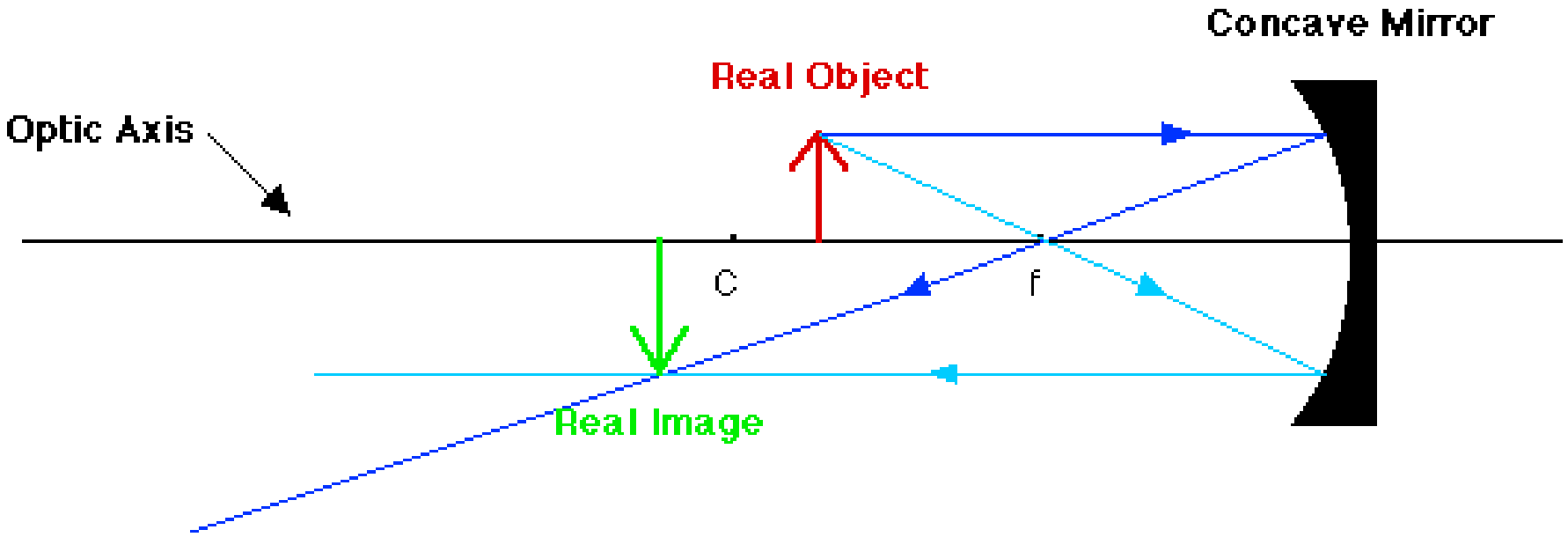


Image formation by concave mirrors

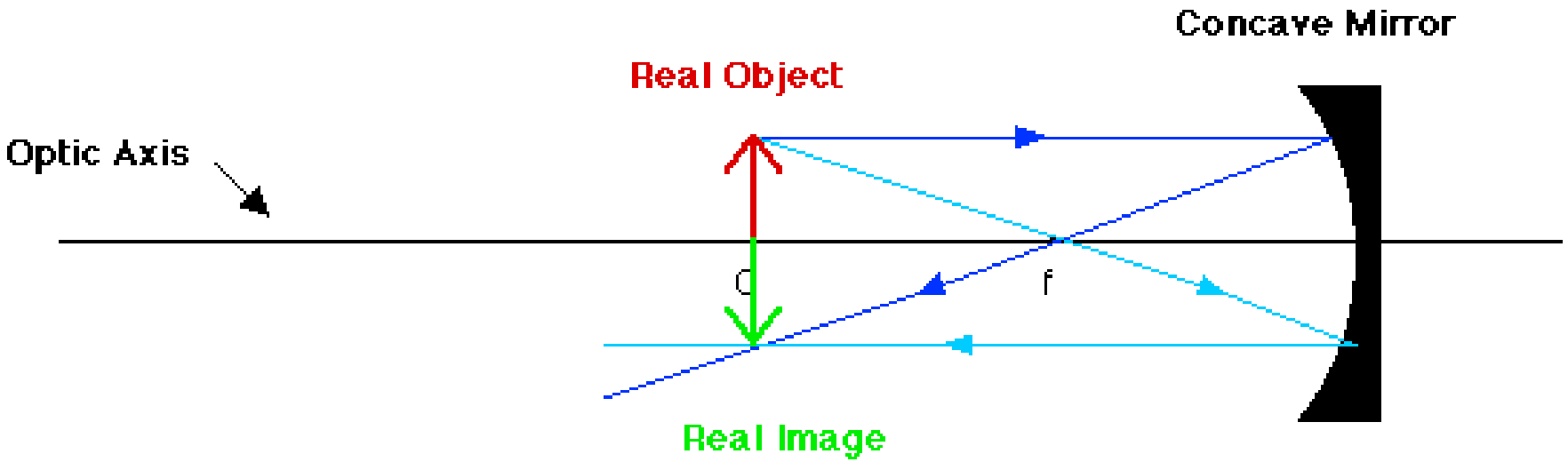
For a real object between f and the mirror, a virtual image is formed behind the mirror. The image is upright and larger than the object.



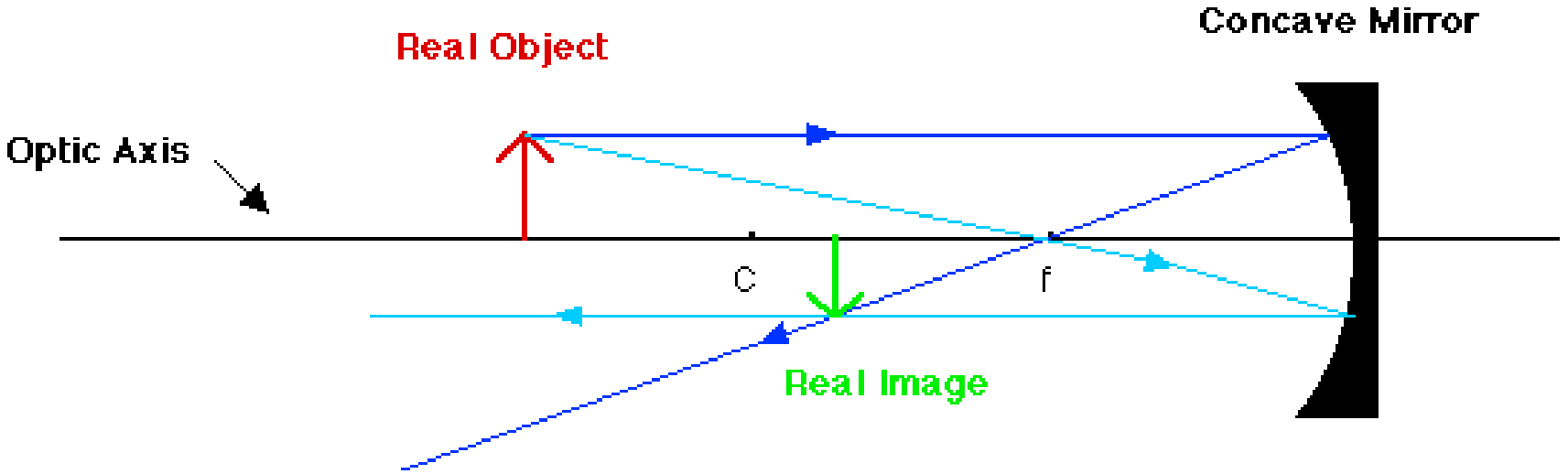
For a real object between C and f, a real image is formed outside of C. The image is inverted and larger than the object.



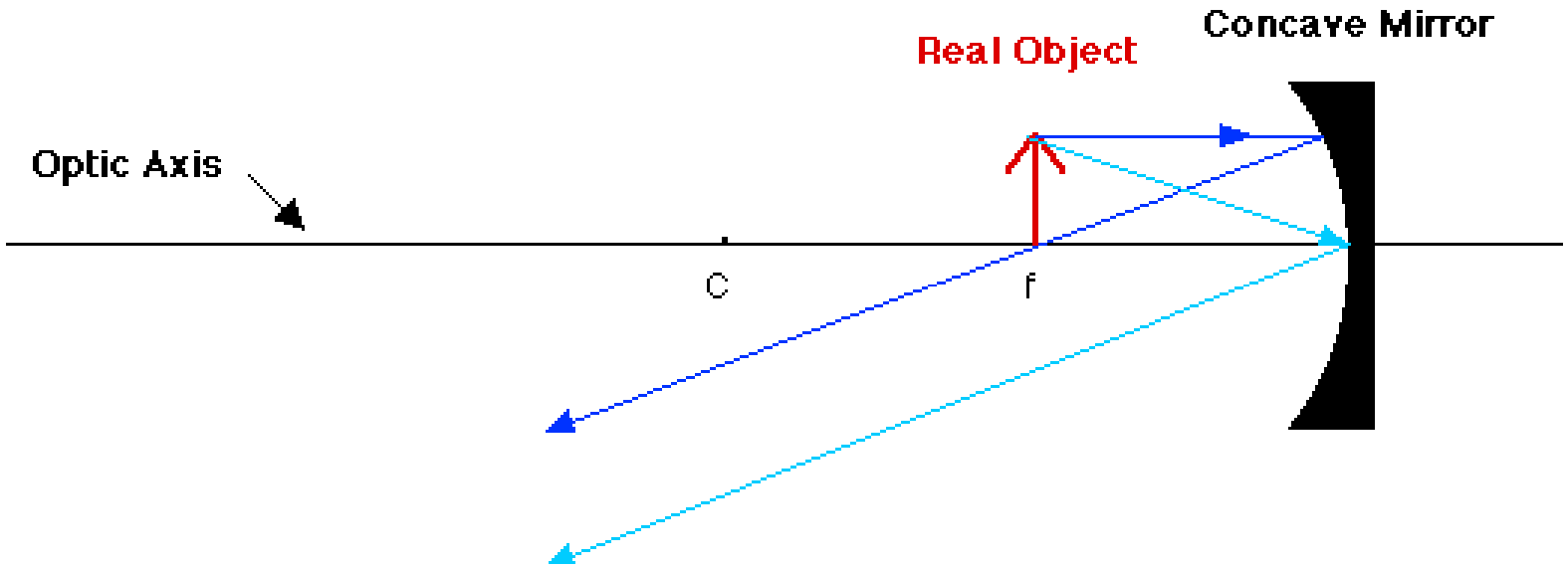
For a real object at C, the real image is formed at C. The image is inverted and the same size as the object.



For a real object close to the mirror but outside of the center of curvature, the real image is formed between C and f. The image is inverted and smaller than the object.



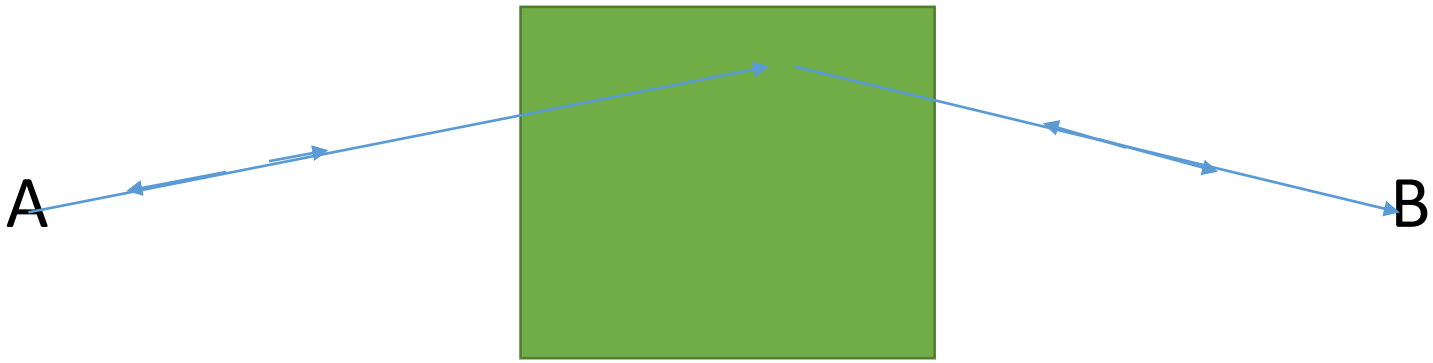
When the object is at the focal poin, f.



For a real object at f, no image is formed. The reflected rays are parallel and never converge.

The principle of reversibility of light

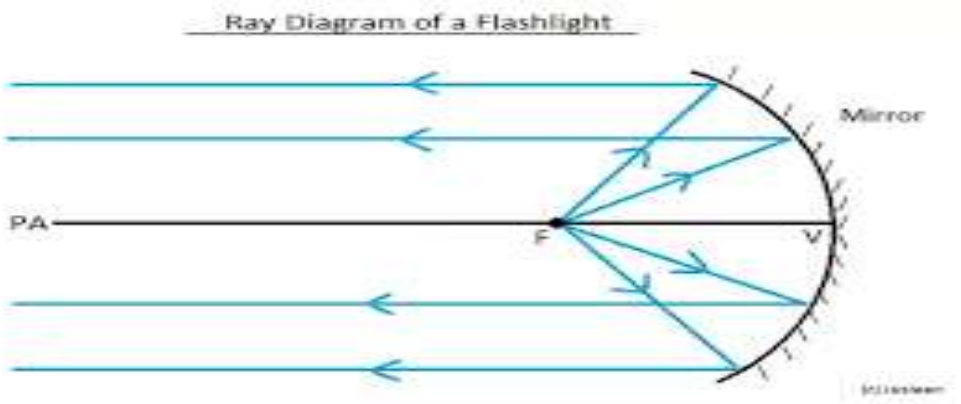
- If the direction of light is reversed, it will follow the same path irrespective of the number of times it is reflected or refracted.



Applications of Reflection From Concave mirrors/surfaces

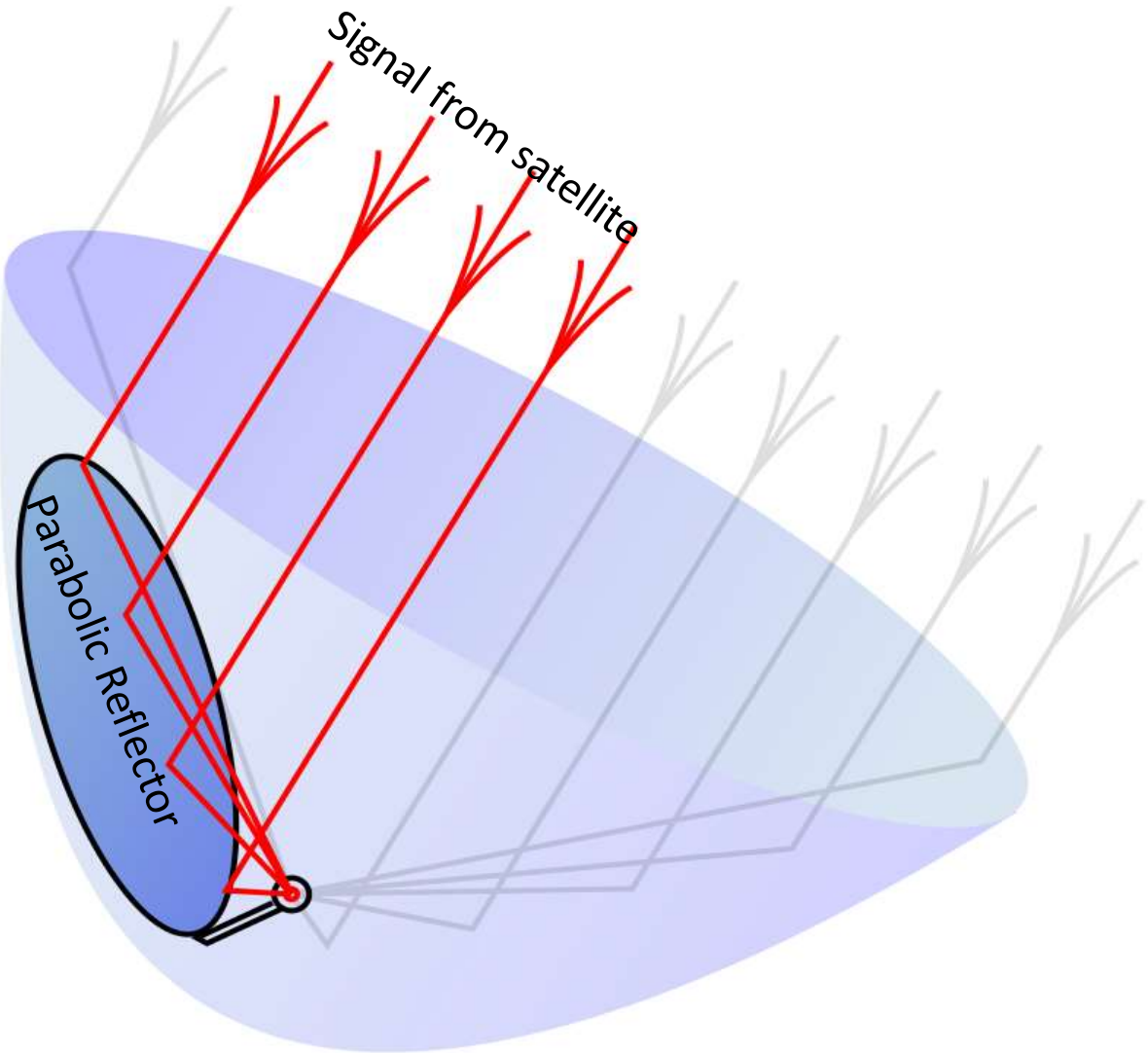
- Torches/ Searchlights, Security lights
- Headlights of vehicles
- Reflecting Telescopes
- Satellite Dishes
- Solar cookers

Searchlight





Television satellite Dish



Convex (Diverging) Mirror

- Curves outward
- Reduces images
- Virtual images
- Use: Rear view mirrors, store security…



CAUTION! Objects are closer than they appear!

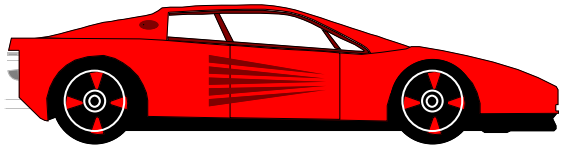
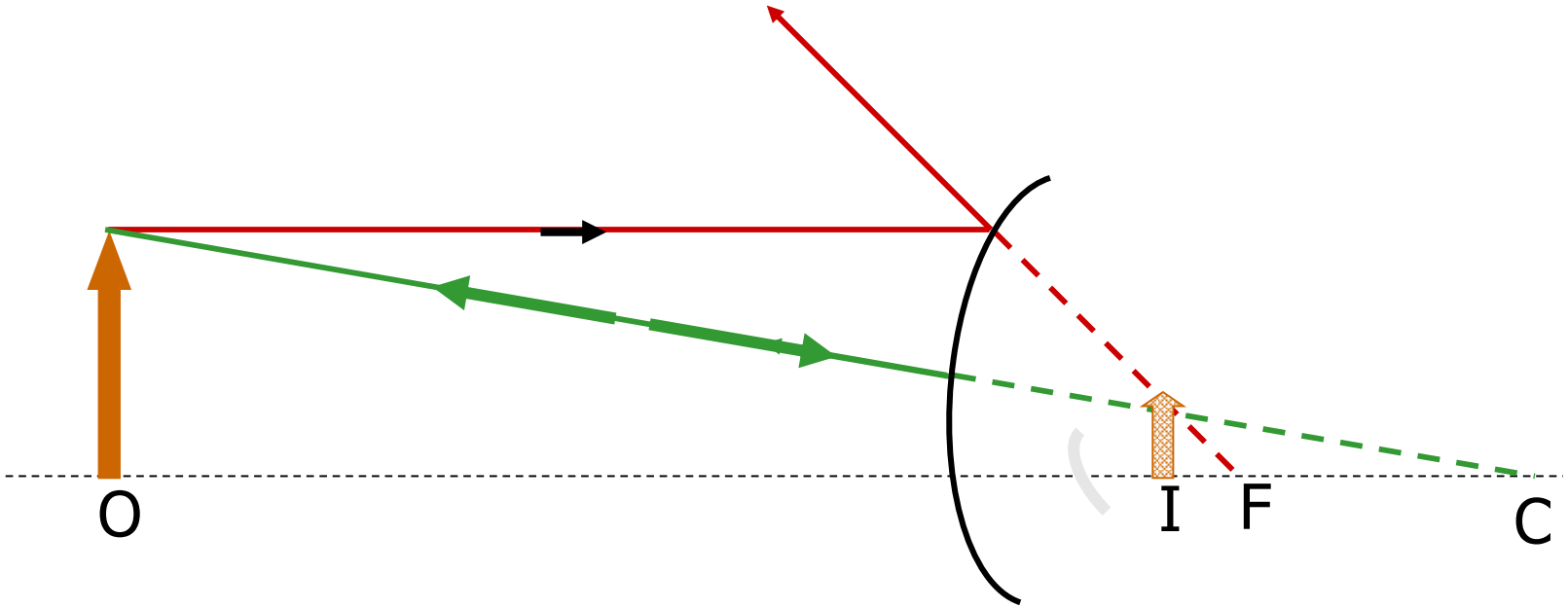


Image Formation By Convex Mirror

Image Formation By Convex Mirrors



- image is **virtual**, **upright**, and **smaller** than object

Mirror Formula

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

u= object distance
V = image distance
f= focal length

Real is positive sign convention for Mirror Formula

u: always +ve
v: +ve for real image and
-ve for virtual image
f: +ve for a converging (i.e. concave) mirror and -ve for a diverging (i.e. convex) mirror

Linear magnification,
M=v/u
M= height of image/height of object.

Worked Example

- Where should an object be placed along the principal axis of a converging mirror of radius of curvature 30 cm such that the real image formed is three times its size?
- Solution:
- f = 30/2 = 15 m, u = ?, v =?

But M= v/u = 3; v= 3u

Using Real is positive sign convention, f is +ve, v is also +ve

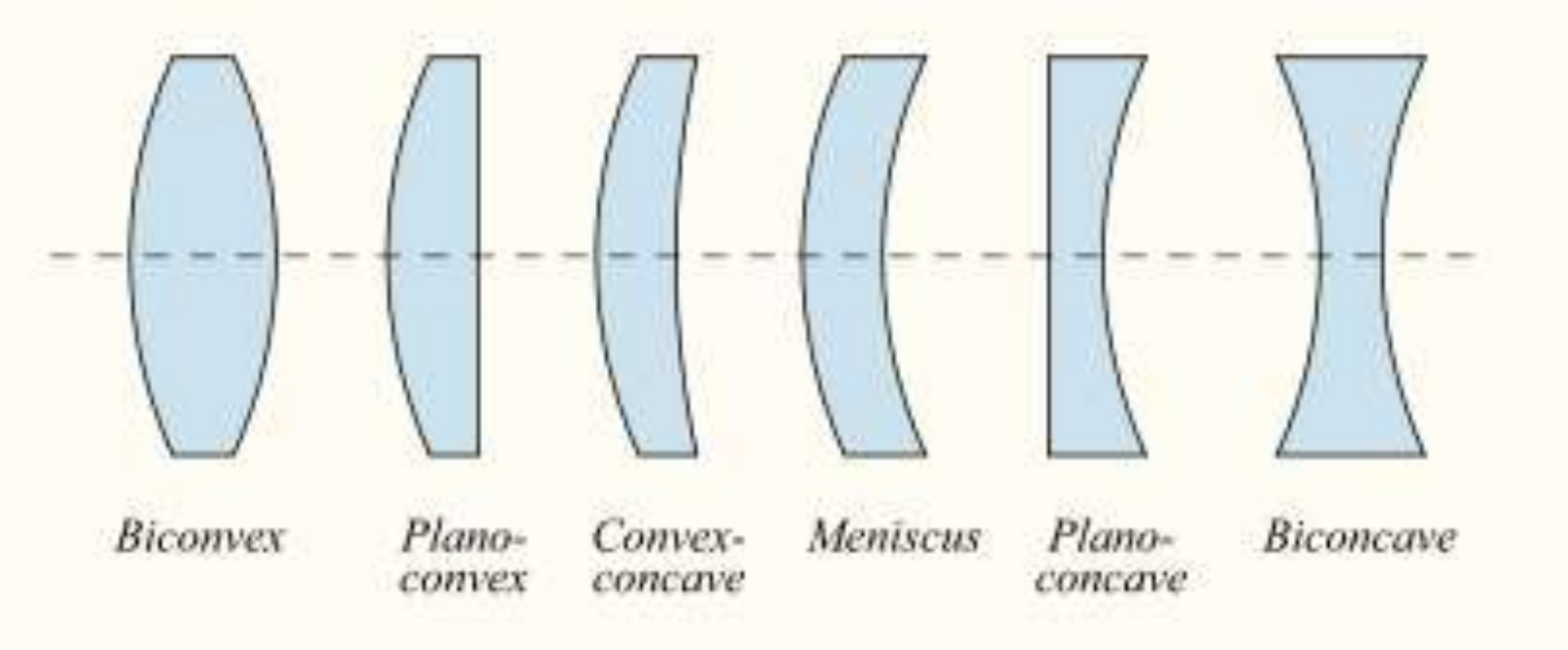
1/u +1/v = 1/f; 1/u +1/3u = 1/15

u = 20 cm

Exercises

- A convex rear view car mirror has a radius of curvature of 40 cm. Determine the location of the image and its magnification for an object 10 m from the mirror.
- An object of height is 1.8 cm) is placed 100 cm away from a concave mirror. The image height is 0.9 cm and the image is inverted. What is the focal length of the mirror ?

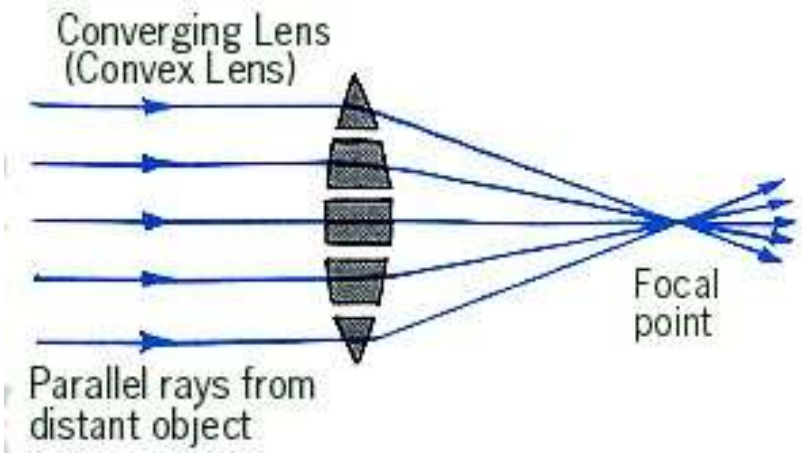
Lenses


Biconvex
Plano-convex
Convex-concave
Meniscus
Plano-concave
Biconcave

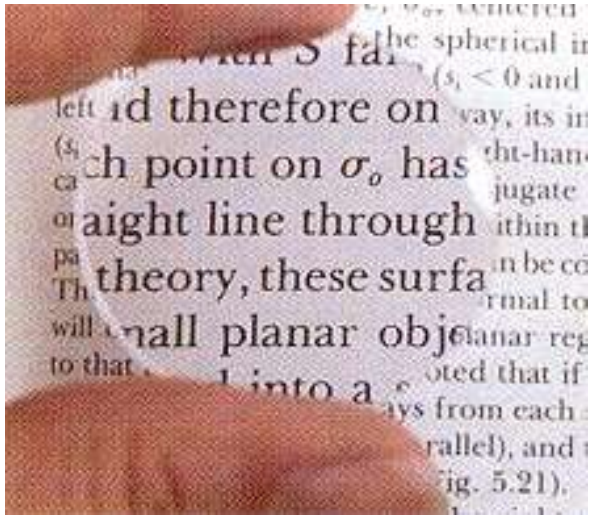
Convex (Converging) Lenses

Thicker in the center than edges.

- Lens that converges (brings together) light rays.
- Forms real images and virtual images depending on position of the object



The Magnifier



Converging Lens

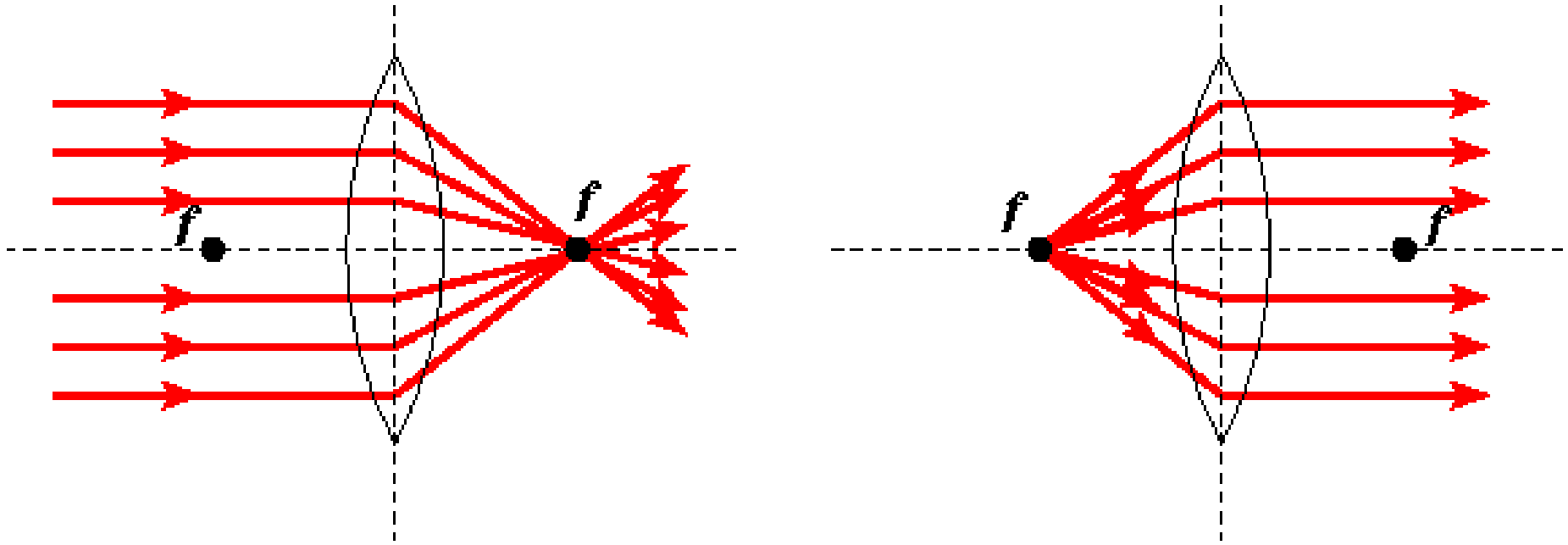
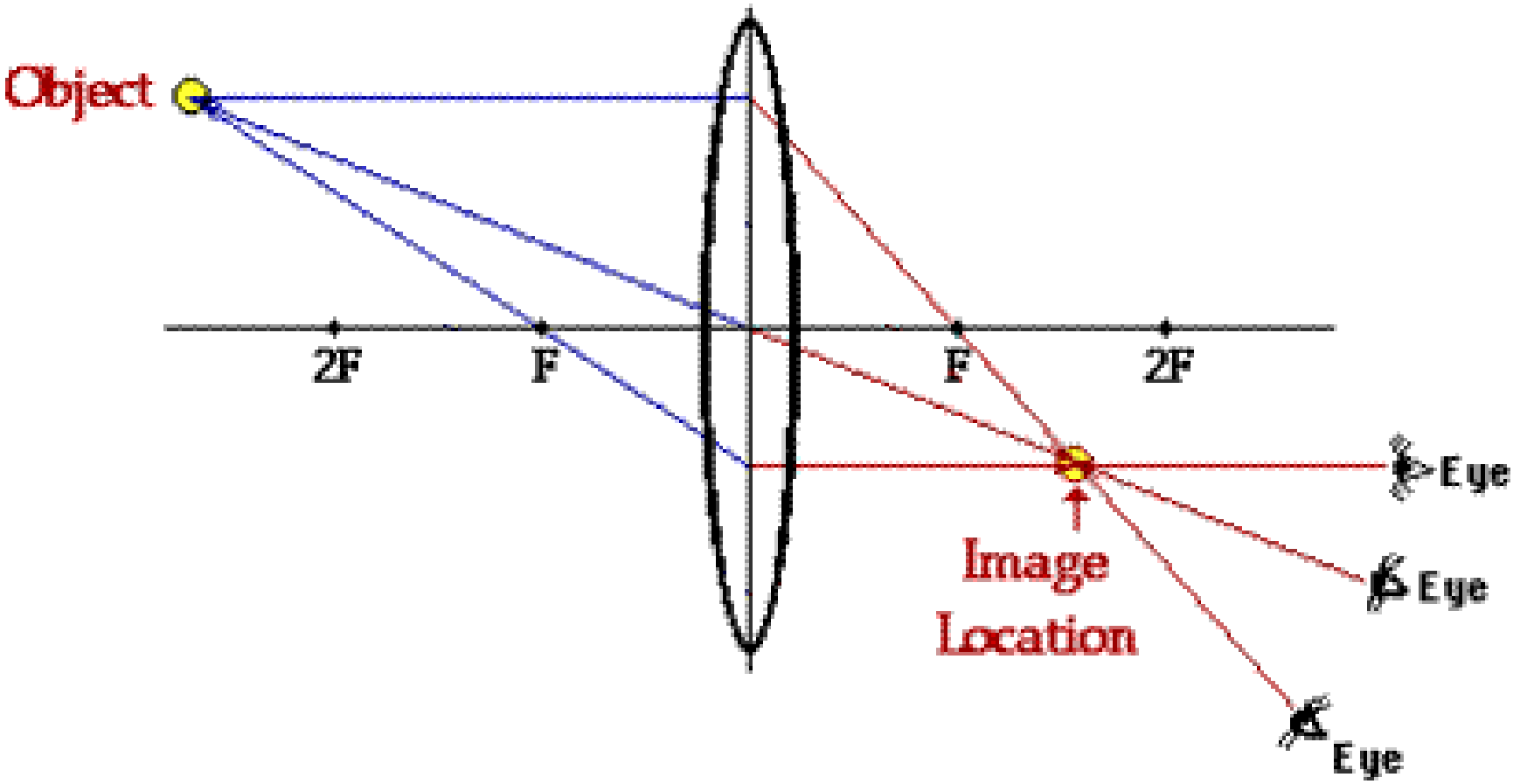


Image Formation by Converging Lens



- 1. Object within *F′*:
  - The image is formed on the same side of the lens.
  - Nature of the image: erect, magnified, virtual

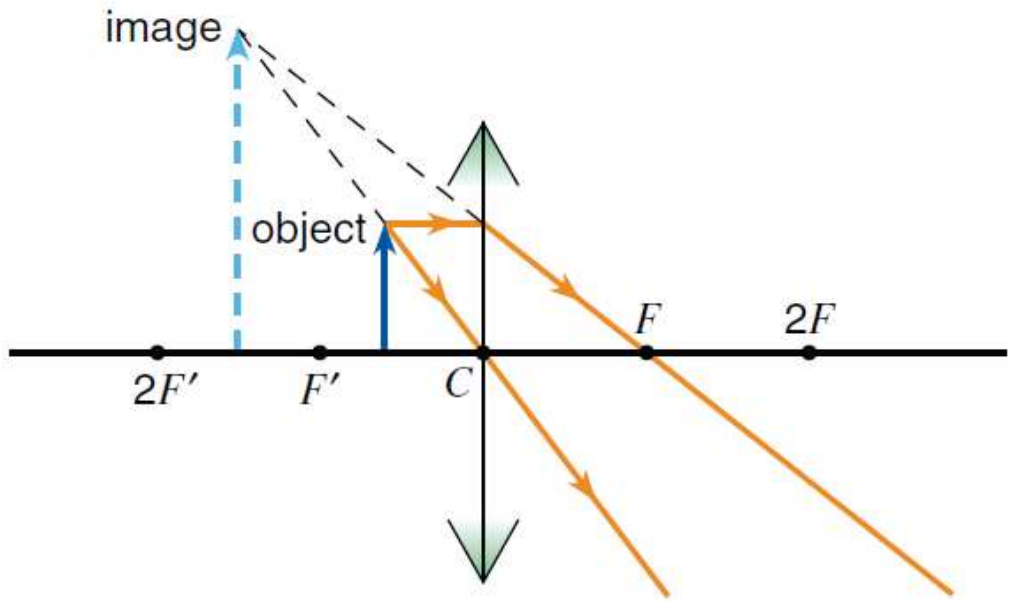
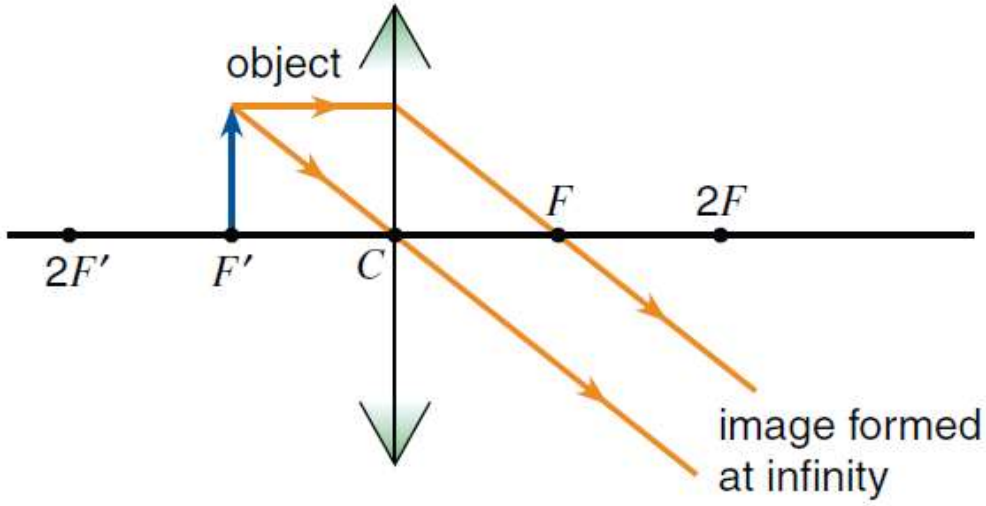
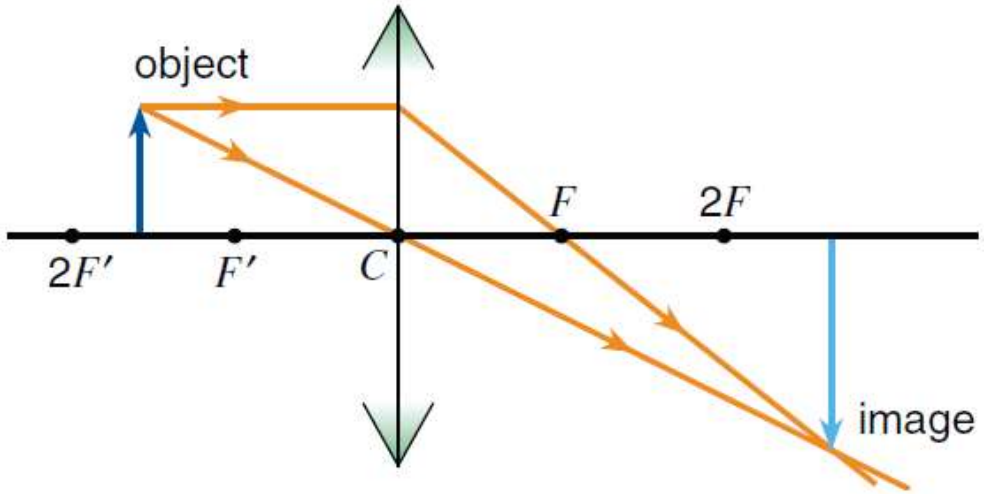


Image formed by convex lens

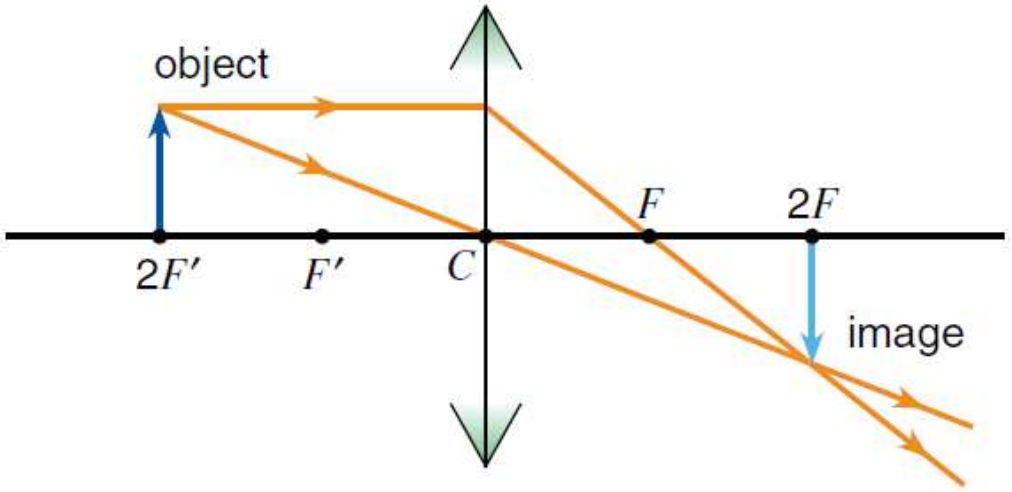
- 2. Object at *F′*:
  - The image is formed at infinity.



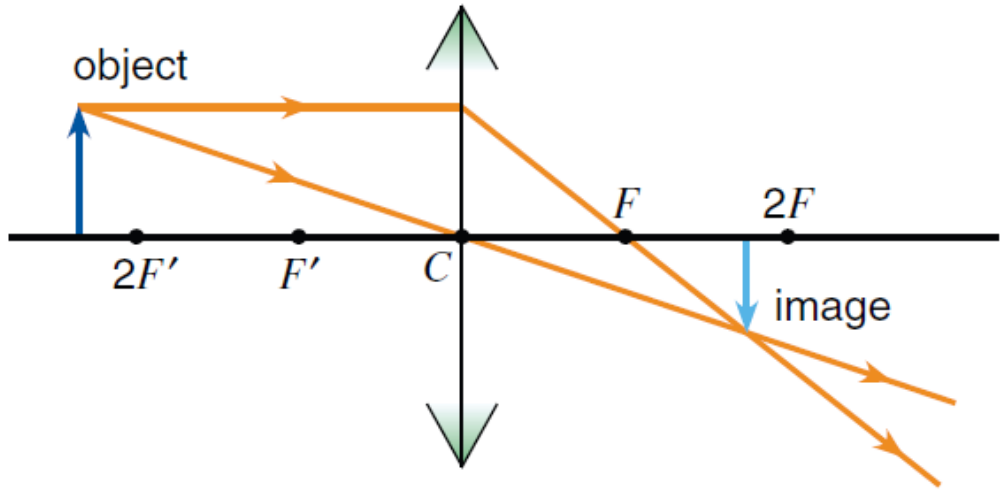
- 3. Object between *F′* and 2*F′*:
  - The image is formed on the other side of the lens beyond 2*F*.
  - Nature of the image: inverted, magnified, real



- 4. Object at 2*F′*:
  - The image is formed on the other side of the lens at 2*F*.
  - Nature of the image: inverted, same size as the object, real

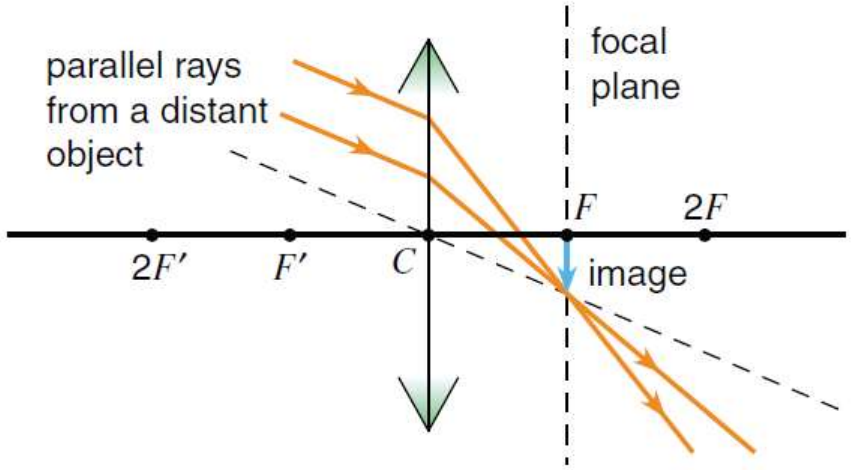


- 5. Object beyond 2*F*′:
  - The image is formed on the other side of the lens between *F* and 2*F*.
  - Nature of the image: inverted, diminished, real

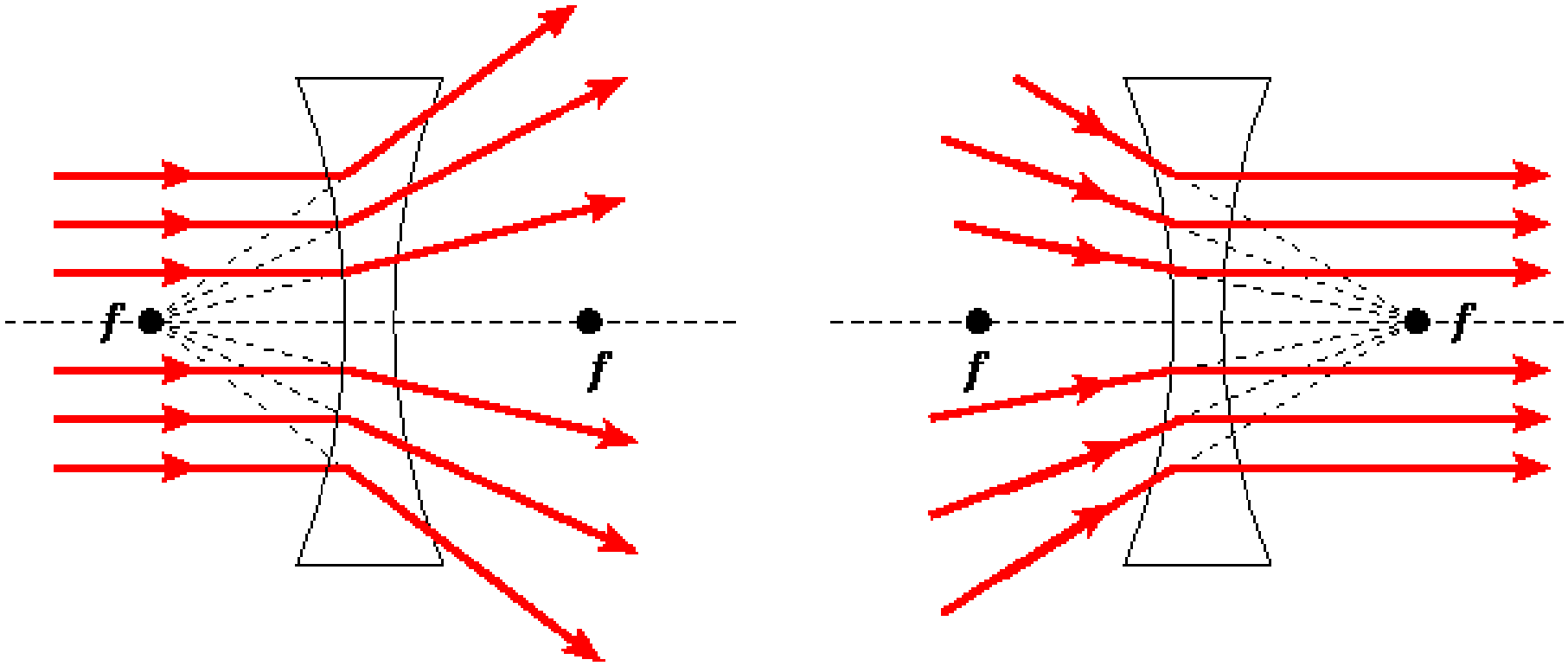


Nature of images formed by convex lenses

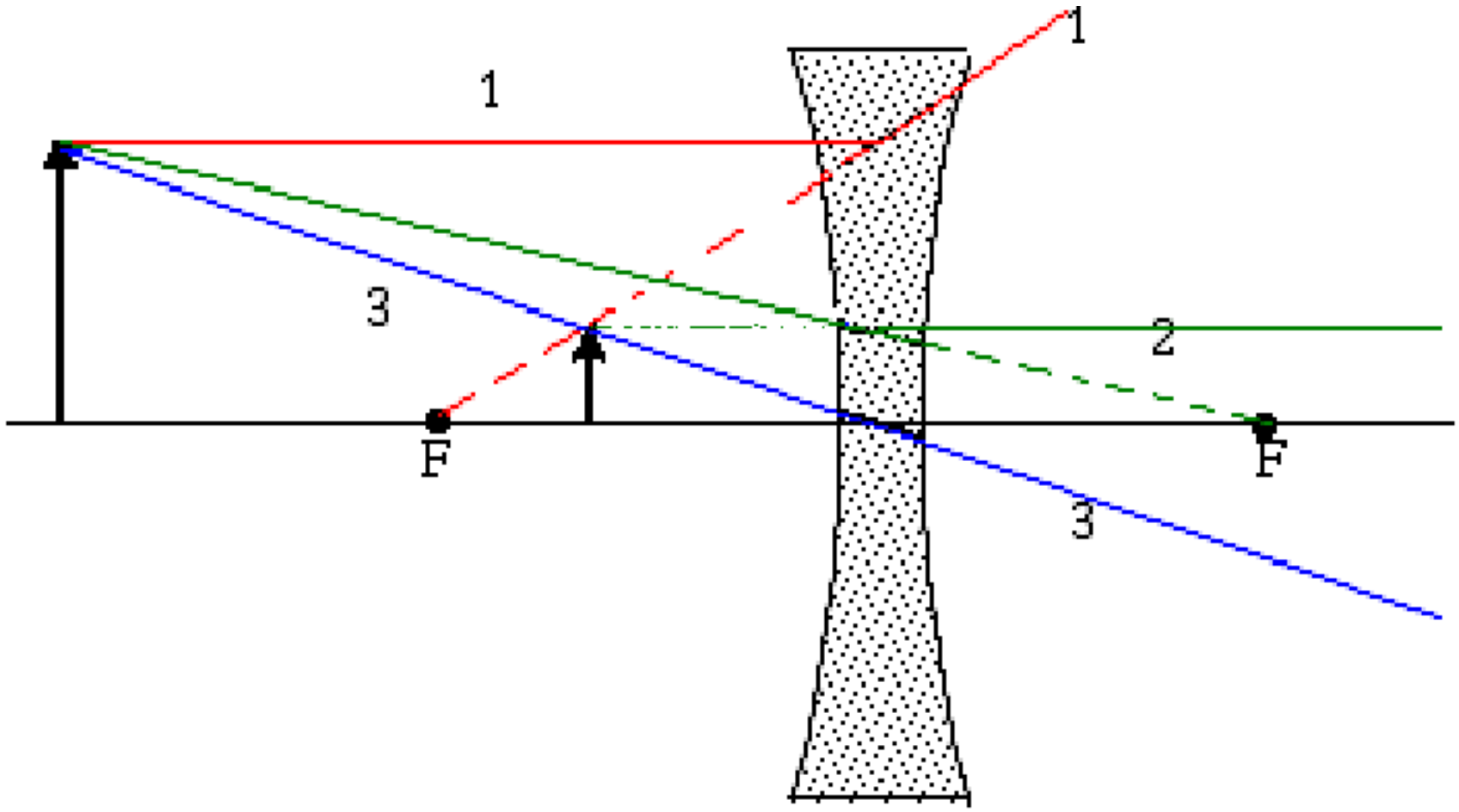
- 6. Distant object:
  - The image is formed on the focal plane on the other side of the lens.
  - Nature of the image: inverted, diminished, real



Concave (Diverging) Lens



Diverging Lens Image Formation



Always Virtual, Smaller, and Right-Side Up

Definition of Terms

- Refractive Power of a Lens: This is defined as the inverse of the focal length of a lens. It is the ability of a lens to bend light rays that pass through it.
- Unit – dioptre (D)
- Power = 1/f.
- Spherical aberration: An optical defect in which a converging lens or mirror is unable to a common focus, parallel rays that are close to its principal axis.
- Cause of spherical aberration: imperfect shape of a mirror or lens

- Chromatic Aberration: Inability of a converging lens to bring to a common focus light rays of different wavelengths.
- Cause: A glass has different refractive indices for lights of different wavelengths

Lens Formula

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

u= object distance
V = image distance
f= focal length

Real is positive sign convention for Lens Formula

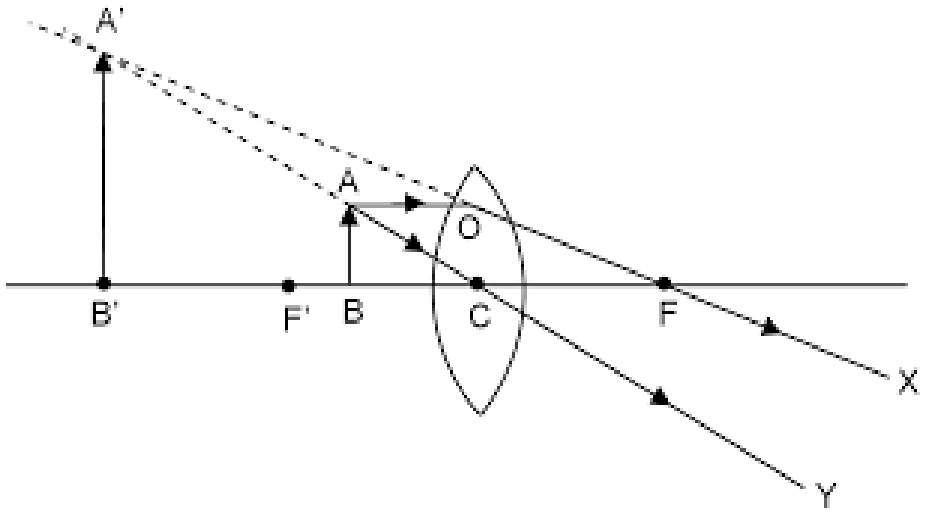
u: always +ve
v: +ve for real image and
-ve for virtual image
f: +ve for a converging (i.e. convex) lens and -ve for a diverging (i.e. concave) lens

Linear magnification,
M=v/u
M= height of image/height of object.

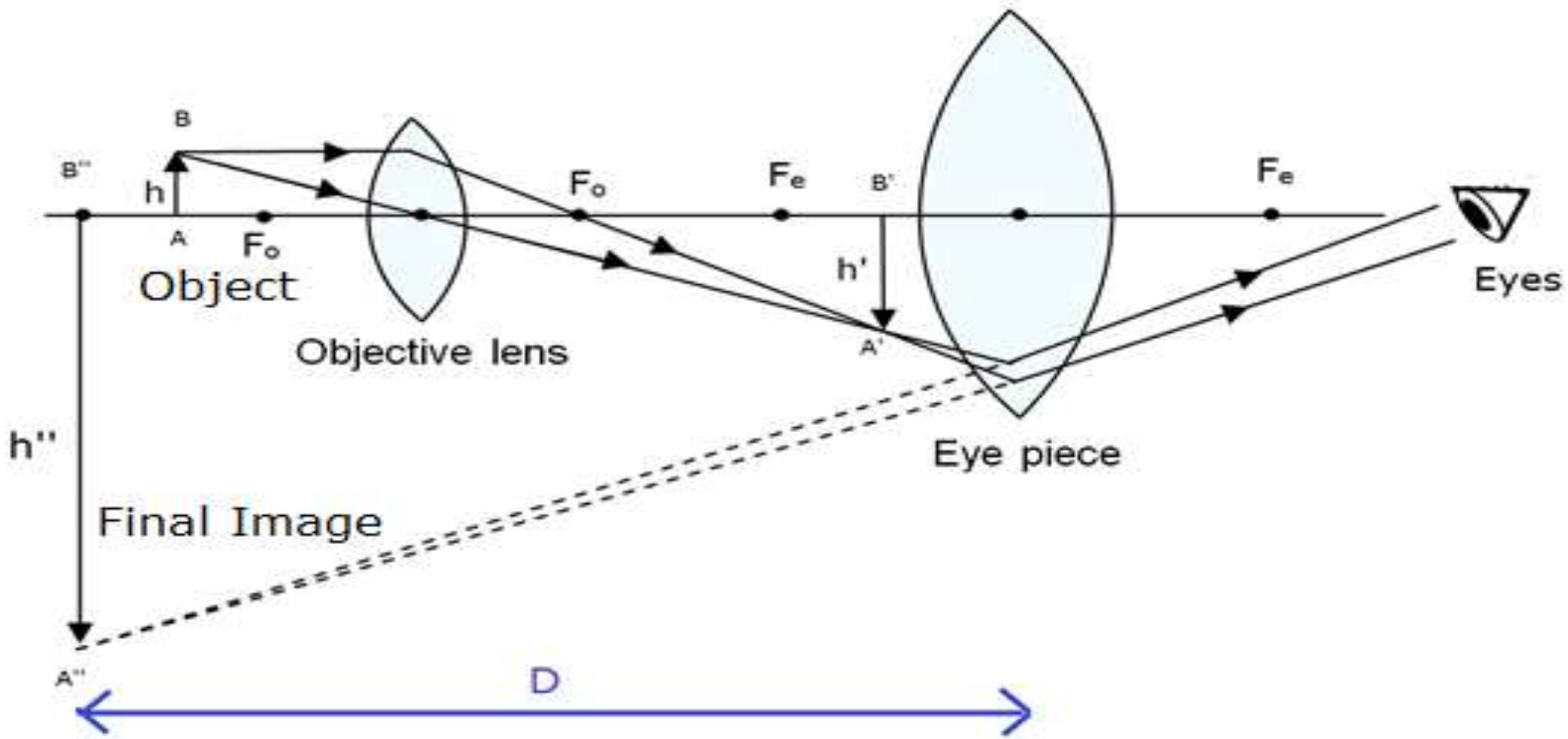
Applications of Refraction Through Lenses: Optical Instruments

- Simple Microscopes.
- Compound Microscopes.
- Telescopes

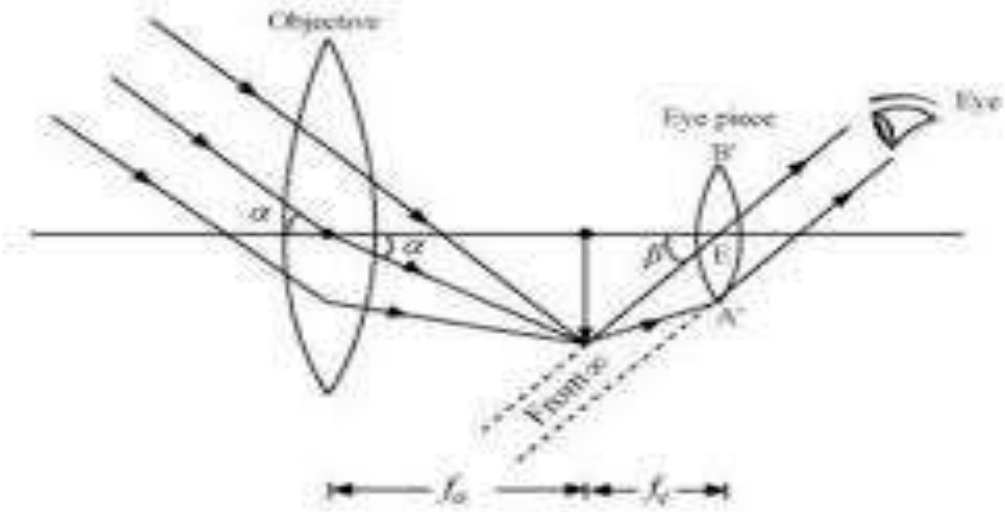
Simple Microscope (Hand Lens)



Compound Microscope



Astronomical Telescope



Magnifying power, $m = -\frac{f_o}{f_e}$

Lensmakers Formula

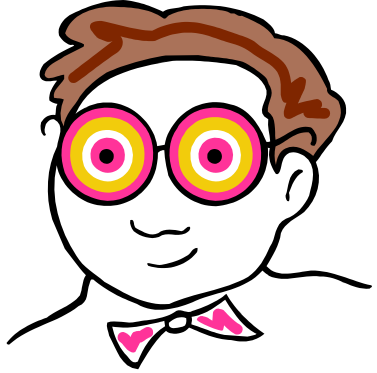
- For a thin lens of radii of curvatures R1 and R2, placed in a medium of refractive index n*.

The Lensmakers formula is given by:

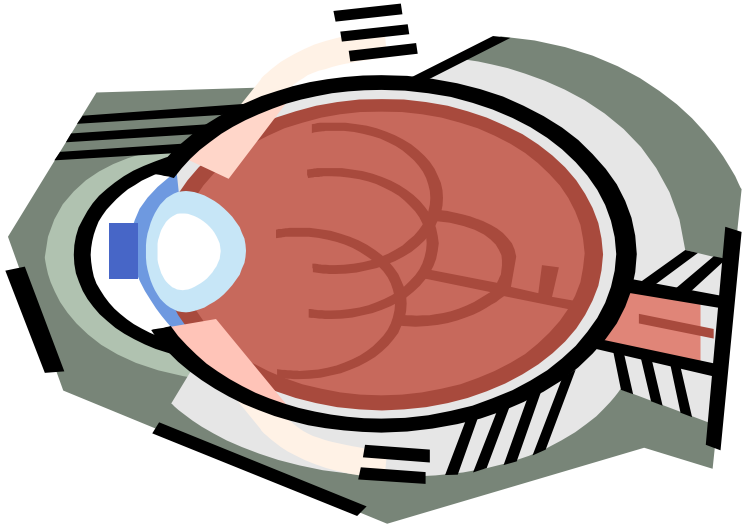
$$\frac{1}{f} = \left(\frac{n}{n^*} - 1\right)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

n= refractive index of the material of the lens

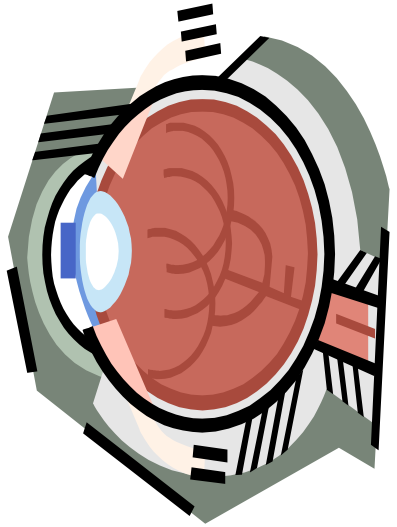
- R is +ve for a convex surface and –ve for a concave surface



How You See

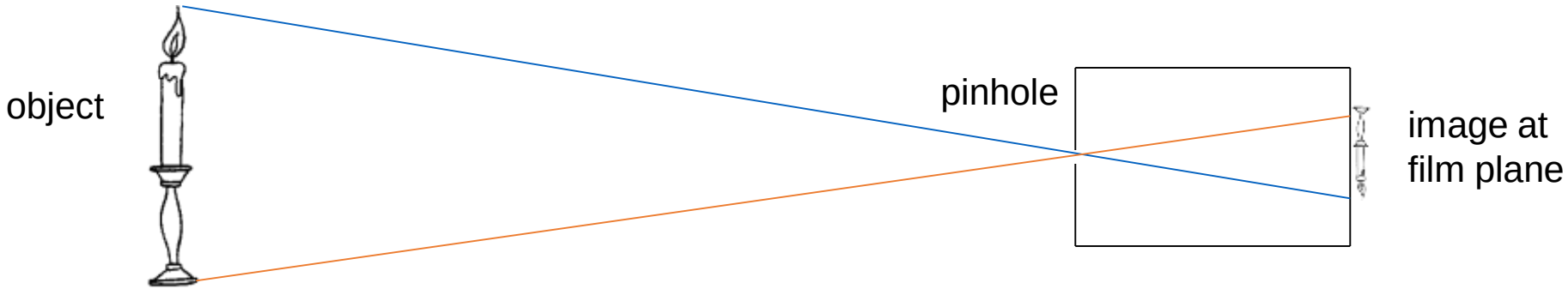


- Near Sighted – Eyeball is too long and image focuses in front of the retina
  - Near Sightedness – Concave lenses expand focal length

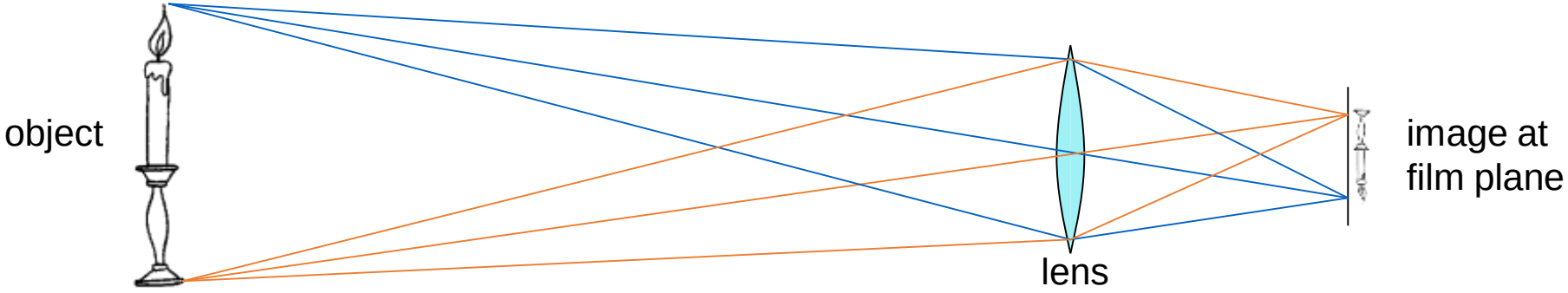


- Far Sighted – Eyeball is too short so image is focused behind the retina.
  - Far Sightedness – Convex lense shortens the focal length.

Cameras, in brief



In a pinhole camera, the hole is so small that light hitting any particular point on the film plane must have come from a particular direction outside the camera



In a camera with a lens, the same applies: that a point on the film plane more-or-less corresponds to a direction outside the camera. Lenses have the important advantage of collecting more light than the pinhole admits